

Package ‘fiscal’

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Title Nonprofit Fiscal Health Metrics

Version 0.0.2.0000

Description

Calculate standard fiscal health accounting metrics for nonprofits from IRS 990 or 990-EZ data.

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URL <https://nonprofit-open-data-collective.github.io/fiscal/>, <https://github.com/Nonprofit-Open-Data-Collective/fiscal>

BugReports <https://github.com/Nonprofit-Open-Data-Collective/fiscal/issues>

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apply_normalization	<i>Apply a fitted normalization to a numeric vector</i>
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Description

Given a `normalize_x_fit` object produced by `find_best_normalization()`, re-winsorizes `x` using the same settings, applies the fitted transformation, and standardizes using the fitted center and scale.

Using the fitted ECDF / parameters rather than re-fitting ensures that new observations (e.g. from a different year) are scored on the same scale as the training data.

Usage

```
apply_normalization(x, fit, verbose = FALSE)
```

Arguments

<code>x</code>	Numeric vector to score.
<code>fit</code>	A <code>normalize_x_fit</code> object returned by <code>find_best_normalization()</code> .
<code>verbose</code>	Logical. Print a brief summary. Default FALSE.

Value

A numeric vector of the same length as `x` with transformation metadata attached as attributes (`transform_type`, `transform_label`, `transform_note`, `guess_reason`, `center`, `scale`, `range`, `winsorize`, `offset`, `fit_object`).

See Also

[find_best_normalization\(\)](#), [normalize_x\(\)](#)

Examples

```
x_train <- c( rnorm(200), rep(0, 30) )
x_new    <- rnorm(50)

# fit on training data, score new data with same parameters
fit <- find_best_normalization( x_train, range = "np", verbose = FALSE )
z   <- apply_normalization( x_new, fit = fit )
```

`apply_transformations` *Apply winsorization, normalization, and percentile ranking to a ratio vector*

Description

`apply_transformations()` is the central post-computation step called by every `get_*`() ratio function. It produces four versions of a ratio:

- **raw** (`raw`; the unsuffixed metric column): the unmodified computed ratio.
- **winsorized** (`_w`): outliers clipped to bounds determined by the `range` argument and the `winsorize` proportion, via [winsorize_x\(\)](#).
- **normalized** (`_z`): a distribution-appropriate transformation of the winsorized values, robustly standardized. The transformation is selected and fitted via [find_best_normalization\(\)](#) (on the stable interior of non-NA, non-sentinel observations; the "rank_normal" reference distribution also keeps the winsorized tails), then scored on the full vector via [apply_normalization\(\)](#).
- **percentile** (`_p`): integer percentile rank (1-100) based on the raw values, via `dplyr::ntile()`.

Usage

```
apply_transformations(
  x,
  winsorize = 0.98,
  offset = 0.001,
  range = "np",
  normalize_type = NULL
)
```

Arguments

<code>x</code>	Numeric vector (the computed ratio, before any transformation).
<code>winsorize</code>	Winsorization proportion between 0 and 1 (default 0.98, which clips at the 1st and 99th percentiles for "np" range).
<code>offset</code>	Sentinel offset applied to fixed bounds (default 0.001). Observations clipped to a fixed bound are stored just outside it (a lower bound minus offset, an upper bound plus offset) so they remain identifiable in the <code>_w</code> column.
<code>range</code>	Character string describing the theoretical range of the ratio. Controls how the lower and upper winsorization bounds are determined: "np" Negative to positive (unbounded both directions). Winsorizes symmetrically at the $(1 - \text{winsorize})/2$ and $1 - (1 - \text{winsorize})/2$ percentiles. Default behaviour. "zp" Zero to positive. The lower bound is fixed at <code>-offset</code> (flagging truncated-at-zero values) and the upper bound is the <code>winsorize</code> percentile of the full distribution. "zo" Zero to one. Both bounds are fixed (<code>-offset</code> and <code>1+offset</code>), flagging values outside <code>[0, 1]</code> . No percentile-based clipping is applied. "nz" Negative to zero. The upper bound is fixed at <code>+offset</code> and the lower bound is the <code>1 - winsorize</code> percentile. "lo;hi" Custom numeric range, e.g. "0;10". The lower bound is fixed at <code>lo - offset</code> and the upper bound at <code>hi + offset</code> .
<code>normalize_type</code>	Transformation type override passed to find_best_normalization() . One of NULL (auto-detect among "asinh", "logit", and "rank_normal"), "asinh", "logit", "rank_normal", or "hurdle" (never auto-selected). Default NULL.

Details**Winsorization**

Delegated entirely to [winsorize_x\(\)](#), which handles all range codes, computes sentinel flags, and returns the winsorized vector alongside diagnostic metadata.

Normalization (`_z` column)

[find_best_normalization\(\)](#) is called first to fit transformation parameters (type, scale constant, center, spread) on the stable interior of the winsorized distribution. [apply_normalization\(\)](#) then scores the full original vector using those fitted parameters. This two-step design means the fitted model can be reused on new data if needed. Centering and scaling use median/MAD, falling back to mean/SD when the MAD is zero. With fewer than three non-missing values no model is fitted and `z` is returned as NA.

Percentile rank (`_p` column)

Integer percentile rank from 1 to 100 based on the raw (pre-winsorized) values, computed with [dplyr::ntile\(\)](#).

Value

A named list with elements `raw`, `winsorized`, `z`, `pctile`.

See Also

[winsorize_x\(\)](#), [find_best_normalization\(\)](#), [apply_normalization\(\)](#), [normalize_x\(\)](#), [plot_normalize_x\(\)](#)

Examples

```
library( fiscal )
data( dat10k )

# winsorize and normalize the debt-to-assets ratio
ratio <- dat10k$F9_10_LIAB_TOT_EOY / dat10k$F9_10_ASSET_TOT_EOY
out  <- apply_transformations( ratio, winsorize = 0.98, range = "zo" )
names( out )
summary( out$z )
```

coerce_numeric	<i>Coerce selected columns to numeric</i>
----------------	---

Description

Converts specified columns in a data frame to numeric, with a guard against non-digit content. Stops if letters are detected in any column; warns if silent coercion was needed.

Usage

```
coerce_numeric(d, vars)
```

Arguments

- d A data.frame.
- vars Character vector of column names to coerce.

Details

bit64 integer64 columns are converted to double. Columns that hold integer64 bytes with the class attribute dropped (every nonzero value is smaller than 1e-300 in absolute value) are detected and repaired, with a warning.

Value

The data frame with the specified columns coerced to numeric.

compute_all	<i>Compute all fiscal health metrics in one call</i>
-------------	--

Description

Applies every get_*() ratio function in the fiscal package to a data frame in a single call. The data frame is sanitized once up front using [sanitize_financials\(\)](#), then each function is called with sanitize = FALSE to avoid redundant imputation passes. A metric that cannot be computed (for example, because one of its input columns is missing from df) is skipped with a warning.

Usage

```
compute_all(  
  df,  
  metrics = c("ratio", "w", "z", "p"),  
  append_to_df = TRUE,  
  winsorize = 0.98,  
  verbose = TRUE  
)
```

Arguments

df	A data.frame containing IRS 990 efile financial fields.
metrics	Character vector of metric variants to return. Any combination of "ratio", "w", "z", "p". Defaults to all four.
append_to_df	Logical. If TRUE (default), returns the original df with metric columns appended. If FALSE, returns only .IDVARS identifier columns plus the selected metric columns.
winsorize	Winsorization proportion (default 0.98).
verbose	Logical. If TRUE (default), prints progress messages.

Value

A data.frame.

Examples

```
library( fiscal )  
data( dat10k )  
d <- compute_all( dat10k )  
dim( d )
```

compute_all_panel	<i>Compute all fiscal health metrics year-by-year across a panel</i>
-------------------	--

Description

A panel-safe wrapper around `compute_all()` that splits the data by year, computes every fiscal health metric independently within each year's slice, then stacks the results. This ensures that winsorization bounds, normalization parameters, and percentile ranks are all estimated from the cross-sectional distribution of a **single year** rather than from the pooled multi-year distribution.

Why year-by-year computation matters:

`compute_all()` passes each raw ratio vector through `apply_transformations()`, which fits winsorization bounds and a normalization model on the data it receives. On a pooled panel those distributions are a mixture of N years, so:

- A ratio at the 95th percentile in 2021 may rank at the 80th percentile in the pooled distribution if 2022—2024 values are higher.
- Winsorization clips to pooled extremes, not year-specific extremes.

- Z-scores are centred and scaled on pooled moments.

Splitting by year before computing avoids all three problems: every `_w`, `_z`, and `_p` column is interpretable as a within-year percentile or standardised value, exactly as it would be for a single-year call to `compute_all()`.

Usage

```
compute_all_panel(
  df,
  year = "TAX_YEAR",
  metrics = c("ratio", "w", "z", "p"),
  append_to_df = TRUE,
  winsorize = 0.98,
  verbose = TRUE
)
```

Arguments

<code>df</code>	A <code>data.frame</code> containing IRS 990 efile financial fields for multiple tax years. Must contain the column named by <code>year</code> .
<code>year</code>	Name of the column that identifies the tax year. Default "TAX_YEAR".
<code>metrics</code>	Character vector of metric variants to return. Any combination of "ratio", "w", "z", "p". Default: all four.
<code>append_to_df</code>	Logical. If TRUE (default), returns the original <code>df</code> with metric columns appended. If FALSE, returns only <code>.IDVARS</code> identifier columns plus the selected metric columns.
<code>winsorize</code>	Winsorization proportion (default 0.98).
<code>verbose</code>	Logical. If TRUE (default), prints per-year progress from <code>compute_all()</code> .

Details

Implementation:

Internally the function:

1. Validates that `year` is present and that the panel contains more than one year (a single-year panel works but a warning is issued since `compute_all()` would give identical results).
2. Splits the row indices of `df` by year with `base::split()`. Rows with a missing year are set aside with a warning.
3. Calls `compute_all()` on each year's slice with `verbose` passed through so per-year messages are visible (or suppressed) as requested. If a year fails, its rows are kept with NA metric columns.
4. Tags each result with the original row positions, stacks the per-year results (and any rows without a year) with `data.table::rbindlist()`, which fills columns missing from some years with NA, and restores the input row order.

Panel size considerations:

On large panels (e.g. 5 years – 50 000 orgs) the per-year loop runs `compute_all()` five times on ~50 000-row slices rather than once on a 250 000-row frame. Each slice call is faster and uses less peak memory; the `data.table::rbindlist()` stack is $O(N)$ with no copies.

Value

A `data.frame` with the same row order as `df` and the same metric columns that `compute_all()` would produce, but with `_w` / `_z` / `_p` values computed within each year's cross-sectional distribution. Rows whose year is NA are kept, with NA metric columns.

See Also

`compute_all()`, `get_panel()`, `deduplicate()`, `panel_smooth()`

Examples

```
## Not run:
panel <- get_panel( years = 2019:2022 )
panel <- deduplicate( panel )

# Year-safe metric computation
panel_ratios <- compute_all_panel( panel )

# Compare: without year splitting (pooled --- incorrect for _w/_z/_p)
panel_ratios_pooled <- compute_all( panel )

# Check that percentile ranks are within-year:
library( data.table )
setDT( panel_ratios )
panel_ratios[ , range(debt_assets_p), by = TAX_YEAR ]
# All years should show 1 to 100.

## End(Not run)
```

dat10k

*IRS 990 Efile Dataset - 2023 Tax Year (10,000 Organizations)***Description**

A random sample of 10,000 nonprofit organizations drawn from the 2023 IRS 990 efile database, combining financial data from Form 990 Parts I, VIII, IX, and X with organizational metadata from the IRS Business Master File (BMF). The sample carries every efile field that the `get_*`() metric functions use by default, so `compute_all()` runs the full battery offline.

Usage

```
dat10k
```

Format

A data frame with 10,000 rows. The dataset includes both full 990 filers and 990-EZ filers; the `RETURN_TYPE` column identifies which form each organization filed. Dollar amounts are whole dollars stored as plain doubles (never bit64 integer64). Variables are organized into the following groups:

Organization identifiers:

- EIN2: Employer Identification Number ("EIN-XX-XXXXXXX")

- OBJECTID: Efile record identifier
- ORG_EIN: Organization EIN (integer)
- ORG_NAME_L1: Organization name (line 1)
- ORG_NAME_L2: Organization name (line 2, if present)

BMF classification and geography:

As returned by `retrieve_efile_data( include_bmf = TRUE )`.

- ntee_code_clean: Normalized three-character NTEE code
- ntee_code_major_group: NTEE major group label
- nteev2: NTEE version 2 code, [SUBSECTOR]–[NTEE]–[ORGTTYPE]
- nteev2_subsector: NTEE v2 subsector (e.g. "HEL", "EDU", "UNU")
- nteev2_org_type: NTEE v2 organization type (e.g. "RG" for regular)
- subsection_code: IRS 501(c) subsection code
- foundation_code: IRS foundation type code
- ruling_year: Year of the IRS exemption ruling
- filing_requirement_code: IRS 990 filing requirement code
- asset_amount, income_amount, revenue_amount: Most recent asset, income, and revenue amounts reported to the BMF (stored as character)
- geo_state_abbr: Two-letter state abbreviation
- geo_county: County name
- geo_metro_area: Metropolitan area name (blank outside metro areas)
- bmf_vintage_ym: BMF release used for the merge ("YYYY-MM")

Return header fields (Part 00):

- RETURN_AMENDED_X: Amended return indicator
- RETURN_GROUP_X: Group return indicator
- RETURN_PARTIAL_X: Partial return indicator
- RETURN_TAXPER_DAYS: Length of tax period in days
- RETURN_TIME_STAMP: Return timestamp
- RETURN_TYPE: Form type filed: "990" or "990EZ"
- TAX_PERIOD_BEGIN_DATE: Start of tax period
- TAX_PERIOD_END_DATE: End of tax period
- TAX_YEAR: Tax year (fiscal year start)
- URL: URL of the source XML filing on AWS
- VERSION: 990 form schema version
- F9_00_EXEMPT_STAT_4947A1_X: 4947(a)(1) trust indicator
- F9_00_EXEMPT_STAT_501C_X: 501(c) exempt status indicator
- F9_00_EXEMPT_STAT_501C3_X: 501(c)(3) indicator
- F9_00_TYPE_ORG_ASSOC_X: Association type indicator
- F9_00_TYPE_ORG_CORP_X: Corporation type indicator
- F9_00_TYPE_ORG_OTH_X: Other type indicator
- F9_00_TYPE_ORG_TRUST_X: Trust type indicator
- F9_00_YEAR_FORMATION: Year of formation

Part I - Summary:

- F9_01_EXP_REV_LESS_EXP_CY: Revenues less expenses, current year (Part I, line 19)
- F9_01_EXP_TOT_CY: Total expenses, current year (Part I, line 18)

- F9_01_NAFB_ASSET_TOT_EOY: Total assets, EOY (Part I, line 20)
- F9_01_NAFB_LIAB_TOT_EOY: Total liabilities, EOY (Part I, line 21)
- F9_01_NAFB_TOT_BOY: Net assets or fund balances, BOY (Part I, line 22)
- F9_01_NAFB_TOT_EOY: Net assets or fund balances, EOY (Part I, line 22)
- F9_01_REV_PROG_TOT_CY: Program service revenue, current year (Part I, line 9)
- F9_01_REV_TOT_CY: Total revenue, current year (Part I, line 12)

Part VIII - Revenue:

- F9_08_REV_CONTR_FED_CAMP: Federated campaigns (Part VIII, line 1a)
- F9_08_REV_CONTR_MEMBERSHIP_DUE: Membership dues (Part VIII, line 1b)
- F9_08_REV_CONTR_FUNDR_EVNT: Fundraising events (Part VIII, line 1c)
- F9_08_REV_CONTR_RLTD_ORG: Related organizations (Part VIII, line 1d)
- F9_08_REV_CONTR_GOVT_GRANT: Government grants (Part VIII, line 1e)
- F9_08_REV_CONTR_TOT: Total contributions (Part VIII, line 1h)
- F9_08_REV_PROG_TOT_TOT: Program service revenue total (Part VIII, line 2g)
- F9_08_REV_OTH_INVEST_INCOME_TOT: Investment income (Part VIII, line 3)
- F9_08_REV_OTH_INVEST_BOND_TOT: Income from investment of tax-exempt bond proceeds (Part VIII, line 4)
- F9_08_REV_OTH_ROY_TOT: Royalties (Part VIII, line 5)
- F9_08_REV_OTH_RENT_NET_TOT: Net rental income (Part VIII, line 6d)
- F9_08_REV_OTH_SALE_GAIN_NET_TOT: Net gain or (loss) from sales of assets other than inventory (Part VIII, line 7d)
- F9_08_REV_OTH_FUNDR_NET_TOT: Net income from fundraising events (Part VIII, line 8c)
- F9_08_REV_MISC_OTH_TOT: Other miscellaneous revenue (Part VIII, line 11d)
- F9_08_REV_TOT_TOT: Total revenue (Part VIII, line 12A)

Part IX - Expenses (column A, total, unless noted):

- F9_09_EXP_GRANT_US_ORG_TOT: Grants to domestic organizations and governments (Part IX, line 1)
- F9_09_EXP_GRANT_US_INDIV_TOT: Grants and assistance to domestic individuals (Part IX, line 2)
- F9_09_EXP_GRANT_FRGN_TOT: Grants to foreign organizations, governments, and individuals (Part IX, line 3)
- F9_09_EXP_BEN_PAID_MEMB_TOT: Benefits paid to or for members (Part IX, line 4)
- F9_09_EXP_COMP_DTK_TOT: Compensation of current officers, directors, trustees, and key employees (Part IX, line 5)
- F9_09_EXP_COMP_DSQ_PERS_TOT: Compensation of disqualified persons (Part IX, line 6)
- F9_09_EXP_OTH_SAL_WAGE_TOT: Other salaries and wages (Part IX, line 7)
- F9_09_EXP_PENSION_CONTR_TOT: Pension plan accruals and contributions (Part IX, line 8)
- F9_09_EXP_OTH_EMPL_BEN_TOT: Other employee benefits (Part IX, line 9)
- F9_09_EXP_PAYROLL_TAX_TOT: Payroll taxes (Part IX, line 10)
- F9_09_EXP_FEE_SVC_MGMT_TOT: Fees for services - management (Part IX, line 11a)
- F9_09_EXP_FEE_SVC_LEGAL_TOT: Fees for services - legal (Part IX, line 11b)
- F9_09_EXP_FEE_SVC_ACC_TOT: Fees for services - accounting (Part IX, line 11c)
- F9_09_EXP_FEE_SVC_LOB_TOT: Fees for services - lobbying (Part IX, line 11d)
- F9_09_EXP_FEE_SVC_FUNDR_TOT: Professional fundraising services (Part IX, line 11e)
- F9_09_EXP_FEE_SVC_INVEST_TOT: Investment management fees (Part IX, line 11f)

- F9_09_EXP_FEE_SVC_OTH_TOT: Fees for services - other (Part IX, line 11g)
- F9_09_EXP_PAY_AFFIL_TOT: Payments to affiliates (Part IX, line 21)
- F9_09_EXP_DEPREC_TOT: Depreciation, depletion, and amortization (Part IX, line 22A)
- F9_09_EXP_TOT_FUNDR: Fundraising expenses (Part IX, line 25D)
- F9_09_EXP_TOT_MGMT: Management and general expenses (Part IX, line 25C)
- F9_09_EXP_TOT_PROG: Program service expenses (Part IX, line 25B)
- F9_09_EXP_TOT_TOT: Total functional expenses (Part IX, line 25A)

Part X - Balance Sheet:

- F9_10_ASSET_CASH_BOY: Cash, beginning of year (Part X, line 1A)
- F9_10_ASSET_CASH_EOY: Cash, end of year (Part X, line 1B)
- F9_10_ASSET_SAVING_EOY: Savings and temporary cash investments, EOY (Part X, line 2B)
- F9_10_ASSET_PLEDGE_NET_EOY: Net pledges and grants receivable, EOY (Part X, line 3B)
- F9_10_ASSET_ACC_NET_EOY: Accounts receivable, net, EOY (Part X, line 4B)
- F9_10_ASSET_INV_SALE_EOY: Inventories for sale or use, EOY (Part X, line 8B)
- F9_10_ASSET_EXP_PREPAID_EOY: Prepaid expenses and deferred charges, EOY (Part X, line 9B)
- F9_10_ASSET_LAND_BLDG: Land, buildings, and equipment, cost or other basis (Part X, line 10a)
- F9_10_ASSET_LAND_BLDG_NET_EOY: Net land, buildings, and equipment, EOY (Part X, line 10C)
- F9_10_ASSET_INVEST_SEC_EOY: Investments - publicly traded securities, EOY (Part X, line 11B)
- F9_10_ASSET_INVEST_SEC_OTH_EOY: Investments - other securities, EOY (Part X, line 12B)
- F9_10_ASSET_TOT_EOY: Total assets, EOY (Part X, line 16B)
- F9_10_LIAB_ACC_PAYABLE_EOY: Accounts payable and accrued expenses, EOY (Part X, line 17B)
- F9_10_LIAB_GRANT_PAYABLE_EOY: Grants and similar amounts payable, EOY (Part X, line 18B)
- F9_10_LIAB_MTG_NOTE_EOY: Secured mortgages and notes payable, EOY (Part X, line 23B)
- F9_10_LIAB_NOTE_UNSEC_EOY: Unsecured notes and loans payable, EOY (Part X, line 24B)
- F9_10_LIAB_TOT_EOY: Total liabilities, EOY (Part X, line 26B)
- F9_10_NAFB_FOLLOW_SFAS117_X: Organization follows SFAS 117 / ASC 958 (Part X, check box above line 27)
- F9_10_NAFB_NO_FOLLOW_SFAS117_X: Organization does not follow SFAS 117 / ASC 958 (Part X, check box above line 30)
- F9_10_NAFB_UNRESTRICT_BOY: Net assets without donor restrictions, beginning of year (Part X, line 27A)
- F9_10_NAFB_UNRESTRICT_EOY: Net assets without donor restrictions, end of year (Part X, line 27B)
- F9_10_NAFB_RESTRICT_EOY: Net assets with donor restrictions, end of year (Part X, line 28B)
- F9_10_NAFB_CAP_STCK_EOY: Capital stock or trust principal, or current funds, EOY (Part X, line 30B)
- F9_10_NAFB_CAP_SURPLUS_EOY: Paid-in or capital surplus, EOY (Part X, line 31B)
- F9_10_NAFB_EARNING_RETAINED_EOY: Retained earnings, endowment, accumulated income, or other funds, EOY (Part X, line 32B)
- F9_10_NAFB_TOT_BOY: Total net assets, beginning of year (Part X, line 33A)
- F9_10_NAFB_TOT_EOY: Total net assets, end of year (Part X, line 33B)

Details

The dataset was built from the 2023 tax-year efile tables (F9-P00-T00-HEADER, F9-P01-T00-SUMMARY, F9-P08-T00-REVENUE, F9-P09-T00-EXPENSES, and F9-P10-T00-BALANCE-SHEET) with BMF fields (NTEE codes, geographic identifiers, and organizational metadata) merged on EIN, using `retrieve_efile_data()`. Filings were deduplicated to one per organization with `deduplicate()` before 10,000 organizations were sampled at random from filings with non-negative total revenue and a usable NTEE code. The BMF marks organizations without one as "UNDEFINED", "INVALID", or "Z99"; those are excluded, so every row has a valid `ntee_code_clean` and no `nteev2_subsector` is "UNU". The sample includes a mix of full 990 and 990EZ filers; see `RETURN_TYPE` to distinguish them.

Most Part VIII, IX, and X detail exists only on the full 990 (PC scope; see `get_pc_fields()`), so 990EZ filers have NA in those columns. Part I summary fields (F9_01_*) and the lines that have a 990EZ counterpart, such as total assets and total net assets (PZ scope; see `get_pz_fields()`), are populated for both forms. Use `sanitize_financials()` to impute zero for genuine zero-value fields before computing ratios.

Source

IRS 990 efile data (2023 tax year) via the Nonprofit Open Data Collective. <https://nonprofit-open-data-collective.github.io/irs990efile/>

Examples

```
data( dat10k )
dim( dat10k )
table( dat10k$RETURN_TYPE )

# Compute the government grant ratio for all organizations
d <- get_grants_govt_ratio( df = dat10k )
summary( d$grants_govt )

# Sanitize first, then compute a batch of ratios
dat_clean <- sanitize_financials( dat10k )
dat_clean <- get_grants_govt_ratio( dat_clean )
dat_clean <- get_program_expenses_ratio( dat_clean )
dat_clean <- get_debt_assets_ratio( dat_clean )
```

deduplicate

Select one filing per organization-year

Description

Compatibility wrapper around `panel990::panel_deduplicate()`.

Usage

```
deduplicate(
  df,
  by_id = "EIN2",
  by_year = "TAX_YEAR",
  col_group = "RETURN_GROUP_X",
  col_partial = "RETURN_PARTIAL_X",
  col_amended = "RETURN_AMENDED_X",
```

```
col_stamp = "RETURN_TIME_STAMP",
verbose = TRUE
)
```

Arguments

df	Filing data frame.
by_id	Organization identifier column.
by_year	Filing-year column.
col_group	Group-return flag column.
col_partial	Partial-return flag column.
col_amended	Amended-return flag column.
col_stamp	Filing timestamp column.
verbose	Print a summary.

Value

A data frame with one filing per organization-year.

detect_ez_rows	<i>Detect 990-EZ filer rows in an efile dataset</i>
----------------	---

Description

Returns a logical vector marking rows that belong to 990-EZ filers. Uses the RETURN_TYPE column when present; otherwise infers filer type from field availability (rows with Part I data but missing Part VIII total revenue are treated as 990-EZ filers).

Usage

```
detect_ez_rows(df)
```

Arguments

df	A data.frame containing efile data.
----	-------------------------------------

Value

A logical vector of length nrow(df).

Examples

```
data( dat10k )
table( detect_ez_rows( dat10k ) )
```

efile_tables	<i>List known efile table names</i>
--------------	-------------------------------------

Description

List known efile table names

Usage

```
efile_tables(cardinality = "all")
```

Arguments

cardinality One of "all", "1x1", "1xm", or "supplemental".

Value

Character vector of known canonical table names.

find_best_normalization	<i>Fit normalization parameters on the stable interior of a distribution</i>
-------------------------	--

Description

Winsorizes x , isolates the stable (non-sentinel, non-NA) interior, selects a transformation type, fits its parameters and centering/scaling constants, and returns a `normalize_x_fit` object that can be reused via [apply_normalization\(\)](#).

Separating fitting from scoring allows the same fitted model to be applied to new data (e.g. a holdout year) without re-estimating parameters, and ensures that sentinel observations do not distort the center and spread estimates used for standardization. The exception is "rank_normal", whose reference distribution (and therefore its center and spread) keeps the winsorized tail observations so that they are scored at their true quantile.

Usage

```
find_best_normalization(
  x,
  range,
  vtype = NULL,
  winsorize = 0.98,
  offset = 0.001,
  standardize = TRUE,
  robust = TRUE,
  zero_tol = 1e-08,
  one_tol = 1e-08,
  boundary_mass_cutoff = 0.1,
  hurdle_trans = c("rank", "logit"),
  hurdle_range = c(1, 100),
  verbose = TRUE
)
```

Arguments

<code>x</code>	Numeric vector to fit on.
<code>range</code>	Character range code ("np", "zp", "zo", "nz", or "lo;hi"). Required.
<code>vtype</code>	Optional transformation type override. One of NULL (auto-detect), "asinh", "logit", "rank_normal", "hurdle".
<code>winsorize</code>	Winsorization proportion. Default 0.98.
<code>offset</code>	Sentinel offset for fixed bounds. Default 0.001.
<code>standardize</code>	Logical. If TRUE, fit centering and scaling constants. Default TRUE.
<code>robust</code>	Logical. If TRUE, use median/MAD; otherwise mean/SD. Default TRUE.
<code>zero_tol</code>	Tolerance for zero detection. Default 1e-8.
<code>one_tol</code>	Tolerance for near-one detection. Default 1e-8.
<code>boundary_mass_cutoff</code>	Share of values at zero (or one) above which a [0, 1] variable is given "rank_normal" instead of "logit". Default 0.10.
<code>hurdle_trans</code>	Positive-part transformation for hurdle variables. "rank" (default) or "logit".
<code>hurdle_range</code>	Rescaled output range for hurdle positive component. Default c(1, 100).
<code>verbose</code>	Logical. If TRUE, print a diagnostic summary. Default TRUE.

Value

An object of class `normalize_x_fit` — a named list containing transformation type, parameters, centering/scaling constants, and diagnostics. Pass this object to [apply_normalization\(\)](#).

See Also

[apply_normalization\(\)](#), [normalize_x\(\)](#)

Examples

```
x <- c( rnorm(200), rep(0, 30) )

# fit on training data
fit <- find_best_normalization( x, range = "np", verbose = FALSE )
fit$transform_type
fit$center
fit$scale
```

fiscal_diagnostics	<i>Diagnose candidate fiscal-health indicators</i>
--------------------	--

Description

Computes variable-level diagnostics used to distinguish indicators that participate in a shared multivariate structure from redundant, standalone, or noisy indicators. Correlation summaries exclude each variable's correlation with itself.

Usage

```
fiscal_diagnostics(
  data,
  variables = NULL,
  corr_cutoff = 0.9,
  kmo_low = 0.5,
  communality_low = 0.2,
  mean_abs_cor_low = 0.1,
  smc_high = 0.9,
  vif_high = 10,
  near_zero_sd = 1e-06,
  nfactors_comm = NULL,
  rotate = "oblimin",
  fm = "minres",
  use = "pairwise.complete.obs"
)
```

Arguments

<code>data</code>	A data frame containing candidate indicators.
<code>variables</code>	Character vector of columns to diagnose. By default, all numeric columns are used.
<code>corr_cutoff</code>	Absolute correlation used to flag redundancy.
<code>kmo_low</code>	KMO/MSA threshold used to flag weak shared variance.
<code>communality_low</code>	Threshold used to flag low communality.
<code>mean_abs_cor_low</code>	Threshold used to flag weak connectedness.
<code>smc_high</code>	Squared multiple correlation threshold.
<code>vif_high</code>	Variance inflation factor threshold.
<code>near_zero_sd</code>	Standard-deviation threshold for near-zero variance.
<code>nfactors_comm</code>	Number of factors in the preliminary communality screen. The default is between one and three, based on the number of usable indicators.
<code>rotate</code>	Rotation passed to <code>psych::fa()</code> .
<code>fm</code>	Factoring method passed to <code>psych::fa()</code> .
<code>use</code>	Missing-value rule passed to <code>stats::cor()</code> .

Details

`psych` is used, when installed, for KMO/MSA and a preliminary common-factor communality screen. The remaining diagnostics are always computed. If `psych` is unavailable or the factor screen fails, the relevant fields are returned as NA and the reason is recorded in warnings.

Value

An object of class `fiscal_diagnostics`. Its `variables` component is a variable-level data frame; other components contain the correlation matrix, redundancy pairs, thresholds, preliminary model, and warnings.

fiscal_dimensions	<i>Discover candidate fiscal-health dimensions</i>
-------------------	--

Description

Uses PCA or exploratory factor analysis to propose empirical dimensions from indicators retained by `fiscal_triage()`. This function discovers and describes candidate groupings; it does not name dimensions, orient scores, or construct a composite fiscal-health index.

Usage

```
fiscal_dimensions(
  data,
  triage = NULL,
  variables = NULL,
  method = c("pca", "efa"),
  nfactors = NULL,
  rotate = "oblimin",
  fm = "minres",
  missing = c("complete", "median"),
  primary_loading = 0.4,
  cross_loading = 0.3,
  min_indicators = 2L,
  parallel_iter = 100L,
  seed = NULL
)
```

Arguments

<code>data</code>	A data frame containing the candidate indicators.
<code>triage</code>	An optional <code>fiscal_triage</code> object. When supplied, only its <code>selected_for_dimensions</code> indicators enter the model and its standalone indicators are carried into the result.
<code>variables</code>	Optional character vector of indicators. Supply either <code>triage</code> or <code>variables</code> , not both.
<code>method</code>	Dimension-discovery method: principal components analysis ("pca") or exploratory factor analysis ("efa").
<code>nfactors</code>	Number of dimensions. If NULL, parallel analysis is used.
<code>rotate</code>	Rotation passed to <code>psych::principal()</code> or <code>psych::fa()</code> .
<code>fm</code>	Factoring method used when <code>method = "efa"</code> .
<code>missing</code>	Missing-data rule. "complete" uses complete rows; "median" replaces missing values with variable medians for this model.
<code>primary_loading</code>	Minimum absolute loading for assignment to a candidate dimension.
<code>cross_loading</code>	Minimum absolute secondary loading used to flag a cross-loading indicator.
<code>min_indicators</code>	Minimum assigned indicators for a proposed cluster to be treated as a multi-indicator dimension rather than a singleton.
<code>parallel_iter</code>	Number of simulated datasets used by parallel analysis when <code>nfactors</code> is NULL.
<code>seed</code>	Optional seed for reproducible parallel analysis.

Value

An object of class `fiscal_dimensions` containing the fitted model, loading and assignment tables, proposed clusters, singleton and unassigned indicators, triage standalones, and the complete modeling specification.

<code>fiscal_health</code>	<i>Fit, score, and optionally aggregate fiscal health</i>
----------------------------	---

Description

A convenience wrapper around `fiscal_health_fit()`, `fiscal_health_score()`, `fiscal_weights()`, and `fiscal_health_index()`. It deliberately does not automate diagnostics, triage, or dimension discovery because those stages require review and substantive judgment.

Usage

```
fiscal_health(
  data,
  dimensions = NULL,
  fit = NULL,
  weights = NULL,
  index = TRUE,
  fit_args = list(),
  score_args = list(),
  weight_args = list(),
  index_args = list()
)
```

Arguments

<code>data</code>	A reference or scoring data frame.
<code>dimensions</code>	A reviewed <code>fiscal_dimensions</code> object or named list of indicator groups. Required when <code>fit</code> is not supplied.
<code>fit</code>	An existing <code>fiscal_health_fit</code> object for scoring new data.
<code>weights</code>	A <code>fiscal_weights</code> object, a named numeric vector of custom weights, or <code>NULL</code> for equal weights.
<code>index</code>	Construct the composite fiscal-health index.
<code>fit_args</code>	Named list of additional arguments for <code>fiscal_health_fit()</code> . Used only when fitting a new model.
<code>score_args</code>	Named list of additional arguments for <code>fiscal_health_score()</code> .
<code>weight_args</code>	Named list of additional arguments for <code>fiscal_weights()</code> when <code>weights</code> is not already a <code>fiscal_weights</code> object.
<code>index_args</code>	Named list of additional arguments for <code>fiscal_health_index()</code> .

Value

An object of class `fiscal_health_result` with `data`, `fit`, `weights`, and `audit` components.

fiscal_health_fit	<i>Fit a reusable fiscal-health measurement model</i>
-------------------	---

Description

Converts reviewed candidate dimensions into a reusable scoring specification. The fitted object stores indicator orientation, centers, scales, missing-value treatment, dimension coefficients, and dimension-score standardization parameters. It can therefore be applied unchanged to later panel years by a scoring function.

Usage

```
fiscal_health_fit(
  data,
  dimensions,
  dimension_names = NULL,
  scoring = "standardized_mean",
  standalone = NULL,
  indicator_direction = NULL,
  missing = c("median", "complete"),
  min_indicators = 2L,
  fm = "minres",
  strict_direction = FALSE
)
```

Arguments

data	Reference data used to estimate the measurement model.
dimensions	A <code>fiscal_dimensions</code> object or a named list whose elements are character vectors of indicators.
dimension_names	Optional named character vector mapping discovered dimension names to substantive names.
scoring	Dimension scoring method. Supply one value for every dimension or a named vector. Supported methods are "standardized_mean", "pca", and "factor".
standalone	Optional character vector of standalone indicators. The default uses standalones carried by a <code>fiscal_dimensions</code> object.
indicator_direction	Optional named vector overriding indicator directions with "higher", "lower", or "context". Other directions are inferred from <code>fiscal_metrics()</code> when possible.
missing	Training-data missing-value rule: "median" or "complete". Median replacement here is statistical missing-value treatment, not e-file data normalization.
min_indicators	Minimum number of indicators required in a modeled dimension.
fm	Factoring method passed to <code>psych::fa()</code> for factor scores.
strict_direction	If TRUE, error when an indicator is context dependent or absent from the metric registry without an explicit override.

Value

An object of class `fiscal_health_fit` containing dimension and standalone scoring specifications, metadata, warnings, and the source discovery object.

<code>fiscal_health_index</code>	<i>Construct a composite fiscal-health index</i>
----------------------------------	--

Description

Aggregates scores produced by `fiscal_health_score()` using a `fiscal_weights()` specification.

Usage

```
fiscal_health_index(  
  data,  
  weights,  
  score_prefix = "fh_",  
  index_name = "fiscal_health",  
  coverage_name = paste0(index_name, "_coverage"),  
  append = TRUE,  
  overwrite = FALSE  
)
```

Arguments

<code>data</code>	A data frame, normally returned by <code>fiscal_health_score()</code> .
<code>weights</code>	A <code>fiscal_weights</code> object.
<code>score_prefix</code>	Prefix used on the component score columns.
<code>index_name</code>	Name of the composite index column.
<code>coverage_name</code>	Name of its weighted-coverage column.
<code>append</code>	If TRUE, append the index to data; otherwise return only the index and coverage columns.
<code>overwrite</code>	Allow replacement of existing output columns.

Value

A data frame containing the composite index and weighted coverage.

fiscal_health_score	<i>Apply a fitted fiscal-health measurement model</i>
---------------------	---

Description

Scores reference or new data using the transformations and coefficients stored by `fiscal_health_fit()`. No model parameters are re-estimated.

Usage

```
fiscal_health_score(
  data,
  fit,
  missing = c("fit", "median", "complete"),
  min_coverage = 0,
  prefix = "fh_",
  include_standalone = TRUE,
  include_coverage = TRUE,
  append = TRUE,
  overwrite = FALSE
)
```

Arguments

<code>data</code>	A data frame containing every indicator required by <code>fit</code> .
<code>fit</code>	A <code>fiscal_health_fit</code> object.
<code>missing</code>	Missing-value rule. "fit" reuses the rule selected during fitting; "median" uses the reference-sample medians stored in <code>fit</code> ; "complete" returns a missing dimension score if any indicator in that dimension is missing.
<code>min_coverage</code>	Minimum proportion of a dimension's source indicators that must be observed before a score is reported. This applies even when reference medians are used.
<code>prefix</code>	Prefix applied to generated score columns.
<code>include_standalone</code>	Include standardized standalone indicators.
<code>include_coverage</code>	Add an observation-level coverage column for every generated measure.
<code>append</code>	If TRUE, append scores to data; otherwise return score and coverage columns only.
<code>overwrite</code>	Allow generated columns to replace existing columns.

Value

A data frame of class `fiscal_health_scores`. Its attributes retain the fitted model, scoring audit, and generated column names.

fiscal_metrics	<i>Fiscal metric registry</i>
----------------	-------------------------------

Description

Returns the canonical relationship between calculation functions, output prefixes, conceptual categories, and score direction. `direction` is "higher", "lower", or "context"; context-dependent composition measures should not be included in a directional score without an explicit scoring rule.

Usage

```
fiscal_metrics()
```

Value

A data frame with one row per fiscal metric.

fiscal_triage	<i>Triage candidate fiscal-health indicators</i>
---------------	--

Description

Converts `fiscal_diagnostics()` results into an auditable recommendation. Standalone indicators are excluded from dimension estimation but retained in the final measurement system. Recommendations can be overridden explicitly.

Usage

```
fiscal_triage(
  diagnostics,
  keep_from_redundant = c("higher_kmo", "higher_mean_abs_cor", "first"),
  overrides = NULL
)
```

Arguments

<code>diagnostics</code>	A <code>fiscal_diagnostics</code> object.
<code>keep_from_redundant</code>	Rule for choosing a provisional representative from a connected redundancy group.
<code>overrides</code>	Optional named character vector assigning indicators to "retain", "standalone", "noise", "redundant", or "review".

Value

An object of class `fiscal_triage` containing the decision table, redundancy groups, dimension candidates, standalone indicators, excluded indicators, and thresholds inherited from the `diagnostics`.

fiscal_weights	<i>Define weights for a composite fiscal-health index</i>
----------------	---

Description

Creates a validated weighting specification for already-scored dimensions and standalone indicators. It does not discover dimensions or fit a measurement model.

Usage

```
fiscal_weights(
  fit = NULL,
  weights = NULL,
  include_standalone = TRUE,
  normalize = TRUE,
  missing = c("reweight", "complete"),
  min_coverage = 1,
  name = NULL
)
```

Arguments

fit	Optional <code>fiscal_health_fit</code> object identifying the available measures and their types.
weights	Optional named numeric vector of custom weights. If omitted, all included measures receive equal weight.
include_standalone	Include standalone indicators from <code>fit</code> in an automatically generated or validated specification.
normalize	Normalize weights to sum to one.
missing	Composite-index missing-score rule: "reweight" reallocates weight across available scores; "complete" requires every positively weighted score.
min_coverage	Minimum weighted coverage required to report the index.
name	Optional descriptive name for the weighting specification.

Value

An object of class `fiscal_weights`.

get_assets_revenue_ratio	<i>Asset Revenue Ratio</i>
--------------------------	----------------------------

Description

Measures the size of an organization's asset base relative to its annual revenue.

Formula:

$$\text{arr} = \text{total_assets} / \text{total_revenue}$$

Definitional Range

Bounded below at zero; unbounded above. The ratio is undefined when revenue is zero. Typical operating nonprofits show values in the [0.5, 5.0] range. Endowed organizations and capital-intensive nonprofits may show values of 10 or higher.

Benchmarks and rules of thumb

- No universal benchmark; most meaningful for within-subsector comparisons.
- **Below 1.0:** Annual revenue exceeds total assets – common for lean service organizations with minimal physical assets.
- **Above 10:** Typically indicates a capital-heavy asset base or a small revenue base relative to accumulated assets.

Calculated For: 990 + 990EZ filers.

Usage

```
get_assets_revenue_ratio( df,
  total_assets = "F9_10_ASSET_TOT_EOY",
  total_revenue = c( "F9_08_REV_TOT_TOT", "F9_01_REV_TOT_CY" ),
  winsorize = 0.98 ,
  range      = "zp",
  sanitize   = TRUE,
  summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
total_assets	Total assets, EOY.
total_revenue	Total revenue. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority over 990EZ.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The asset revenue ratio measures how many dollars of assets are held per dollar of annual revenue. It is an asset intensity measure: capital-intensive organizations (hospitals, universities, housing providers with large real estate portfolios) show high ratios; lean operating nonprofits show low ratios. It can also be interpreted as an approximate measure of how long the organization could theoretically operate on its asset base - though this is not a direct liquidity measure.

A related interpretation is efficiency: a lower ratio may indicate more efficient use of assets to generate revenue, though this is not universally true for nonprofits where asset accumulation may reflect reserve-building rather than operational inefficiency.

Formula variations and their sources:

Total assets EOY / total revenue. The inverse (revenue / assets) is sometimes called the asset turnover ratio and is more common in commercial analysis. For nonprofits, the assets-to-revenue direction is more intuitive because it expresses asset intensity in terms of revenue multiples. Some studies use average assets $((BOY + EOY)/2)$ in the denominator to account for mid-year asset changes, but the EOY value is used here for consistency and data availability.

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good: Executive compensation in nonprofit organizations. *Policy and Society*, 20(4), 94-112.
- Bowman, W. (2011). Financial capacity and sustainability of ordinary nonprofits. *Nonprofit Management and Leadership*, 22(1), 37-51.

Variables used::

- F9_10_ASSET_TOT_EOY: Total assets, end of year (total_assets, 990)
- F9_08_REV_TOT_TOT: Total revenue from Part VIII (total_revenue, 990)
- F9_01_REV_TOT_CY: Total revenue from Part I (total_revenue, 990EZ fallback)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``assets_rev`` - asset revenue ratio (raw)
- ``assets_rev_w`` - winsorized version
- ``assets_rev_z`` - standardized z-score (based on winsorized values)
- ``assets_rev_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_assets_revenue_ratio( df = dat10k )
head( d[ , c( "assets_rev", "assets_rev_w", "assets_rev_z", "assets_rev_p" ) ] )
```

get_bmf_vars	<i>Return the list of default BMF variables</i>
--------------	---

Description

Returns the character vector of BMF-derived and BMF-retained variables appended by default when `retrieve_efile_data()` is called with `include_bmf = TRUE`.

Usage

```
get_bmf_vars()
```

Details

These include native normalized NTEE classifications, BMF administrative fields, recent financial amounts, geocoded location and quality fields, and source/vintage provenance from the current BMF master.

Value

A character vector of column names.

Examples

```
get_bmf_vars()
```

get_bottom_n	<i>Compute the lower winsorization threshold for a given range specification</i>
--------------	--

Description

Compute the lower winsorization threshold for a given range specification

Usage

```
get_bottom_n(x, range, winsorize = 0.98, offset = 0.001)
```

Arguments

<code>x</code>	Numeric vector (NAs are ignored).
<code>range</code>	Character range code ("np", "zp", "zo", "nz", or "lo;hi").
<code>winsorize</code>	Winsorization proportion (default 0.98).
<code>offset</code>	Sentinel offset for fixed bounds (default 0.001).

Value

A single numeric threshold value, or NA when no lower clipping applies.

get_cash_assets_ratio *Cash and Savings to Assets Ratio*

Description

Cash and savings as a share of total assets; measures the liquid composition of the asset base.

Formula:

$$\text{cash_assets} = (\text{cash} + \text{savings}) / \text{total_assets}$$

Definitional Range

Bounded [0, 1]: zero means no liquid assets; one means the entire asset base is cash and savings. In practice most nonprofits fall in the [0.05, 0.50] range. Values above 0.60 are uncommon for operating nonprofits and may indicate excess cash hoarding.

Benchmarks and rules of thumb

- No universally accepted benchmark. Values below 0.05 suggest minimal liquid cushion; values above 0.60 may indicate under-deployment of assets toward mission.
- Foundations and endowed institutions naturally show higher ratios than operating nonprofits.

Calculated For: 990 filers only.

Usage

```
get_cash_assets_ratio( df,
    cash                = "F9_10_ASSET_CASH_EOY",
    total_assets        = "F9_10_ASSET_TOT_EOY",
    savings              = "F9_10_ASSET_SAVING_EOY",
    winsorize           = 0.98 ,
    range               = "zp",
    sanitize            = TRUE,
    summarize           = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
cash	Cash on hand, EOY.
total_assets	Total assets, EOY.
savings	Savings and temporary cash investments, EOY.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Primary uses and key insights:

The cash and savings to assets ratio measures what fraction of an organization's total asset base is held in immediately liquid form. It is an asset composition indicator rather than a pure liquidity test: a high ratio means the organization holds a large share of its resources in cash (potentially signalling under-investment in mission-related assets), while a low ratio may indicate heavy investment in fixed assets or program-related investments.

This metric is distinct from `get_cash_liquidity_ratio()`, which compares the same cash+savings numerator to current liabilities (a coverage test), and from `get_cash_on_hand()`, which reports the dollar amount rather than a proportion.

Formula variations and their sources:

The Stata-derived version of this ratio (`cs_ta`) used in nonprofit financial structure studies uses exactly this formulation: $(\text{cash} + \text{savings}) / \text{total assets EOY}$. Some balance sheet decomposition studies include pledges receivable and accounts receivable in the numerator to capture all near-liquid items, but this implementation follows the narrower two-item definition to maintain comparability with the cash ratio literature.

Why this formula was chosen:

Cash (line 1B) and savings (line 2B) are the two most reliable, narrowly liquid items on the 990 balance sheet. Total assets (line 16B) is the standard denominator for asset composition ratios. This pairing cleanly answers the question: what share of total resources is in immediately spendable form?

Canonical citations:

- Bowman, W. (2011). Financial capacity and sustainability of ordinary nonprofits. *Nonprofit Management and Leadership*, 22(1), 37-51.
- Frumkin, P. & Keating, E.K. (2001). The price of doing good: Executive compensation in nonprofit organizations. *Policy and Society*, 20(4), 94-112.

Variables used::

- `F9_10_ASSET_CASH_EOY`: Cash on hand, EOY (cash)
- `F9_10_ASSET_SAVING_EOY`: Savings and temporary cash investments, EOY (savings)
- `F9_10_ASSET_TOT_EOY`: Total assets, EOY (`total_assets`)

Value

The original data frame with four columns appended: `cash_assets`, `cash_assets_w`, `cash_assets_z`, `cash_assets_p`.

Examples

```
library( fiscal )
data( dat10k )
d <- get_cash_assets_ratio( df = dat10k )
head( d[ , c( "cash_assets", "cash_assets_w", "cash_assets_z", "cash_assets_p" ) ] )
```

get_cash_burn_ratio	<i>Burn Rate Ratio</i>
---------------------	------------------------

Description

End-of-year cash relative to beginning-of-year cash; measures whether an organization is drawing down or accumulating its cash position.

Formula:

$$\text{cash_burn} = \text{cash_eoy} / \text{cash_boy}$$

Definitional Range

Bounded below at zero (cash cannot be negative). Values below 1.0 indicate cash depletion; values above 1.0 indicate cash accumulation. Ratios near zero indicate near-complete cash exhaustion. The ratio is undefined (NA) when beginning-of-year cash is zero, which occurs in the first year of operation or after a complete cash drawdown.

Benchmarks and rules of thumb

- A ratio consistently below 1.0 over multiple years is a warning sign; a single year below 1.0 may reflect planned reserve spending.
- **Below 0.50** in a single year (cash halved) warrants investigation.
- A ratio consistently above 1.0 indicates accumulating liquidity reserves.

Calculated For: 990 filers only.

Usage

```
get_cash_burn_ratio( df,
  cash_eoy           = "F9_10_ASSET_CASH_EOY",
  cash_boy           = "F9_10_ASSET_CASH_BOY",
  months_in_period = 12,
  winsorize         = 0.98 ,
  range             = "zp",
  sanitize          = TRUE,
  summarize         = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
cash_eoy	Cash on hand, end of year.
cash_boy	Cash on hand, beginning of year.
months_in_period	Deprecated and ignored. Retained for backward compatibility with the earlier dollar-per-month formulation; the EOY/BOY ratio does not depend on period length. Supplying it raises a warning.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.

range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The burn rate ratio compares end-of-year cash to beginning-of-year cash, measuring the rate at which an organization is accumulating or depleting its cash position over a single fiscal year. A ratio below 1.0 means the organization ended the year with less cash than it started with (burning cash); a ratio above 1.0 means it accumulated cash. It is most useful for detecting multi-year cash erosion trends before they reach a crisis point.

Formula variations and their sources:

The term "burn rate" originates in startup finance, where it refers to monthly cash outflows. The nonprofit adaptation here is an annual version: EOY cash divided by BOY cash. An alternative formulation computes the dollar change (EOY - BOY) divided by annual expenses to express the burn as a fraction of the operating budget, but that version requires an additional variable and is better captured by the days/months of cash functions ([get_days_cash_operations\(\)](#), [get_months_cash_operations\(\)](#)).

Why this formula was chosen:

The EOY/BOY ratio is the simplest formulation and requires only two fields from the Part X balance sheet. It directly answers "is the cash position improving or deteriorating year-over-year?" without requiring expense data, making it calculable even for organizations where Part IX data is incomplete.

Canonical citations:

- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley. - Discusses cash trend analysis for nonprofits.
- Nonprofit Finance Fund. (Annual). *State of the Nonprofit Sector Survey*.
 - Annual survey tracks cash position changes as a sector-wide indicator.

Variables used::

- F9_10_ASSET_CASH_EOY: Cash on hand, end of year (cash_eoy)
- F9_10_ASSET_CASH_BOY: Cash on hand, beginning of year (cash_boy)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `cash\_burn` - EOY cash / BOY cash (raw)
- `cash\_burn\_w` - winsorized version
- `cash\_burn\_z` - standardized z-score (based on winsorized values)
- `cash\_burn\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_cash_burn_ratio( df = dat10k )
head( d[ , c( "cash_burn", "cash_burn_w", "cash_burn_z", "cash_burn_p" ) ] )
```

```
get_cash_liquidity_ratio
```

Cash Ratio

Description

Measures immediate liquidity using only cash and cash equivalents against current liabilities.

Formula:

```
casr = ( cash + savings ) / current_liabilities

current_liabilities = accounts_payable + grants_payable
```

Definitional Range

Bounded below at zero and unbounded above. A ratio of 1.0 means cash exactly covers current payables. The empirical range for nonprofits is approximately [0, 20], with most values between 0.1 and 5.0. Very high values are common for grant-making foundations and endowed organizations that carry minimal accounts payable.

Benchmarks and rules of thumb

- **Below 0.5:** Organization cannot cover half of near-term payables from cash alone; reliant on receivables collection or borrowing.
- **1.0 or above:** Full cash coverage of current liabilities.
- Because this is a highly conservative measure, values somewhat below 1.0 are normal if receivables are healthy.

Calculated For: 990 filers only.

Usage

```
get_cash_liquidity_ratio( df,
  cash           = "F9_10_ASSET_CASH_EOY",
  savings        = "F9_10_ASSET_SAVING_EOY",
  accounts_payable = "F9_10_LIAB_ACC_PAYABLE_EOY",
  grants_payable  = "F9_10_LIAB_GRANT_PAYABLE_EOY",
  winsorize       = 0.98 ,
  range          = "zp",
  sanitize        = TRUE,
  summarize       = FALSE )
```

Arguments

<code>df</code>	A <code>data.frame</code> containing the fields required for computing the metric.
<code>cash</code>	Cash on hand, EOY.
<code>savings</code>	Short-term investments (savings), EOY.
<code>accounts_payable</code>	Accounts payable and accrued expenses, EOY.
<code>grants_payable</code>	Grants and similar amounts payable, EOY.
<code>winsorize</code>	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
<code>summarize</code>	Logical. If TRUE, prints a <code>summary()</code> of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details**Primary uses and key insights:**

The cash liquidity ratio is the most conservative liquidity test: it asks whether an organization could cover its most immediate obligations using only cash and savings, without collecting any receivables or selling any investments. It is sometimes called the absolute liquidity ratio or cash coverage ratio. For nonprofits with long or uncertain receivables collection cycles (e.g., government reimbursement contracts, multi-year pledge collections), it provides a pessimistic but useful lower bound on liquidity.

This ratio is stricter than both the quick ratio (`get_quick_ratio()`) and the current ratio (`get_current_ratio()`), which both include receivables in the numerator.

Formula variations and their sources:

The commercial equivalent (sometimes called the "super-quick" ratio) uses only cash and marketable securities. The nonprofit version here uses cash (line 1B) plus savings/temporary investments (line 2B) as the numerator, and accounts payable plus grants payable as the denominator. This follows the same liability proxy used across the package (Tuckman & Chang 1991).

Why this formula was chosen:

Cash and savings are the two fields on the 990 that most unambiguously represent immediately available funds. The denominator (accounts payable + grants payable) represents the most pressing and legally enforceable near-term obligations. Together this pair provides the most conservative 990-based liquidity measure available.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.

- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley.

Variables used::

- F9_10_ASSET_CASH_EOY: Cash on hand, EOY (cash)
- F9_10_ASSET_SAVING_EOY: Savings and temporary cash investments, EOY (savings)
- F9_10_LIAB_ACC_PAYABLE_EOY: Accounts payable and accrued expenses, EOY (accounts_payable)
- F9_10_LIAB_GRANT_PAYABLE_EOY: Grants and similar amounts payable, EOY (grants_payable)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `cash\_liq` - cash ratio (raw)
- `cash\_liq\_w` - winsorized version
- `cash\_liq\_z` - standardized z-score (based on winsorized values)
- `cash\_liq\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_cash_liquidity_ratio( df = dat10k )
head( d[ , c( "cash_liq", "cash_liq_w", "cash_liq_z", "cash_liq_p" ) ] )
```

get_cash_on_hand	<i>Cash on Hand</i>
------------------	---------------------

Description

Absolute dollar amount of cash and liquid savings held at end of year.

Formula:

$\text{coh} = \text{cash} + \text{savings}$

Definitional Range

Cash on hand is bounded below at zero (cannot be negative on a properly prepared balance sheet) and unbounded above. Unlike ratio measures, the scale is in dollars, so comparison across organizations of different sizes requires normalization. Use [get_months_cash_operations\(\)](#) or [get_days_cash_operations\(\)](#) for size-adjusted comparisons.

Benchmarks and rules of thumb

- As an absolute measure, benchmarks depend on the organization's expense base. A common practitioner rule is at least 60-90 days of operating expenses in liquid cash.
- The Nonprofit Finance Fund recommends a minimum of three months of operating expenses in accessible reserves.

Calculated For: 990 filers only.

Usage

```
get_cash_on_hand( df,
  cash      = "F9_10_ASSET_CASH_EOY",
  savings   = "F9_10_ASSET_SAVING_EOY",
  winsorize = 0.98 ,
  range     = "zp",
  sanitize  = TRUE,
  summarize = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
cash	Cash on hand, EOY.
savings	Short-term investments (savings), EOY.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details**Primary uses and key insights:**

Cash on hand is a simple dollar-denominated liquidity measure rather than a ratio. It answers the most direct version of the liquidity question: how much immediately spendable money does the organization have right now? It is most useful for communicating financial position to non-technical audiences (boards, funders), for absolute comparison against known fixed costs (monthly payroll, rent), and as the numerator input for ratio-based measures such as months of cash ([get_months_cash_operations\(\)](#)) and days of cash ([get_days_cash_operations\(\)](#)).

Formula variations and their sources:

Some practitioners define cash on hand as cash only (Part X line 1B), excluding savings. Others include marketable securities or investments held for sale. This implementation uses cash plus savings (lines 1B + 2B) as the most common definition in the nonprofit financial management literature (Zietlow et al. 2007), since savings accounts and money market funds are effectively as liquid as operating cash for most organizations.

Why this formula was chosen:

Lines 1B (cash) and 2B (savings and temporary cash investments) are the two most reliably liquid items on the 990 balance sheet and are the least subject to valuation uncertainty. Pledges and receivables are excluded because their timing and collectibility are uncertain. This definition is consistent with the "cash and cash equivalents" concept in nonprofit financial management practice.

Canonical citations:

- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley. - Standard practitioner reference for cash management in nonprofits.
- Calabrese, T.D. (2013). Running on empty: The operating reserves of U.S. nonprofit organizations. *Nonprofit Management and Leadership*, 23(3), 281-302. - Examines operating cash and reserves as components of financial sustainability.

Variables used::

- F9_10_ASSET_CASH_EOY: Cash on hand, EOY (cash)
- F9_10_ASSET_SAVING_EOY: Savings and temporary cash investments, EOY (savings)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``cash_on_hand`` - cash on hand in dollars (raw)
- ``cash_on_hand_w`` - winsorized version
- ``cash_on_hand_z`` - standardized z-score (based on winsorized values)
- ``cash_on_hand_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_cash_on_hand( df = dat10k )
head( d[ , c( "cash_on_hand", "cash_on_hand_w", "cash_on_hand_z", "cash_on_hand_p" ) ] )
```

get_clean_ntee

Normalize legacy NTEE codes

Description

Converts raw NTEE codes into a normalized 3-character form used by all downstream classification functions. Handles padding, blank codes, specialty nonprofits (digits 2-3 of 01-19), and 5-character legacy codes.

Usage

```
get_clean_ntee(x)
```

Arguments

`x` Character vector of raw NTEE codes.

Value

A character vector of normalized 3-character NTEE codes.

Examples

```
get_clean_ntee( c("B29", "B0129", "B8443", "E0521", "", "A115") )
```

get_current_ratio	<i>Current Ratio</i>
-------------------	----------------------

Description

Measures short-term liquidity by comparing current assets to current liabilities.

Formula:

$$cr = \text{current_assets} / \text{current_liabilities}$$

$$\begin{aligned} \text{current_assets} &= \text{cash} + \text{savings} + \text{pledges_receivable} + \text{accounts_receivable} \\ &\quad + \text{inventories} + \text{prepaid_expenses} \\ \text{current_liabilities} &= \text{accounts_payable} + \text{grants_payable} \end{aligned}$$

Definitional Range

The current ratio is bounded below at zero and unbounded above. Values below 1.0 mean current liabilities exceed current assets. The empirical range for nonprofits is approximately [0, 15], with most organizations clustered between 0.5 and 5.

Benchmarks and rules of thumb

- **1.0:** Conventional adequacy threshold – liabilities exactly covered.
- **Below 0.5:** Indicates acute short-term risk (Tuckman & Chang 1991).
- **Above 3.0-4.0:** May indicate excess cash or slow receivables collection.
- Sector medians generally fall in the [1.0, 2.5] range (Zietlow et al. 2007).

Calculated For: 990 filers only.

Usage

```
get_current_ratio( df,
  cash              = "F9_10_ASSET_CASH_EOY",
  savings           = "F9_10_ASSET_SAVING_EOY",
  pledges_receivable = "F9_10_ASSET_PLEDGE_NET_EOY",
  accounts_receivable = "F9_10_ASSET_ACC_NET_EOY",
  inventories       = "F9_10_ASSET_INV_SALE_EOY",
  prepaid_expenses  = "F9_10_ASSET_EXP_PREPAID_EOY",
  accounts_payable  = "F9_10_LIAB_ACC_PAYABLE_EOY",
  grants_payable    = "F9_10_LIAB_GRANT_PAYABLE_EOY",
  numerator = NULL, denominator = NULL, winsorize = 0.98 ,
  range       = "zp",
  sanitize    = TRUE,
  summarize   = FALSE,
  investment_sales = NULL )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
cash	Cash on hand, EOY.
savings	Short-term investments (savings), EOY.

pledges_receivable	Net pledges and grants receivable, EOY.
accounts_receivable	Accounts receivable, net, EOY.
inventories	Inventories for sale or use, EOY (Part X line 8).
prepaid_expenses	Prepaid expenses and deferred charges, EOY.
accounts_payable	Accounts payable and accrued expenses, EOY.
grants_payable	Grants and similar amounts payable, EOY.
numerator	Optional. A pre-aggregated column name for current assets. When supplied, the asset component arguments are ignored (it is an error to also set one of them explicitly).
denominator	Optional. A pre-aggregated column name for current liabilities. When supplied, accounts_payable and grants_payable are ignored (it is an error to also set one of them explicitly).
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.
investment_sales	Deprecated alias for inventories, retained for backward compatibility. The field it defaulted to, F9_10_ASSET_INV_SALE_EOY, is inventories, not investments.

Details

Primary uses and key insights:

The current ratio is the most widely used measure of short-term liquidity in both commercial and nonprofit finance. It asks whether an organization has sufficient near-term assets to meet its near-term obligations without requiring asset sales, additional borrowing, or emergency fundraising. For nonprofits it is particularly important as a stress-test, because grant revenue often arrives in irregular cycles that do not align with payroll and vendor payment schedules.

Unlike the quick ratio ([get_quick_ratio\(\)](#)), the current ratio includes prepaid expenses and inventories, making it a somewhat more generous (and less conservative) measure of liquidity. In practice for nonprofits, the two ratios often move together since prepaid expenses are rarely a large share of assets.

Formula variations and their sources:

The standard commercial definition (current assets / current liabilities) uses balance sheet categories that are explicitly labelled as current. The 990 balance sheet (Part X) does not use current vs. non-current labels. This implementation follows the operationalization in Tuckman & Chang (1991) and subsequent studies by defining:

- **Current assets:** cash + savings + pledges receivable + accounts receivable + inventories for sale or use + prepaid expenses (Part X lines 1, 2, 3, 4, 8, 9).
- **Current liabilities:** accounts payable + grants payable (Part X lines 17, 18). Mortgage and note obligations are excluded as long-term.

Some analysts also include publicly traded securities (Part X line 11) in current assets as marketable investments; they are excluded here.

Why this formula was chosen:

The asset components selected are the most reliably current items available on the 990 balance sheet. The liability components exclude long-term debt, which matches the intent of the current ratio (near-term coverage) while working within the structural constraints of the 990. This formulation is consistent with the majority of nonprofit financial health studies and the NCCS financial indicators project.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.
- Greenlee, J.S. & Trussel, J.M. (2000). Predicting the financial vulnerability of charitable organizations. *Nonprofit Management and Leadership*, 11(2), 199-210.
- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley.

Variables used::

- F9_10_ASSET_CASH_EOY: Cash on hand, EOY (cash)
- F9_10_ASSET_SAVING_EOY: Savings and short-term investments, EOY (savings)
- F9_10_ASSET_PLEDGE_NET_EOY: Net pledges receivable, EOY (pledges_receivable)
- F9_10_ASSET_ACC_NET_EOY: Accounts receivable, net, EOY (accounts_receivable)
- F9_10_ASSET_INV_SALE_EOY: Inventories for sale or use, EOY (Part X line 8) (inventories)
- F9_10_ASSET_EXP_PREPAID_EOY: Prepaid expenses and deferred charges, EOY (prepaid_expenses)
- F9_10_LIAB_ACC_PAYABLE_EOY: Accounts payable and accrued expenses, EOY (accounts_payable)
- F9_10_LIAB_GRANT_PAYABLE_EOY: Grants and similar amounts payable, EOY (grants_payable)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `current` - current ratio (raw)
- `current\_w` - winsorized version
- `current\_z` - standardized z-score (based on winsorized values)
- `current\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_current_ratio( df = dat10k )
head( d[ , c( "current", "current_w", "current_z", "current_p" ) ] )
```

get_days_cash_investments

Days of Operating Cash and Investments

Description

Measures how many days an organization can operate using its liquid assets plus its investment securities.

Formula:

$$\text{docl} = (\text{liquid_assets} + \text{investments}) / \text{daily_expenses}$$

```
liquid_assets = cash + savings + pledges_receivable + accounts_receivable
investments   = publicly_traded_securities + other_securities
daily_expenses = ( total_expenses - depreciation ) / 365
```

Definitional Range

Zero or positive for ordinary filings and unbounded above, expressed in days. Values above 365 indicate more than one year of coverage. The metric adds investments to the numerator of [get_days_cash_operations\(\)](#), so it is at least as large whenever the reported investment balances are non-negative. Negative values come only from filing anomalies: negative receivable or investment balances, or depreciation larger than total expenses.

Benchmarks and rules of thumb

- **180-365 days:** Considered a solid reserve position for endowed organizations.
- A large gap between this metric and [get_days_cash_operations\(\)](#) indicates liquidity concentrated in investments rather than accessible cash.

Calculated For: 990 filers only.

Usage

```
get_days_cash_investments( df,
  cash           = "F9_10_ASSET_CASH_EOY",
  savings        = "F9_10_ASSET_SAVING_EOY",
  pledges_receivable = "F9_10_ASSET_PLEDGE_NET_EOY",
  accounts_receivable = "F9_10_ASSET_ACC_NET_EOY",
  investments      = c( "F9_10_ASSET_INVEST_SEC_EOY",
                        "F9_10_ASSET_INVEST_SEC_OTH_EOY" ),
  total_expenses    = "F9_09_EXP_TOT_TOT",
  depreciation      = "F9_09_EXP_DEPREC_TOT",
  numerator = NULL, denominator = NULL,
```

```
winsorize = 0.98 ,
range      = "zp",
sanitize   = TRUE,
summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
cash	Cash on hand, EOY.
savings	Savings and temporary cash investments, EOY.
pledges_receivable	Net pledges and grants receivable, EOY.
accounts_receivable	Accounts receivable, net, EOY.
investments	Investment securities, EOY. One or more column names whose values are summed (unlike most column arguments, which coalesce a 990 and a 990-EZ field). Defaults to publicly traded securities (Part X line 11) plus other securities (line 12).
total_expenses	Total functional expenses.
depreciation	Depreciation, depletion, and amortization.
numerator	Optional. A pre-aggregated column for liquid assets plus investments. When supplied, cash, savings, pledges_receivable, accounts_receivable, and investments are ignored; it is an error to also set one of them explicitly.
denominator	Optional. A pre-aggregated column for annual non-depreciation expenses (the function divides this by 365 internally). When supplied, total_expenses, depreciation are ignored; it is an error to also set one of them explicitly.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

Days of cash and investments extends [get_days_cash_operations\(\)](#) by adding investment securities to the liquidity numerator. It asks: if the organization drew on its cash, receivables, and investment portfolio, how many days of operations could it fund? This broader measure captures organizations that hold significant reserves in investment portfolios rather than bank accounts.

It is most relevant for endowed organizations, foundations, and mature nonprofits that hold investment portfolios. For organizations with no investments, [get_days_cash_operations\(\)](#) and this metric are identical, so the difference between the two is the coverage that investments add.

Formula variations and their sources:

(Cash + savings + pledges receivable + accounts receivable + publicly traded securities + other securities) / ((total expenses - depreciation) / 365), using Part X lines 1, 2, 3, 4, 11, and 12 and Part IX line 25 less line 22. Rating agencies use a similar "days cash on hand" measure that adds unrestricted cash and investments; this version keeps the receivables in [get_days_cash_operations\(\)](#) so the two metrics nest. Program-related investments (line 13) are excluded because they are not held for liquidity.

The Form 990 does not separate donor-restricted investments, such as permanent endowment, from unrestricted ones, so the numerator can overstate the resources available for general operations. For organizations with large restricted endowments, read this metric alongside [get_netassets_composition_ra](#)

Earlier versions started from unrestricted net assets, added investments, and subtracted both net fixed assets and mortgages. That counted investments twice (they are already part of net assets) and treated mortgage debt as reducing liquidity a second time, so organizations with mortgaged buildings often scored negative. Earlier versions also used F9_10_ASSET_INV_SALE_EOY (Part X line 8, inventories) as investments.

Canonical citations:

- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley.
- Calabrese, T.D. (2013). Running on empty: The operating reserves of U.S. nonprofit organizations. *Nonprofit Management and Leadership*, 23(3), 281-302.

Variables used::

- F9_10_ASSET_CASH_EOY: Cash, non-interest bearing, EOY (Part X line 1) (cash)
- F9_10_ASSET_SAVING_EOY: Savings and temporary cash investments, EOY (Part X line 2) (savings)
- F9_10_ASSET_PLEDGE_NET_EOY: Pledges and grants receivable, net, EOY (Part X line 3) (pledges_receivable)
- F9_10_ASSET_ACC_NET_EOY: Accounts receivable, net, EOY (Part X line 4) (accounts_receivable)
- F9_10_ASSET_INVEST_SEC_EOY: Investments, publicly traded securities, EOY (Part X line 11) (investments)
- F9_10_ASSET_INVEST_SEC_OTH_EOY: Investments, other securities, EOY (Part X line 12) (investments)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)
- F9_09_EXP_DEPREC_TOT: Depreciation and amortization (depreciation)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `days\_cash\_inv` - days of operating cash and investments (raw)
- `days\_cash\_inv\_w` - winsorized version
- `days\_cash\_inv\_z` - standardized z-score (based on winsorized values)
- `days\_cash\_inv\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_days_cash_investments( df = dat10k )
head( d[, c( "days_cash_inv", "days_cash_inv_w", "days_cash_inv_z", "days_cash_inv_p" ) ] )
```

get_days_cash_operations

Days of Operating Cash on Hand

Description

Measures how many days an organization can operate using available liquid assets.

Formula:

$$\text{doch} = \text{liquid_assets} / \text{daily_expenses}$$

$$\begin{aligned} \text{liquid_assets} &= \text{cash} + \text{savings} + \text{pledges_receivable} + \text{accounts_receivable} \\ \text{daily_expenses} &= (\text{total_expenses} - \text{depreciation}) / 365 \end{aligned}$$
Definitional Range

Bounded below at zero and unbounded above, expressed in days. The typical range for nonprofits is approximately [0, 365]. Values above 365 (more than one year of cash) indicate very large liquid reserves relative to expenses, which is uncommon for operating nonprofits but typical for foundations.

Benchmarks and rules of thumb

- **Below 30 days:** Generally considered a distress indicator.
- **60-90 days:** Commonly cited minimum threshold (Nonprofit Finance Fund).
- **90-180 days:** Considered healthy for most operating nonprofits.
- Healthcare nonprofits are typically benchmarked at 150+ days due to longer receivables cycles.

Calculated For: 990 filers only.

Usage

```
get_days_cash_operations( df,
  cash           = "F9_10_ASSET_CASH_EOY",
  savings        = "F9_10_ASSET_SAVING_EOY",
  pledges_receivable = "F9_10_ASSET_PLEDGE_NET_EOY",
  accounts_receivable = "F9_10_ASSET_ACC_NET_EOY",
  total_expenses    = "F9_09_EXP_TOT_TOT",
  depreciation     = "F9_09_EXP_DEPREC_TOT",
  numerator = NULL, denominator = NULL, winsorize = 0.98 ,
  range       = "zp",
  sanitize    = TRUE,
  summarize   = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
cash	Cash on hand, EOY.
savings	Short-term investments (savings), EOY.
pledges_receivable	Net pledges and grants receivable, EOY.
accounts_receivable	Accounts receivable, net, EOY.
total_expenses	Total functional expenses.
depreciation	Depreciation, depletion, and amortization.
numerator	Optional. A pre-aggregated column name for liquid assets. When supplied, cash, savings, pledges_receivable, accounts_receivable are ignored; it is an error to also set one of them explicitly.
denominator	Optional. A pre-aggregated column name for annual non-depreciation expenses (the function divides this by 365 internally). When supplied, total_expenses, depreciation are ignored; it is an error to also set one of them explicitly.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

Days of cash on hand expresses liquidity in an operationally intuitive unit: how many days of expenses could the organization cover if all revenue stopped today? It converts the raw dollar amount of liquid assets into a time-based runway measure that is easier for non-financial audiences (boards, program staff, funders) to interpret than a ratio. It is widely used in nonprofit financial management, healthcare finance, and government financial reporting.

Formula variations and their sources:

The denominator in this implementation is cash operating expenses per day, computed as (total expenses - depreciation) / 365. Depreciation is subtracted because it is a non-cash charge that does not represent actual cash outflow. Some formulations use total expenses without the depreciation adjustment, which is more conservative (produces lower days values). The monthly equivalent is [get_months_cash_operations\(\)](#), which uses /12 instead of /365.

The numerator includes cash, savings, pledges receivable, and accounts receivable - the same four items used in the quick ratio ([get_quick_ratio\(\)](#)). Some stricter formulations use only cash and

savings (excluding receivables); some broader ones add investments held for sale. The four-item version follows the Healthcare Financial Management Association (HFMA) standard, which is the most commonly cited benchmark source.

Why this formula was chosen:

The HFMA definition is the most widely benchmarked version and is increasingly applied to nonprofits outside healthcare. Subtracting depreciation from the denominator follows standard financial management practice (Zietlow et al. 2007) because the ratio is intended to measure cash coverage of cash expenses.

Canonical citations:

- Healthcare Financial Management Association (HFMA). *Key Hospital Financial Statistics and Ratio Medians*. Annual report. - Source for the most widely cited days-of-cash benchmarks; originally healthcare-focused but increasingly applied to nonprofits broadly.
- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley. - Adopts the HFMA approach for nonprofits.
- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.

Variables used::

- F9_10_ASSET_CASH_EOY: Cash on hand (cash)
- F9_10_ASSET_SAVING_EOY: Savings (savings)
- F9_10_ASSET_PLEDGE_NET_EOY: Net pledges receivable (pledges_receivable)
- F9_10_ASSET_ACC_NET_EOY: Accounts receivable (accounts_receivable)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)
- F9_09_EXP_DEPREC_TOT: Depreciation and amortization (depreciation)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `days\_cash\_ops` - days of operating cash on hand (raw)
- `days\_cash\_ops\_w` - winsorized version
- `days\_cash\_ops\_z` - standardized z-score (based on winsorized values)
- `days\_cash\_ops\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_days_cash_operations( df = dat10k )
head( d[ , c( "days_cash_ops", "days_cash_ops_w", "days_cash_ops_z", "days_cash_ops_p" ) ] )
```

get_debt_assets_ratio *Debt to Asset Ratio*

Description

Proportion of total assets financed through liabilities.

Formula:

$$\text{dar} = \text{total_liabilities} / \text{total_assets}$$

Definitional Range

Theoretically bounded [0, 1] when net assets are positive: zero means no debt; one means liabilities equal total assets (zero net assets). Values above 1.0 occur when total liabilities exceed total assets, i.e., the organization has negative net assets (accumulated deficits exceed equity). This is a distress indicator but not uncommon in capital-intensive nonprofits (hospitals, housing) carrying large long-term debt.

Benchmarks and rules of thumb

- **Below 0.50:** Generally considered financially stable.
- **0.50-0.70:** Moderate leverage; manageable but warrants monitoring.
- **Above 0.70:** High leverage; common vulnerability threshold (Tuckman & Chang 1991).
- **Above 1.0:** Negative net assets; acute financial risk.

Calculated For: 990 + 990EZ filers.

Usage

```
get_debt_assets_ratio( df,
  debt   = c( "F9_10_LIAB_TOT_EOY", "F9_01_NAFB_LIAB_TOT_EOY" ),
  assets = c( "F9_10_ASSET_TOT_EOY", "F9_01_NAFB_ASSET_TOT_EOY" ),
  winsorize = 0.98 ,
  range     = "zo",
  sanitize  = TRUE,
  summarize = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
debt	Total liabilities, EOY. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority over 990EZ.
assets	Total assets, EOY. Accepts one or two column names.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").

sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The debt to asset ratio (also called the leverage ratio or debt ratio) measures what fraction of total assets is financed by liabilities rather than equity (net assets). It is the most fundamental solvency indicator in both commercial and nonprofit finance: a ratio approaching 1.0 means the organization is almost entirely debt-financed and has minimal equity cushion; a ratio near 0 means assets are almost entirely owned free of debt.

For nonprofits this ratio is particularly important because: (1) they cannot raise equity capital by issuing stock, so debt is the primary external financing mechanism; and (2) funders, lenders, and rating agencies commonly use this ratio to assess creditworthiness and organizational stability.

Formula variations and their sources:

The commercial formula is identical: total liabilities / total assets. Nonprofit applications use the same ratio but may define "total liabilities" differently. This implementation uses total liabilities from the 990 Part X (line 26B; 990EZ Part II line 26), falling back to the 990 Part I summary field (line 21), which includes all short- and long-term obligations reported on the balance sheet.

Some studies use only long-term debt in the numerator to focus on structural leverage rather than near-term obligations. The full liabilities version is used here as the most comprehensive and commonly cited measure.

Why this formula was chosen:

Total liabilities / total assets is the most universally understood and reported leverage measure. Using total liabilities rather than a subset avoids definitional ambiguity about which obligations to include. The PZ scope (both 990 and 990EZ) maximizes coverage across all filing types.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.
- Greenlee, J.S. & Trussel, J.M. (2000). Predicting the financial vulnerability of charitable organizations. *Nonprofit Management and Leadership*, 11(2), 199-210.
- Keating, E.K., Fischer, M., Gordon, T.P. & Greenlee, J. (2005). Assessing financial vulnerability in the nonprofit sector. *Harvard Business School Working Paper 04-016*.

Variables used::

- F9_10_LIAB_TOT_EOY: Total liabilities, EOY (debt; 990 Part X line 26, 990EZ Part II line 26)
- F9_01_NAFB_LIAB_TOT_EOY: Total liabilities from Part I line 21 (debt, fallback)
- F9_10_ASSET_TOT_EOY: Total assets, EOY (assets; 990 Part X line 16, 990EZ Part II line 25)
- F9_01_NAFB_ASSET_TOT_EOY: Total assets from Part I line 20 (assets, fallback)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `debt\_assets` - debt to asset ratio (raw)
- `debt\_assets\_w` - winsorized version
- `debt\_assets\_z` - standardized z-score (based on winsorized values)
- `debt\_assets\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_debt_assets_ratio( df = dat10k )
head( d[ , c( "debt_assets", "debt_assets_w", "debt_assets_z", "debt_assets_p" ) ] )
```

get_debt_equity_ratio *Debt to Equity Ratio*

Description

Compares total liabilities to unrestricted net assets.

Formula:

$$\text{der} = \text{total_liabilities} / \text{unrestricted_net_assets}$$
Definitional Range

Bounded below at zero when liabilities are non-negative and unrestricted net assets are positive. Unbounded above when unrestricted net assets are very small. Values below zero occur when unrestricted net assets are negative (accumulated deficits exceed unrestricted equity), an acute financial distress signal. The ratio is undefined (NA) when unrestricted net assets equal zero.

Benchmarks and rules of thumb

- Values below 1.0 mean total liabilities are less than the unrestricted equity base – generally considered strong.
- Values between 1.0 and 3.0 indicate moderate leverage.
- Values above 5.0 indicate very high leverage relative to the equity base and are a common vulnerability threshold.
- Negative values indicate negative unrestricted net assets, which almost always signals significant financial stress.

Calculated For: 990 filers only.

Usage

```
get_debt_equity_ratio( df,
  debt   = c( "F9_10_LIAB_TOT_EOY",      "F9_01_NAFB_LIAB_TOT_EOY" ),
  equity = "F9_10_NAFB_UNRESTRICT_EOY",
  restricted_net_assets = "F9_10_NAFB_RESTRICT_EOY",
  total_net_assets      = "F9_10_NAFB_TOT_EOY",
  winsorize = 0.98 ,
  range      = "np",
  sanitize   = TRUE,
  summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
debt	Total liabilities, EOY. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority.
equity	Unrestricted net assets, EOY. Accepts one or two column names; if two are provided they are coalesced with the first taking priority.
restricted_net_assets	Restricted net assets, EOY. Used with total_net_assets to detect filers that do not follow SFAS 117; see resolve_unrestricted_net_assets() .
total_net_assets	Total net assets, EOY. Substituted for unrestricted net assets when unrestricted and restricted net assets are both zero. Set to NULL to disable the fallback.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details**Primary uses and key insights:**

The debt to equity ratio compares total obligations to the organization's equity cushion - its unrestricted net assets. In the nonprofit context, unrestricted net assets represent the accumulated surplus that the organization controls without donor restriction: the purest measure of its financial equity. A high ratio signals that liabilities substantially exceed the equity base, meaning a relatively small revenue shortfall could impair the ability to meet obligations.

This ratio is closely related to [get_debt_assets_ratio\(\)](#) (DAR), but uses unrestricted net assets as the denominator instead of total assets. It is a stricter solvency measure because unrestricted net assets are typically much smaller than total assets, and because restricted assets cannot be used to pay general obligations.

Formula variations and their sources:

The commercial debt/equity ratio uses total debt / shareholders' equity. The nonprofit adaptation substitutes unrestricted net assets for equity, following Tuckman & Chang (1991) and the argument that only unrestricted resources represent true organizational equity available to cover obligations.

An alternative uses total net assets in the denominator (restricted + unrestricted), which is more generous and less analytically precise; see [get_debt_netassets_ratio\(\)](#). A third variant uses only long-term debt in the numerator. This implementation uses total liabilities / unrestricted net assets as the most commonly cited nonprofit version.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.
- Bowman, W. (2011). Financial capacity and sustainability of ordinary nonprofits. *Nonprofit Management and Leadership*, 22(1), 37-51.

Variables used::

- F9_10_LIAB_TOT_EOY: Total liabilities, EOY (debt)
- F9_01_NAFB_LIAB_TOT_EOY: Total liabilities from Part I line 21 (debt, fallback)
- F9_10_NAFB_UNRESTRICT_EOY: Unrestricted net assets, EOY (equity)
- F9_10_NAFB_RESTRICT_EOY: Restricted net assets, EOY (restricted_net_assets)
- F9_10_NAFB_TOT_EOY: Total net assets, EOY (total_net_assets)

Unrestricted net assets fall back to total net assets for filers that do not follow SFAS 117 (see [resolve_unrestricted_net_assets\(\)](#)).

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``debt_equity`` - debt to equity ratio (raw)
- ``debt_equity_w`` - winsorized version
- ``debt_equity_z`` - standardized z-score (based on winsorized values)
- ``debt_equity_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_debt_equity_ratio( df = dat10k )
head( d[ , c( "debt_equity", "debt_equity_w", "debt_equity_z", "debt_equity_p" ) ] )
```

get_debt_netassets_ratio

Debt to Net Assets Ratio

Description

Compares total liabilities to total net assets.

Formula:

$$\text{dnar} = \text{total_liabilities} / \text{total_net_assets}$$

Definitional Range

Bounded below at zero when total net assets are positive; unbounded above as net assets approach zero. Negative values occur when total net assets are negative (liabilities exceed assets), an insolvency signal, so the ratio is not monotone in financial health across zero. The ratio is undefined (NaN) when total net assets equal zero.

Benchmarks and rules of thumb

- Values below 1.0 mean liabilities are smaller than the net asset base.
- Values above 3.0-5.0 are commonly flagged as high leverage in the nonprofit vulnerability literature.
- Always read alongside [get_debt_equity_ratio\(\)](#), which uses only unrestricted net assets: a large gap between the two means much of the equity cushion is donor restricted.

Calculated For: 990 + 990EZ filers.

Usage

```
get_debt_netassets_ratio( df,
  liabilities = c( "F9_10_LIAB_TOT_EOY", "F9_01_NAFB_LIAB_TOT_EOY" ),
  net_assets  = c( "F9_10_NAFB_TOT_EOY", "F9_01_NAFB_TOT_EOY" ),
  winsorize = 0.98 ,
  range      = "np",
  sanitize   = TRUE,
  summarize = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
liabilities	Total liabilities, EOY. Accepts one or two column names; if two are provided they are coalesced with the first taking priority.
net_assets	Total net assets, EOY. Accepts one or two column names; if two are provided they are coalesced with the first taking priority.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").

sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The debt to net assets ratio compares total obligations with the organization's entire equity base, restricted and unrestricted. It is the nonprofit analogue of the commercial debt-to-equity ratio. Because total net assets are reported on both the Form 990 (Part X line 33) and the 990-EZ (Part II line 27), this ratio is available for both filer types.

Formula variations and their sources:

Total liabilities (Part X line 26) / total net assets (Part X line 33). A stricter variant, [get_debt_equity_ratio\(\)](#), divides by unrestricted net assets only, on the argument that donor-restricted resources cannot be used to meet general obligations. The two ratios are identical for organizations with no restricted net assets. Earlier versions of this function used unrestricted net assets, which made it a duplicate of [get_debt_equity_ratio\(\)](#).

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.

Variables used::

- F9_10_LIAB_TOT_EOY: Total liabilities, EOY (Part X line 26; 990-EZ Part II line 26) (liabilities)
- F9_01_NAFB_LIAB_TOT_EOY: Total liabilities from Part I line 21 (liabilities, fallback)
- F9_10_NAFB_TOT_EOY: Total net assets, EOY (Part X line 33; 990-EZ Part II line 27) (net_assets)
- F9_01_NAFB_TOT_EOY: Net assets from Part I line 22 (net_assets, fallback)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``debt_netassets`` - debt to net assets ratio (raw)
- ``debt_netassets_w`` - winsorized version
- ``debt_netassets_z`` - standardized z-score (based on winsorized values)
- ``debt_netassets_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_debt_netassets_ratio( df = dat10k )
head( d[ , c( "debt_netassets", "debt_netassets_w", "debt_netassets_z", "debt_netassets_p" ) ] )
```

get_debt_secured_ratio

Secured Debt Ratio

Description

Secured mortgages and notes payable as a share of total liabilities.

Formula:

$$\text{debt_secured} = \text{secured_mortgages_notes} / \text{total_liabilities}$$

Definitional Range

Bounded [0, 1]. Most nonprofits have zero secured debt; capital-intensive organizations (housing, healthcare, higher education) can show values above 0.70.

Benchmarks and rules of thumb

- No standard benchmark. Most informative when tracked over time or compared within sub-sectors with similar capital structures.
- For organizations with high secured debt ratios, lenders will scrutinize the collateral value and debt service coverage ratio.

Calculated For: 990 filers only.

Usage

```
get_debt_secured_ratio( df,
    secured_mortgages_notes = "F9_10_LIAB_MTG_NOTE_EOY",
    total_liabilities        = "F9_10_LIAB_TOT_EOY",
    winsorize = 0.98 ,
    range     = "zo",
    sanitize  = TRUE,
    summarize = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
secured_mortgages_notes	Secured mortgages and notes payable, EOY.
total_liabilities	Total liabilities, EOY.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Primary uses and key insights:

The secured debt ratio measures what share of total liabilities consists of mortgages and notes payable secured by collateral. It is a liability composition indicator: a high ratio means the organization's debt is predominantly secured (usually real estate mortgages), while a low ratio means most debt is unsecured.

This metric is useful for understanding the nature of an organization's leverage, since secured debt typically has lower interest rates but pledges specific assets as collateral, limiting flexibility. It is particularly relevant for housing nonprofits, healthcare organizations, and educational institutions.

Formula variations and their sources:

Secured mortgages and notes payable (Part X line 23B) / total liabilities (line 26B). The complement is approximately [get_debt_unsecured_ratio\(\)](#) (line 24B / line 26B), though the two do not sum to 1.0 because there are other liability categories.

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good: Executive compensation in nonprofit organizations. *Policy and Society*, 20(4), 94-112.

Variables used::

- F9_10_LIAB_MTG_NOTE_EOY: Secured mortgages and notes payable, EOY (secured_mortgages_notes)
- F9_10_LIAB_TOT_EOY: Total liabilities, EOY (total_liabilities)

Value

The original data.frame with four columns appended: debt_secured, debt_secured_w, debt_secured_z, debt_secured_p.

Examples

```
library( fiscal )
data( dat10k )
d <- get_debt_secured_ratio( df = dat10k )
head( d[ , c( "debt_secured", "debt_secured_w", "debt_secured_z", "debt_secured_p" ) ] )
```

```
get_debt_shortterm_ratio
```

Short Term Debt Ratio

Description

Measures short-term obligations as a share of total net assets.

Formula:

```
stdr = short_term_liabilities / net_assets
```

```
short_term_liabilities = accounts_payable + grants_payable
```

Definitional Range

Bounded below at zero when net assets are positive. Negative values occur when net assets are negative (accumulated deficit), which makes the ratio uninterpretable as a burden measure. Unbounded above when net assets approach zero. Typical values for financially stable nonprofits are in the [0, 0.30] range.

Benchmarks and rules of thumb

- Values below 0.10 indicate modest short-term payables relative to the equity base – a comfortable position.
- Values above 0.30 suggest near-term obligations are placing meaningful pressure on net assets.

Calculated For: 990 filers only.

Usage

```
get_debt_shortterm_ratio( df,
  accounts_payable = "F9_10_LIAB_ACC_PAYABLE_EOY",
  grants_payable   = "F9_10_LIAB_GRANT_PAYABLE_EOY",
  net_assets       = "F9_10_NAFB_TOT_EOY",
  numerator = NULL, denominator = NULL, winsorize = 0.98 ,
  range         = "zp",
  sanitize      = TRUE,
  summarize     = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
accounts_payable	Accounts payable and accrued expenses, EOY.
grants_payable	Grants and similar amounts payable, EOY.
net_assets	Total net assets, EOY.
numerator	Optional. A pre-aggregated column for short-term liabilities. When supplied, accounts_payable, grants_payable are ignored; it is an error to also set one of them explicitly.
denominator	Optional. A pre-aggregated column for the denominator. When supplied, net_assets is ignored; it is an error to also set it explicitly.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The short term debt ratio measures what share of total net assets is represented by near-term payables (accounts payable plus grants payable). It is a measure of the pressure that immediate obligations place on the equity base. A high ratio means current payables are large relative to net assets, which can signal cash flow stress even for organizations that appear solvent on a long-term basis.

Note that unlike most debt ratios, the denominator here is net assets rather than total liabilities or total assets. This measures the short-term burden relative to the equity cushion.

Formula variations and their sources:

Some formulations use total current liabilities (including all short-term items) / total net assets. This implementation uses accounts payable plus grants payable as the best available proxy for current liabilities on the 990, consistent with the approach used across this package's liquidity measures. Total net assets (Part X line 33B) is used as the denominator rather than unrestricted net assets, for broader applicability.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.

Variables used::

- F9_10_LIAB_ACC_PAYABLE_EOY: Accounts payable and accrued expenses, EOY (accounts_payable)
- F9_10_LIAB_GRANT_PAYABLE_EOY: Grants and similar amounts payable, EOY (grants_payable)
- F9_10_NAFB_TOT_EOY: Total net assets, EOY (net_assets)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``debt_shortterm`` - short term debt ratio (raw)
- ``debt_shortterm_w`` - winsorized version
- ``debt_shortterm_z`` - standardized z-score (based on winsorized values)
- ``debt_shortterm_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_debt_shortterm_ratio( df = dat10k )
head( d[ , c( "debt_shortterm", "debt_shortterm_w", "debt_shortterm_z", "debt_shortterm_p" ) ] )
```

get_debt_unsecured_ratio

Unsecured Debt Ratio

Description

Unsecured notes and loans payable as a share of total liabilities.

Formula:

$$\text{debt_unsecured} = \text{unsecured_notes_loans} / \text{total_liabilities}$$

Definitional Range

Bounded [0, 1]. Most nonprofits carry no unsecured notes, so the distribution is highly concentrated at zero with a long right tail.

Benchmarks and rules of thumb

- No established benchmark. Presence of unsecured debt is not inherently negative – many healthy nonprofits use lines of credit for cash flow management – but a large and growing ratio warrants scrutiny.
- Trend analysis is more informative than a single year's value.

Calculated For: 990 filers only.

Usage

```
get_debt_unsecured_ratio( df,
    unsecured_notes_loans    = "F9_10_LIAB_NOTE_UNSEC_EOY",
    total_liabilities        = "F9_10_LIAB_TOT_EOY",
    winsorize = 0.98 ,
    range      = "zo",
    sanitize   = TRUE,
    summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
unsecured_notes_loans	Unsecured notes and loans payable, EOY.
total_liabilities	Total liabilities, EOY.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Primary uses and key insights:

The unsecured debt ratio measures what share of total liabilities consists of unsecured notes and loans payable (Part X line 24B), typically including lines of credit and unsecured bank loans. Used alongside `get_debt_secured_ratio()`, it provides a decomposition of the liability structure.

Formula variations and their sources:

Unsecured notes and loans payable (Part X line 24B) / total liabilities (line 26B). A direct read from two 990 balance sheet fields.

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good. *Policy and Society*, 20(4), 94-112.

Variables used::

- F9_10_LIAB_NOTE_UNSEC_EOY: Unsecured notes and loans payable, EOY (unsecured_notes_loans)
- F9_10_LIAB_TOT_EOY: Total liabilities, EOY (total_liabilities)

Value

The original `data.frame` with four columns appended: `debt_unsecured`, `debt_unsecured_w`, `debt_unsecured_z`, `debt_unsecured_p`.

Examples

```
library( fiscal )
data( dat10k )
d <- get_debt_unsecured_ratio( df = dat10k )
head( d[ , c( "debt_unsecured", "debt_unsecured_w", "debt_unsecured_z", "debt_unsecured_p" ) ] )
```

get_donations_revenue_ratio

Donation/Grant Dependence Ratio

Description

Measures reliance on contributions and fundraising as a share of total revenue.

Formula:

$$\text{dgdr} = \text{donation_revenue} / \text{total_revenue}$$

$$\text{donation_revenue} = \text{contributions} + \text{fundraising_revenue}$$

Definitional Range

Bounded [0, 1]. Values near 1.0 characterize traditional charitable organizations with minimal earned income; values near 0 characterize fee-based service providers.

Benchmarks and rules of thumb

- Values above 0.70-0.80 warrant monitoring of donor concentration and retention.
- Organizations above 0.90 have almost no earned income buffer if philanthropic support declines.

Calculated For: 990 filers only.

Usage

```
get_donations_revenue_ratio( df,
  contributions      = "F9_08_REV_CONTR_TOT",
  fundraising_revenue = "F9_08_REV_OTH_FUNDR_NET_TOT",
  total_revenue      = "F9_08_REV_TOT_TOT",
  numerator = NULL, denominator = NULL, winsorize = 0.98 ,
  range      = "zo",
  sanitize   = TRUE,
  summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
contributions	Total contributions, EOY.
fundraising_revenue	Net fundraising event revenue.
total_revenue	Total revenue.
numerator	Optional. A pre-aggregated column name for donation revenue. When supplied, contributions, fundraising_revenue are ignored; it is an error to also set one of them explicitly.
denominator	Optional. A pre-aggregated column name for the denominator. When supplied, total_revenue is ignored; it is an error to also set it explicitly.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details**Primary uses and key insights:**

The donations and grant dependence ratio measures the combined share of total revenue from contributions (individual donations, foundation grants, corporate giving, federated campaigns) and net fundraising event income. It captures the overall dependence on philanthropic and voluntary support.

A high ratio indicates the organization relies heavily on the goodwill of donors and the fundraising environment rather than earned or contractual income. This creates vulnerability to donor fatigue, economic downturns (when charitable giving declines), and changes in donor priorities.

Formula variations and their sources:

(Total contributions + net fundraising event revenue) / total revenue (Part VIII lines 1h + 8c / line 12A). Government grants are included in total contributions (line 1h) in this formulation. For a pure private philanthropy measure that excludes government grants, combine this ratio with [get_grants_govt_ratio\(\)](#).

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290.
- Froelich, K.A. (1999). Diversification of revenue strategies. *Nonprofit and Voluntary Sector Quarterly*, 28(3), 246-268.

Variables used::

- F9_08_REV_CONTR_TOT: Total contributions (contributions)
- F9_08_REV_OTH_FUNDR_NET_TOT: Net fundraising event revenue (fundraising_revenue)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``donations_rev`` - donation/grant dependence ratio (raw)
- ``donations_rev_w`` - winsorized version
- ``donations_rev_z`` - standardized z-score (based on winsorized values)
- ``donations_rev_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_donations_revenue_ratio( df = dat10k )
head( d[, c( "donations_rev", "donations_rev_w", "donations_rev_z", "donations_rev_p" ) ] )
```

get_earned_income_ratio

Earned Income Dependency Ratio

Description

Share of total revenue derived from earned (non-donation) income sources.

Formula:

$$\text{eidr} = \text{earned_revenue} / \text{total_revenue}$$

$$\text{earned_revenue} = \text{program_service_revenue} + \text{royalties} + \text{other_revenue}$$

Definitional Range

Bounded [0, 1]. Organizations heavily reliant on philanthropy show values near zero; fee-based service providers may show values above 0.90.

Benchmarks and rules of thumb

- No universal threshold – context matters: an advocacy organization is expected to show a low ratio; a hospital is expected to show a high one.
- A low ratio combined with high donation dependence indicates philanthropic concentration risk.

Calculated For: 990 filers only.

Usage

```
get_earned_income_ratio( df,
  program_service_rev = "F9_08_REV_PROG_TOT_TOT",
  royalties           = "F9_08_REV_OTH_ROY_TOT",
  other_revenue       = "F9_08_REV_MISC_OTH_TOT",
  total_revenue       = "F9_08_REV_TOT_TOT",
  numerator = NULL, denominator = NULL, winsorize = 0.98 ,
  range          = "zo",
  sanitize       = TRUE,
  summarize      = FALSE,
  membership_dues = NULL )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
program_service_rev	Program service revenue.
royalties	Royalties.
other_revenue	Other miscellaneous revenue.
total_revenue	Total revenue.
numerator	Optional. A pre-aggregated column for earned revenue. When supplied, program_service_rev, royalties, other_revenue are ignored; it is an error to also set one of them explicitly.
denominator	Optional. A pre-aggregated column for total revenue. When supplied, total_revenue is ignored; it is an error to also set it explicitly.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.
membership_dues	Deprecated and ignored. Membership dues reported on Part VIII line 1b are already part of total contributions (line 1h), so counting them as earned income double counts them against get_donations_revenue_ratio() . Supplying it raises a warning.

Details

Primary uses and key insights:

The earned income dependency ratio measures the combined share of revenue from program services, royalties, and miscellaneous revenue - the sources that flow from the organization's own activities rather than from voluntary contributions. It is the revenue-side complement to [get_donations_revenue_ratio\(\)](#), and a broader version of [get_self_sufficiency_ratio\(\)](#), which compares program revenue to total expenses.

Organizations with high earned income ratios are often considered more financially resilient because earned revenue is tied to service delivery rather than donor preferences, though it also creates exposure to market competition and customer retention challenges.

Formula variations and their sources:

(Program service revenue + royalties + other miscellaneous revenue) / total revenue (Part VIII lines 2g + 5 + 11d / line 12A). This broad definition of earned income follows several studies (Young 2007). A narrower version uses only program service revenue (see [get_revenue_programs_ratio\(\)](#)).

Membership dues are excluded. On the Form 990, dues that are gifts are reported on Part VIII line 1b and are already included in total contributions (line 1h), which [get_donations_revenue_ratio\(\)](#) counts; dues paid in exchange for member services are reported as program service revenue (line 2) and are counted here. Earlier versions added line 1b to earned income, so earned_income and donations_rev could sum to more than 1.

Canonical citations:

- Young, D.R. (2007). *Financing Nonprofits*. AltaMira Press.
- Weisbrod, B.A. (1998). The nonprofit mission and its financing. *Journal of Policy Analysis and Management*, 17(2), 165-174.
- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290.

Variables used::

- F9_08_REV_PROG_TOT_TOT: Program service revenue (program_service_rev)
- F9_08_REV_OTH_ROY_TOT: Royalties (royalties)
- F9_08_REV_MISC_OTH_TOT: Other miscellaneous revenue (other_revenue)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `earned\_income` - earned income dependency ratio (raw)
- `earned\_income\_w` - winsorized version
- `earned\_income\_z` - standardized z-score (based on winsorized values)
- `earned\_income\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_earned_income_ratio( df = dat10k )
head( d[ , c( "earned_income", "earned_income_w", "earned_income_z", "earned_income_p" ) ] )
```

get_equity_ratio	<i>Equity Ratio</i>
------------------	---------------------

Description

Share of total assets financed through net assets (equity).

Formula:

$$er = \text{total_net_assets} / \text{total_assets}$$

Definitional Range

Theoretically bounded [0, 1] when net assets are positive. Values above 1.0 are not possible. Values below zero occur when net assets are negative (accumulated deficit), which is a distress indicator. The empirical range for most nonprofits is approximately [0.10, 0.95].

Benchmarks and rules of thumb

- **Above 0.50:** Equity finances more than half of assets – generally healthy.
- **Below 0.30:** High leverage; common in capital-intensive subsectors.
- Most useful as a trend measure: declining values warrant investigation.

Calculated For: 990 + 990EZ filers.

Usage

```
get_equity_ratio( df,
  net_assets      = "F9_10_NAFB_TOT_EOY",
  total_assets    = "F9_10_ASSET_TOT_EOY",
  winsorize       = 0.98 ,
  range           = "zo",
  sanitize        = TRUE,
  summarize       = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
net_assets	Total net assets, EOY.
total_assets	Total assets, EOY.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The equity ratio measures what share of total assets is financed by net assets (organizational equity) rather than liabilities. It is the arithmetic complement of the debt to asset ratio: equity ratio + debt ratio = 1.0 (when net assets are positive). A higher equity ratio indicates greater financial independence and resilience; the organization owns more of its assets outright.

For nonprofits, the equity ratio is particularly meaningful because net assets represent the accumulated result of mission-related financial decisions over time. A declining equity ratio over several years signals that liabilities are growing faster than assets - a potential sustainability warning.

Formula variations and their sources:

Total net assets / total assets. Some formulations use unrestricted net assets in the numerator to focus on the portion of equity truly available for general operations. This implementation uses total net assets (restricted + unrestricted) for maximum coverage across both 990 and 990EZ filers; for a more conservative version, see [get_netassets_composition_ratio\(\)](#).

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.
- Bowman, W. (2011). Financial capacity and sustainability of ordinary nonprofits. *Nonprofit Management and Leadership*, 22(1), 37-51.

Variables used::

- F9_10_NAFB_TOT_EOY: Total net assets, EOY (net_assets)
- F9_10_ASSET_TOT_EOY: Total assets, EOY (total_assets)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``equity`` - equity ratio (raw)
- ``equity_w`` - winsorized version
- ``equity_z`` - standardized z-score (based on winsorized values)
- ``equity_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_equity_ratio( df = dat10k )
head( d[ , c( "equity", "equity_w", "equity_z", "equity_p" ) ] )
```

get_expenses_admin_ratio

Administrative Overhead Ratio

Description

Share of total expenses devoted to management and general administration.

Formula:

$$\text{aer} = \text{management_expenses} / \text{total_expenses}$$

Definitional Range

Bounded [0, 1]. In combination with program and fundraising ratios, all three sum to 1.0 by construction. The empirical range for most nonprofits is approximately [0.05, 0.35].

Benchmarks and rules of thumb

- Charity Navigator targets management and general expenses at or below 15% of total expenses for a favorable score.
- Values below 5% may indicate underinvestment in governance systems.

Calculated For: 990 filers only.

Usage

```
get_expenses_admin_ratio( df,
  mgmt_expenses = "F9_09_EXP_TOT_MGMT",
  total_expenses = "F9_09_EXP_TOT_TOT",
  winsorize = 0.98 ,
  range      = "zo",
  sanitize   = TRUE,
  summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
mgmt_expenses	Management and general expenses.
total_expenses	Total functional expenses.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The administrative overhead ratio measures what fraction of total expenses is devoted to management and general administration - the governance, compliance, financial management, and executive functions of the organization. It is the management component of the broader overhead ratio ([get_overhead_ratio\(\)](#)), which adds fundraising expenses.

A high administrative ratio may indicate organizational complexity, compliance burden, or inefficiency; a very low ratio may indicate underinvestment in governance and financial systems. The appropriate level depends heavily on organizational size and mission type.

Formula variations and their sources:

Management and general expenses (Part IX line 25C) / total functional expenses (line 25A). This is the most common operationalization. An alternative includes unallocated costs or uses a broader definition of overhead, but the Part IX column breakdown (program, management, fundraising) is the standard basis.

Canonical citations:

- Lecy, J.D. & Searing, E.A. (2015). Anatomy of the nonprofit starvation cycle. *Nonprofit and Voluntary Sector Quarterly*, 44(3), 539-563.
- Overhead Myth Campaign (GuideStar, BBB Wise Giving, Charity Navigator, 2013).
- Nunnenkamp, P. & Ohler, H. (2012). Throwing foreign aid at HIV/AIDS in developing countries: Missing the target? *World Development*, 40(10), 1978-1994. - Uses administrative ratios in cross-organizational comparisons.

Variables used::

- F9_09_EXP_TOT_MGMT: Management and general expenses (mgmt_expenses)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``expenses_admin`` - administrative overhead ratio (raw)
- ``expenses_admin_w`` - winsorized version
- ``expenses_admin_z`` - standardized z-score (based on winsorized values)
- ``expenses_admin_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_expenses_admin_ratio( df = dat10k )
head( d[, c( "expenses_admin", "expenses_admin_w", "expenses_admin_z", "expenses_admin_p" ) ] )
```

get_expenses_affiliates_ratio

Payments to Affiliates Ratio

Description

Payments to affiliates as a share of total expenses.

Formula:

$$\text{expenses_affiliates} = \text{payments_to_affiliates} / \text{total_expenses}$$

Definitional Range

Bounded [0, 1]. Near zero for standalone organizations; potentially above 0.50 for federated intermediaries.

Benchmarks and rules of thumb

- Interpret in the context of organizational structure.
- Large payments to affiliates in the absence of a known federated structure warrant review of related-party transaction disclosures.

Calculated For: 990 filers only.

Usage

```
get_expenses_affiliates_ratio( df,
    payments_to_affiliates    = "F9_09_EXP_PAY_AFFIL_TOT",
    total_expenses            = "F9_09_EXP_TOT_TOT",
    winsorize                 = 0.98 ,
    range                     = "zo",
    sanitize                  = TRUE,
    summarize                 = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
payments_to_affiliates	Payments to affiliates (total).
total_expenses	Total functional expenses.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Primary uses and key insights:

The payments to affiliates ratio measures the share of total functional expenses transferred to affiliated organizations. It captures inter-organizational resource flows within nonprofit families (federated structures, supporting organizations, shared-services models).

Formula:

Payments to affiliates (Part IX line 21, Column A) / total expenses (line 25A).

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good. *Policy and Society*, 20(4), 94-112.

Variables used::

- F9_09_EXP_PAY_AFFIL_TOT: Payments to affiliates (payments_to_affiliates)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)

Value

The original data.frame with four new columns:

- expenses_affiliates (raw ratio)
- expenses_affiliates_w (winsorized)
- expenses_affiliates_z (z-score)
- expenses_affiliates_p (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_expenses_affiliates_ratio( df = dat10k )
head( d[, c( "expenses_affiliates", "expenses_affiliates_w", "expenses_affiliates_z", "expenses_affiliates_p" )]
```

```
get_expenses_compensation_ratio
```

Compensation Expense Ratio

Description

Total compensation and employee-related expenses as a share of total functional expenses.

Formula:

```
expenses_compensation = ( officer_comp + disqualified_comp + other_salaries
+ pension_contributions + other_employee_benefits
+ payroll_taxes ) / total_expenses
```

Definitional Range

Bounded [0, 1]. Most service-delivery nonprofits fall in the [0.50, 0.85] range. Grant-making organizations may show ratios below 0.20.

Benchmarks and rules of thumb

- No universal benchmark – labor intensity varies fundamentally by mission type.
- Human services, health, and education nonprofits typically show ratios above 0.60; grant-making organizations may show ratios below 0.30.
- A sharp year-over-year increase may indicate revenue contraction rather than compensation growth.

Calculated For: 990 filers only.

Usage

```
get_expenses_compensation_ratio( df,
    officer_comp          = "F9_09_EXP_COMP_DTK_TOT",
    disqualified_comp       = "F9_09_EXP_COMP_DSQ_PERS_TOT",
    other_salaries         = "F9_09_EXP_OTH_SAL_WAGE_TOT",
    pension_contributions  = "F9_09_EXP_PENSION_CONTR_TOT",
    other_employee_benefits = "F9_09_EXP_OTH_EMPL_BEN_TOT",
    payroll_taxes          = "F9_09_EXP_PAYROLL_TAX_TOT",
    total_expenses         = "F9_09_EXP_TOT_TOT",
    winsorize              = 0.98 ,
    range                  = "zo",
    sanitize               = TRUE,
    summarize              = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
officer_comp	Compensation of current officers, directors, and key employees (total).
disqualified_comp	Compensation of disqualified persons (total).
other_salaries	Other salaries and wages (total).
pension_contributions	Pension plan accruals and contributions (total).
other_employee_benefits	Other employee benefits (total).
payroll_taxes	Payroll taxes (total).
total_expenses	Total functional expenses.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE).
summarize	Logical (default FALSE).

Details

Primary uses and key insights:

The compensation expense ratio measures the share of total functional expenses devoted to all forms of employee compensation: salaries, benefits, payroll taxes, and pension contributions. For most service-delivering nonprofits this is the largest single expense category (60-80\ indicator of labor intensity and human capital investment).

Formula variations and their sources:

Sum of six Part IX compensation fields (lines 5-10, Column A) / total expenses (line 25A): officer compensation, disqualified person compensation, other salaries, pension contributions, other employee benefits, and payroll taxes.

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good. *Policy and Society*, 20(4), 94-112.
- Leete, L. (2001). Whither the nonprofit wage differential? *Journal of Labor Economics*, 19(1), 136-170.

Variables used::

- F9_09_EXP_COMP_DTK_TOT: Officer/director compensation (officer_comp)
- F9_09_EXP_COMP_DSQ_PERS_TOT: Disqualified person compensation (disqualified_comp)
- F9_09_EXP_OTH_SAL_WAGE_TOT: Other salaries and wages (other_salaries)
- F9_09_EXP_PENSION_CONTR_TOT: Pension contributions (pension_contributions)
- F9_09_EXP_OTH_EMPL_BEN_TOT: Other employee benefits (other_employee_benefits)
- F9_09_EXP_PAYROLL_TAX_TOT: Payroll taxes (payroll_taxes)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)

Value

The original data.frame with four new columns:

- expenses_compensation (raw ratio)
- expenses_compensation_w (winsorized)
- expenses_compensation_z (z-score)
- expenses_compensation_p (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_expenses_compensation_ratio( df = dat10k )
head( d[, c( "expenses_compensation", "expenses_compensation_w", "expenses_compensation_z", "expenses_compensation_p" )]
```

get_expenses_feesforservice_ratio

Fees for Services Expense Ratio

Description

Total fees paid for outside services (management, legal, accounting, lobbying, professional fundraising, investment management, and other) as a share of total functional expenses.

Formula:

$$\text{expenses_feesforservice} = \left(\begin{array}{l} \text{mgmt_fees} + \text{legal_fees} + \text{accounting_fees} \\ + \text{lobbying_fees} + \text{prof_fundraising_fees} \\ + \text{investment_mgmt_fees} + \text{other_fees} \end{array} \right) / \text{total_expenses}$$

Definitional Range

Bounded [0, 1]. Most nonprofits show values in the [0.01, 0.20] range.

Benchmarks and rules of thumb

- No universal benchmark; most meaningful within subsector or over time.
- Unusually high investment management fees relative to total expenses may indicate poor endowment management or excessive fee structures.

Calculated For: 990 filers only.

Usage

```
get_expenses_feesforservice_ratio( df,
    mgmt_fees      = "F9_09_EXP_FEE_SVC_MGMT_TOT",
    legal_fees     = "F9_09_EXP_FEE_SVC_LEGAL_TOT",
    accounting_fees = "F9_09_EXP_FEE_SVC_ACC_TOT",
    lobbying_fees   = "F9_09_EXP_FEE_SVC_LOB_TOT",
    prof_fundraising_fees = "F9_09_EXP_FEE_SVC_FUNDR_TOT",
    investment_mgmt_fees = "F9_09_EXP_FEE_SVC_INVEST_TOT",
    other_fees      = "F9_09_EXP_FEE_SVC_OTH_TOT",
    total_expenses  = "F9_09_EXP_TOT_TOT",
    winsorize       = 0.98 ,
    range           = "zo",
    sanitize        = TRUE,
    summarize       = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
mgmt_fees	Management fees (total).
legal_fees	Legal fees (total).
accounting_fees	Accounting fees (total).
lobbying_fees	Lobbying fees (total).

prof_fundraising_fees	Professional fundraising fees (total).
investment_mgmt_fees	Investment management fees (total).
other_fees	Other fees for services (total).
total_expenses	Total functional expenses.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE).
summarize	Logical (default FALSE).

Details

Primary uses and key insights:

The fees for services ratio measures the share of total expenses paid to outside contractors across seven categories: management consultants, legal, accounting, lobbying, professional fundraisers, investment managers, and other. It captures the degree to which the organization outsources professional expertise.

Formula variations and their sources:

Sum of seven Part IX fee fields (lines 11a-11g, Column A) / total expenses (line 25A).

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good. *Policy and Society*, 20(4), 94-112.

Variables used::

- F9_09_EXP_FEE_SVC_MGMT_TOT: Management fees (mgmt_fees)
- F9_09_EXP_FEE_SVC_LEGAL_TOT: Legal fees (legal_fees)
- F9_09_EXP_FEE_SVC_ACC_TOT: Accounting fees (accounting_fees)
- F9_09_EXP_FEE_SVC_LOB_TOT: Lobbying fees (lobbying_fees)
- F9_09_EXP_FEE_SVC_FUNDR_TOT: Professional fundraising fees (prof_fundraising_fees)
- F9_09_EXP_FEE_SVC_INVEST_TOT: Investment management fees (investment_mgmt_fees)
- F9_09_EXP_FEE_SVC_OTH_TOT: Other fees for services (other_fees)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)

Value

The original data.frame with four new columns:

- expenses_feesforservice (raw ratio)
- expenses_feesforservice_w (winsorized)
- expenses_feesforservice_z (z-score)
- expenses_feesforservice_p (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_expenses_feesforservice_ratio( df = dat10k )
head( d[, c( "expenses_feesforservice", "expenses_feesforservice_w", "expenses_feesforservice_z", "expenses_
```

get_expenses_grants_ratio

Grants-to-Others Expense Ratio

Description

Total grants paid to domestic organizations, domestic individuals, and foreign entities as a share of total functional expenses.

Formula:

$$\text{expenses_grants} = \left(\text{us_org_grants} + \text{us_indiv_grants} + \text{foreign_grants} \right) / \text{total_expenses}$$
Definitional Range

Bounded [0, 1]. Near zero for direct-service organizations; near 1.0 for pure pass-through grant-makers.

Benchmarks and rules of thumb

- Subsector comparisons are essential: a 0.80 ratio is expected for a community foundation; the same value would be anomalous for a hospital.
- Rising ratios over time may indicate a strategic shift toward a more redistributive model.

Calculated For: 990 filers only.

Usage

```
get_expenses_grants_ratio( df,
  us_org_grants   = "F9_09_EXP_GRANT_US_ORG_TOT",
  us_indiv_grants = "F9_09_EXP_GRANT_US_INDIV_TOT",
  foreign_grants  = "F9_09_EXP_GRANT_FRGN_TOT",
  total_expenses  = "F9_09_EXP_TOT_TOT",
  winsorize       = 0.98 ,
  range           = "zo",
  sanitize        = TRUE,
  summarize       = FALSE )
```

Arguments

df A data.frame containing the fields required for computing the metric.

us_org_grants Grants to domestic organizations and governments (total).

us_indiv_grants Grants and assistance to domestic individuals (total).

foreign_grants	Grants to foreign organizations and governments (total).
total_expenses	Total functional expenses.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE).
summarize	Logical (default FALSE).

Details

Primary uses and key insights:

The grants-to-others ratio measures the share of total expenses paid out as grants to other organizations, individuals, or foreign entities. It is the defining characteristic of grant-making and pass-through organizations - foundations, federated funders, and community chests - where the primary activity is resource redistribution rather than direct service delivery.

Formula:

(Domestic org grants + domestic individual grants + foreign grants) / total expenses. Note: US organization grants (line 1) are PZ scope; other components are PC scope.

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good. *Policy and Society*, 20(4), 94-112.

Variables used::

- F9_09_EXP_GRANT_US_ORG_TOT: Grants to domestic organizations (us_org_grants)
- F9_09_EXP_GRANT_US_INDIV_TOT: Grants to domestic individuals (us_indiv_grants)
- F9_09_EXP_GRANT_FRGN_TOT: Grants to foreign entities (foreign_grants)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)

Value

The original data.frame with four columns appended: expenses_grants, expenses_grants_w, expenses_grants_z, expenses_grants_p.

Examples

```
library( fiscal )
data( dat10k )
d <- get_expenses_grants_ratio( df = dat10k )
head( d[ , c( "expenses_grants", "expenses_grants_w", "expenses_grants_z", "expenses_grants_p" ) ] )
```

get_expenses_membbenefits_ratio

Member Benefits Expense Ratio

Description

Benefits paid to or for members as a share of total expenses.

Formula:

$$\text{expenses_membbenefits} = \text{member_benefits} / \text{total_expenses}$$

Definitional Range

Bounded [0, 1]. Near zero for most public benefit charities; potentially high (above 0.50) for mutual benefit organizations.

Benchmarks and rules of thumb

- Interpret in the context of the organization's exempt purpose and subsection code.
- Non-zero values for 501(c)(3) organizations should be reviewed for compliance with private benefit restrictions.

Calculated For: 990 filers only.

Usage

```
get_expenses_membbenefits_ratio( df,
    member_benefits      = "F9_09_EXP_BEN_PAID_MEMB_TOT",
    total_expenses       = "F9_09_EXP_TOT_TOT",
    winsorize = 0.98 ,
    range      = "zo",
    sanitize   = TRUE,
    summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
member_benefits	Benefits paid to or for members (total).
total_expenses	Total functional expenses.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Primary uses and key insights:

The member benefits expense ratio measures the share of total functional expenses paid out as direct benefits to members. Primarily relevant for membership associations (unions, trade associations, professional societies) where member benefit payments are a core activity. For most public benefit nonprofits this ratio is near zero.

Formula:

Benefits paid to or for members (Part IX line 4, Column A) / total expenses (line 25A). The numerator field is PZ scope, but the denominator (F9_09_EXP_TOT_TOT) is reported only on the full 990, so the ratio is calculated for 990 filers only.

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good. *Policy and Society*, 20(4), 94-112.

Variables used::

- F9_09_EXP_BEN_PAID_MEMB_TOT: Benefits paid to members (member_benefits)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)

Value

The original data.frame with four new columns:

- expenses_membbenefits (raw ratio)
- expenses_membbenefits_w (winsorized)
- expenses_membbenefits_z (z-score)
- expenses_membbenefits_p (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_expenses_membbenefits_ratio( df = dat10k )
head( d[, c( "expenses_membbenefits", "expenses_membbenefits_w", "expenses_membbenefits_z", "expenses_membbenefits_p" )]
```

```
get_fundraising_efficiency_ratio
```

Fundraising Efficiency Ratio

Description

Cost of raising one dollar of contributions; measures how efficiently fundraising expenses generate donated revenue.

Formula:

fer = fundraising_expenses / total_contributions

Definitional Range

Bounded below at zero; values above 1.0 (spending more than a dollar to raise a dollar) are possible, particularly for young organizations building donor bases or those in difficult fundraising environments. The typical range is approximately [0, 0.50] for most operating nonprofits.

Benchmarks and rules of thumb

- **<= 10 cents per dollar (ratio <= 0.10):** Charity Navigator threshold for a full fundraising efficiency score.
- **<= 35 cents per dollar:** BBB Wise Giving Alliance standard.
- **Above 50 cents:** Commonly flagged as inefficient.

Calculated For: 990 filers only.

Usage

```
get_fundraising_efficiency_ratio( df,
    fundraising_expenses = "F9_09_EXP_TOT_FUNDR",
    total_contributions = "F9_08_REV_CONTR_TOT",
    winsorize = 0.98 ,
    range      = "zp",
    sanitize   = TRUE,
    summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
fundraising_expenses	Fundraising expenses.
total_contributions	Total contributions received.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The fundraising efficiency ratio measures the cost of raising one dollar of contributions. A ratio of 0.20 means the organization spends 20 cents to raise each dollar - a common benchmark in

the charity watchdog literature. Lower values indicate more efficient fundraising; higher values indicate more costly fundraising relative to the contributions raised.

This metric should be interpreted alongside the fundraising strategy: organizations investing in major gift programs or capital campaigns may show temporarily high ratios that reflect future revenue not yet received. Mature direct mail programs with large donor databases tend to show very low ratios.

Formula variations and their sources:

Fundraising expenses (Part IX line 25D) / total contributions (Part VIII line 1h). An alternative uses total revenue rather than contributions in the denominator, which produces a lower ratio but conflates earned and contributed income. The contributions-only denominator is the most direct measure of fundraising productivity.

Note: contributions (Part VIII 1h) include grants and federated campaign revenue alongside individual gifts, which may inflate the denominator for organizations receiving large government or foundation grants with minimal fundraising cost.

Canonical citations:

- Greenlee, J.S. & Trussel, J.M. (2000). Predicting the financial vulnerability of charitable organizations. *Nonprofit Management and Leadership*, 11(2), 199-210.
- Charity Navigator. *Financial Health Methodology*. charitynavigator.org.
 - Uses fundraising efficiency as a rating factor (target: 10 cents per dollar or less).
- Hager, M.A. (2001). Financial vulnerability among arts organizations. *Nonprofit and Voluntary Sector Quarterly*, 30(2), 376-392.

Variables used::

- F9_09_EXP_TOT_FUNDR: Total fundraising expenses (fundraising_expenses)
- F9_08_REV_CONTR_TOT: Total contributions received (total_contributions)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``fundr_eff`` - fundraising efficiency ratio (raw)
- ``fundr_eff_w`` - winsorized version
- ``fundr_eff_z`` - standardized z-score (based on winsorized values)
- ``fundr_eff_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_fundraising_efficiency_ratio( df = dat10k )
head( d[ , c( "fundr_eff", "fundr_eff_w", "fundr_eff_z", "fundr_eff_p" ) ] )
```

get_grants_govt_ratio *Government Grant Ratio*

Description

Share of total revenue originating from government grants.

Formula:

$$ggr = \text{government_grants} / \text{total_revenue}$$

Definitional Range

Bounded [0, 1]. Zero means no government funding; one means all revenue comes from government sources. The empirical distribution varies widely by subsector: social service organizations often show ratios above 0.50; arts organizations typically below 0.20.

Benchmarks and rules of thumb

- Values above 0.50 indicate majority government-funded organizations, which face heightened vulnerability to government budget changes.
- Revenue diversification theory suggests maintaining no single source above 0.33-0.50 for optimal stability (Chang & Tuckman 1994).

Calculated For: 990 filers only.

Usage

```
get_grants_govt_ratio( df,
  government_grants = "F9_08_REV_CONTR_GOVT_GRANT",
  total_revenue     = "F9_08_REV_TOT_TOT",
  winsorize = 0.98 ,
  range      = "zo",
  sanitize   = TRUE,
  summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
government_grants	Government grants and contributions.
total_revenue	Total revenue.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.

`summarize` Logical. If TRUE, prints a `summary()` of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The government grant ratio measures what fraction of total revenue comes from government grants and contributions. It is a revenue diversification indicator specifically focused on public sector dependency. High ratios indicate heavy reliance on government funding, which creates exposure to policy changes, budget cycles, and political priorities that can suddenly alter the funding environment.

Government funding also typically comes with extensive compliance, reporting, and audit requirements that increase administrative costs, making the government grant ratio a useful input to overhead analysis. This ratio is a core component of revenue concentration and diversification studies in the nonprofit literature.

Formula variations and their sources:

Government grants (Part VIII line 1e) / total revenue (Part VIII line 12A). Some studies also include government contracts reported as program service revenue (Part VIII line 2), which would capture government fees-for-service alongside grants, but these cannot be separated from private program revenue on the 990.

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290. - Foundational paper on revenue concentration risk.
- Carroll, D.A. & Stater, K.J. (2009). Revenue diversification in nonprofit organizations: Does it lead to financial stability? *Journal of Public Administration Research and Theory*, 19(4), 947-966.
- Froelich, K.A. (1999). Diversification of revenue strategies: Evolving resource dependence in nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 28(3), 246-268.

Variables used::

- F9_08_REV_CONTR_GOV_TOT: Government grants and contributions (`government_grants`)
- F9_08_REV_TOT_TOT: Total revenue (`total_revenue`)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

```
- `grants_govt` - government grant ratio (raw)
- `grants_govt_w` - winsorized version
- `grants_govt_z` - standardized z-score (based on winsorized values)
- `grants_govt_p` - percentile rank (1-100)
```

Examples

```
library( fiscal )
data( dat10k )

d <- get_grants_govt_ratio( df = dat10k )
head( d[ , c( "grants_govt", "grants_govt_w", "grants_govt_z", "grants_govt_p" ) ] )
```

get_idvars

Return the list of efile record identifier variables

Description

Returns the character vector of column names that identify a filing record across all efile tables. These are retained alongside financial fields in the working subset inside every get_*() function.

Usage

```
get_idvars()
```

Value

A character vector of column names.

Examples

```
get_idvars()
```

get_industry

Derive top-level NTEE industry classification

Description

Maps raw NTEE codes to their 3-letter high-level industry group. Special overrides: B40-B43/B50 ? UNI (Universities), E20-E24 ? HOS (Hospitals).

Usage

```
get_industry(x)
```

Arguments

x Character vector of NTEE codes (raw or cleaned).

Value

Character vector of 3-letter industry codes.

Examples

```
get_industry( c("B29", "B8443", "E0521", "Q12", "Z99") )
```

get_investments_assets_ratio
Investment Assets Ratio

Description

Investment securities as a share of total assets.

Formula:

$$\text{investments_assets} = (\text{pub_traded_securities} + \text{other_securities}) / \text{total_assets}$$

Definitional Range

Bounded [0, 1]. Most operating nonprofits show values near zero; foundations and endowed institutions may show values above 0.80.

Benchmarks and rules of thumb

- No universal benchmark. Operating nonprofits with values above 0.30 are typically holding significant endowment or reserve portfolios.
- A high investment ratio combined with a high debt ratio may indicate borrowing to fund operations while holding investments – worth scrutinizing.

Calculated For: 990 filers only.

Usage

```
get_investments_assets_ratio( df,
    pub_traded_securities    = "F9_10_ASSET_INVEST_SEC_EOY",
    total_assets             = "F9_10_ASSET_TOT_EOY",
    other_securities         = "F9_10_ASSET_INVEST_SEC_OTH_EOY",
    winsorize = 0.98 ,
    range       = "zo",
    sanitize    = TRUE,
    summarize   = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
pub_traded_securities	Investments in publicly traded securities, EOY.
total_assets	Total assets, EOY.
other_securities	Investments in other securities, EOY.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Primary uses and key insights:

The investment assets ratio measures what share of total assets is held in financial securities (publicly traded and other investments). It distinguishes financially oriented organizations (endowments, foundations) from operationally oriented ones. A high ratio indicates significant financial reserves; a low ratio indicates assets are primarily operational. Most informative for endowed organizations and foundations.

Formula variations and their sources:

(Publicly traded securities + other securities) / total assets. Both investment fields (Part X lines 11B and 12B) are PZ scope. A more comprehensive version would include program-related investments (line 13B).

Canonical citations:

- Bowman, W. (2011). Financial capacity and sustainability of ordinary nonprofits. *Nonprofit Management and Leadership*, 22(1), 37-51.
- Calabrese, T.D. (2013). Running on empty. *Nonprofit Management and Leadership*, 23(3), 281-302.

Variables used::

- F9_10_ASSET_INVEST_SEC_EOY: Publicly traded securities, EOY (pub_traded_securities)
- F9_10_ASSET_INVEST_SEC_OTH_EOY: Other securities, EOY (other_securities)
- F9_10_ASSET_TOT_EOY: Total assets, EOY (total_assets)

Value

The original data.frame with four new columns:

- investments_assets (raw ratio)
- investments_assets_w (winsorized)
- investments_assets_z (z-score)
- investments_assets_p (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_investments_assets_ratio( df = dat10k )
head( d[, c( "investments_assets", "investments_assets_w", "investments_assets_z", "investments_assets_p" ) ] )
```

get_investment_income_ratio

Investment Income Dependency Ratio

Description

Measures reliance on investment-related revenue as a share of total revenue.

Formula:

$$iidr = \text{investment_income} / \text{total_revenue}$$

$$\text{investment_income} = \text{invest_income} + \text{bond_proceeds} + \text{rent_income} + \text{asset_sale_income}$$

Definitional Range

Bounded [0, 1] in most years, but values above 1.0 are possible in years with large asset sale gains. Values below zero can result from asset sale losses or negative investment returns.

Benchmarks and rules of thumb

- Values above 0.30 indicate significant financial asset dependency and warrant monitoring of portfolio performance.
- Large year-over-year swings often reflect one-time asset transactions rather than structural revenue changes.

Calculated For: 990 filers only.

Usage

```
get_investment_income_ratio( df,
    invest_income      = "F9_08_REV_OTH_INVEST_INCOME_TOT",
    bond_proceeds      = "F9_08_REV_OTH_INVEST_BOND_TOT",
    rent_income        = "F9_08_REV_OTH_RENT_NET_TOT",
    asset_sale_income  = "F9_08_REV_OTH_SALE_GAIN_NET_TOT",
    total_revenue      = "F9_08_REV_TOT_TOT",
    numerator = NULL, denominator = NULL, winsorize = 0.98 ,
    range            = "np",
    sanitize         = TRUE,
    summarize        = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
invest_income	Investment income.
bond_proceeds	Income from investment of tax-exempt bond proceeds.
rent_income	Net rental income or loss (Part VIII line 6d, column A).
asset_sale_income	Net gain or loss from sales of assets other than inventory (Part VIII line 7d, column A).
total_revenue	Total revenue.

numerator	Optional. A pre-aggregated column for investment income. When supplied, invest_income, bond_proceeds, rent_income, asset_sale_income are ignored; it is an error to also set one of them explicitly.
denominator	Optional. A pre-aggregated column for total revenue. When supplied, total_revenue is ignored; it is an error to also set it explicitly.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The investment income dependency ratio measures the share of total revenue from investment-related sources: interest, dividend income, rental income, bond proceeds, and proceeds from asset sales. It captures financial portfolio dependency - the degree to which the organization's budget relies on endowment returns, rental properties, or investment gains.

A high ratio may be a sign of financial maturity (large endowment generating returns) or financial risk (dependency on volatile investment markets). During periods of market decline or low interest rates, organizations with high investment income ratios face greater revenue volatility.

Formula variations and their sources:

$(\text{Investment income} + \text{bond income} + \text{rental income} + \text{asset sale income}) / \text{total revenue}$ (Part VIII lines 3 + 4 + 6d + 7d / line 12A). Note that this combines recurring investment income with potentially one-time asset sale proceeds; some analysts separate these. The asset sale proceeds (line 7d) are particularly volatile and may distort the ratio in years with large asset disposals.

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290.
- Carroll, D.A. & Stater, K.J. (2009). Revenue diversification in nonprofit organizations. *Journal of Public Administration Research and Theory*, 19(4), 947-966.

Variables used::

- F9_08_REV_OTH_INVEST_INCOME_TOT: Investment income (invest_income)
- F9_08_REV_OTH_INVEST_BOND_TOT: Income from bond proceeds (bond_proceeds)
- F9_08_REV_OTH_RENT_NET_TOT: Net rental income or loss, line 6d (rent_income)
- F9_08_REV_OTH_SALE_GAIN_NET_TOT: Net gain or loss from asset sales, line 7d (asset_sale_income)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``invest_income`` - investment income dependency ratio (raw)
- ``invest_income_w`` - winsorized version
- ``invest_income_z`` - standardized z-score (based on winsorized values)
- ``invest_income_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_investment_income_ratio( df = dat10k )
head( d[ , c( "invest_income", "invest_income_w", "invest_income_z", "invest_income_p" ) ] )
```

```
get_land_assets_gross_ratio
```

Land Asset Ratio

Description

Share of total assets invested in land, buildings, and equipment.

Formula:

$$\text{lar} = \text{land_buildings_equipment} / \text{total_assets}$$
Definitional Range

Usually in [0, 1], but can exceed 1 when accumulated depreciation is large relative to other assets. The distribution is highly right-skewed: most nonprofits with no owned real estate show values near zero, while capital-intensive organizations (healthcare, housing, higher education) may show values above 0.50.

Benchmarks and rules of thumb

- Most informative for within-subsector comparisons.
- High values indicate illiquidity risk: if the majority of assets are fixed property, the organization has limited ability to quickly convert assets to cash.

Calculated For: 990 filers only.

Usage

```
get_land_assets_gross_ratio( df,
  land_buildings = "F9_10_ASSET_LAND_BLDG",
  total_assets   = "F9_10_ASSET_TOT_EOY",
  winsorize      = 0.98 ,
  range         = "zp",
  sanitize      = TRUE,
  summarize     = FALSE )
```

Arguments

<code>df</code>	A <code>data.frame</code> containing the fields required for computing the metric.
<code>land_buildings</code>	Land, buildings, and equipment at cost or other basis, before accumulated depreciation (Part X line 10a).
<code>total_assets</code>	Total assets, EOY.
<code>winsorize</code>	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
<code>summarize</code>	Logical. If TRUE, prints a <code>summary()</code> of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details**Primary uses and key insights:**

The land, buildings, and equipment to assets ratio (gross version) measures what share of total assets is represented by the gross book value of fixed physical assets before accumulated depreciation is netted out. A high ratio indicates a capital-intensive organization with a large physical footprint; a low ratio indicates a lean service organization whose assets are primarily financial.

The gross version uses the cost or other basis of land, buildings, and equipment (Part X line 10a), before accumulated depreciation (line 10b) is netted out. This is distinct from the net version ([get_land_assets_net_ratio\(\)](#)), which uses the net book value after depreciation (Part X line 10c). The gross version gives a better picture of the original investment in fixed assets; the net version better reflects current book value.

Formula variations and their sources:

$F9_{10_ASSET_LAND_BLDG}$ (cost or other basis, line 10a) / total assets. Because total assets carry fixed assets at net book value, the ratio can exceed 1 for heavily depreciated organizations. Earlier versions used accumulated depreciation ($F9_{10_ASSET_LAND_BLDG_DEPREC}$, line 10b), which is not the gross cost.

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good. *Policy and Society*, 20(4), 94-112. - Asset composition analysis for nonprofits.

Variables used::

- $F9_{10_ASSET_LAND_BLDG}$: Land, buildings, and equipment, cost or other basis, line 10a (`land_buildings`)
- $F9_{10_ASSET_TOT_EOY}$: Total assets, EOY (`total_assets`)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `land\_assets\_gross` - land asset ratio (raw)
- `land\_assets\_gross\_w` - winsorized version
- `land\_assets\_gross\_z` - standardized z-score (based on winsorized values)
- `land\_assets\_gross\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_land_assets_gross_ratio( df = dat10k )
head( d[ , c( "land_assets_gross", "land_assets_gross_w", "land_assets_gross_z", "land_assets_gross_p" ) ] )
```

```
get_land_assets_net_ratio
```

Land, Buildings, and Equipment to Assets Ratio (Net)

Description

Net land, buildings, and equipment as a share of total assets.

Formula:

$$\text{land_assets_net} = \text{land_bldg_equip_net} / \text{total_assets}$$
Definitional Range

Bounded [0, 1] in normal conditions. Heavily skewed toward zero because many nonprofits own no real property.

Benchmarks and rules of thumb

- Values below 0.10 are typical for lean service organizations.
- Values above 0.50 indicate that more than half of assets are fixed property, limiting financial flexibility.
- Declining values over time may reflect depreciation outpacing new capital investment rather than strategic deleveraging.

Calculated For: 990 + 990EZ filers.

Usage

```
get_land_assets_net_ratio( df,
  land_bldg_equip_net      = "F9_10_ASSET_LAND_BLDG_NET_EOY",
  total_assets             = "F9_10_ASSET_TOT_EOY",
  winsorize                = 0.98 ,
  range                    = "zo",
  sanitize                 = TRUE,
  summarize                 = FALSE )
```

Arguments

<code>df</code>	A <code>data.frame</code> containing the fields required for computing the metric.
<code>land_bldg equip_net</code>	Net land, buildings, and equipment EOY (after accumulated depreciation).
<code>total_assets</code>	Total assets, EOY.
<code>winsorize</code>	Winsorization proportion between 0 and 1 (default 0.98).
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
<code>summarize</code>	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Primary uses and key insights:

The land, buildings, and equipment to assets ratio (net version) measures what share of total assets is represented by the net book value of fixed physical assets after accumulated depreciation. This is the standard version used in most nonprofit financial analyses, since it reflects current accounting value. High values indicate capital-intensive organizations; low values indicate lean service organizations. Very high values (above 0.60) may suggest illiquidity risk. Available on both 990 and 990EZ forms (PZ scope).

Formula variations and their sources:

Net land, buildings, and equipment (Part X line 10cB) / total assets (line 16B). The gross version ([get_land_assets_gross_ratio\(\)](#)) uses accumulated depreciation instead of net book value.

Canonical citations:

- Frumkin, P. & Keating, E.K. (2001). The price of doing good. *Policy and Society*, 20(4), 94-112.
- Bowman, W. (2011). Financial capacity and sustainability of ordinary nonprofits. *Nonprofit Management and Leadership*, 22(1), 37-51.

Variables used::

- F9_10_ASSET_LAND_BLDG_NET_EOY: Net land, buildings, and equipment, EOY (`land_bldg equip_net`)
- F9_10_ASSET_TOT_EOY: Total assets, EOY (`total_assets`)

Value

The original `data.frame` with four columns appended: `land_assets_net`, `land_assets_net_w`, `land_assets_net_z`, `land_assets_net_p`.

Examples

```
library( fiscal )
data( dat10k )
d <- get_land_assets_net_ratio( df = dat10k )
head( d[ , c( "land_assets_net", "land_assets_net_w", "land_assets_net_z", "land_assets_net_p" ) ] )
```

get_liquid_assets_months

Months of Liquid Unrestricted Net Assets (LUNA)

Description

Number of months an organization can operate using its liquid unrestricted net assets.

Formula:

$$\text{luna} = \text{liquid_unrestricted_net_assets} / \text{monthly_expenses}$$

$$\text{liquid_unrestricted_net_assets} = \text{unrestricted_net_assets} - (\text{net_fixed_assets} - \text{mortgages_payable})$$

$$\text{monthly_expenses} = \text{total_expenses} / 12$$

Definitional Range

Unbounded in both directions, expressed in months. Negative values indicate that fixed property (net of mortgage debt) exceeds unrestricted net assets. The practical range for operating nonprofits is approximately [-12, 36] months.

Benchmarks and rules of thumb

- Same general benchmarks as [get_operating_reserve_ratio\(\)](#) (which is expressed in years rather than months): 3-6 months is considered adequate for most operating nonprofits.
- Negative LUNA values are particularly informative: they indicate unrestricted equity is entirely absorbed by illiquid fixed assets.

Calculated For: 990 filers only.

Usage

```
get_liquid_assets_months( df,
    unrestricted_net_assets = "F9_10_NAFB_UNRESTRICT_EOY",
    net_fixed_assets         = "F9_10_ASSET_LAND_BLDG_NET_EOY",
    mortgages_payable        = "F9_10_LIAB_MTG_NOTE_EOY",
    total_expenses           = "F9_09_EXP_TOT_TOT",
    restricted_net_assets     = "F9_10_NAFB_RESTRICT_EOY",
    total_net_assets         = "F9_10_NAFB_TOT_EOY",
    winsorize = 0.98 ,
    range     = "np",
    sanitize  = TRUE,
    summarize = FALSE )
```

Arguments

df A data.frame containing the fields required for computing the metric.

unrestricted_net_assets Unrestricted net assets, EOY.

net_fixed_assets Net land, buildings, and equipment, EOY.

mortgages_payable	Mortgages and notes payable, EOY.
total_expenses	Total functional expenses.
restricted_net_assets	Restricted net assets, EOY. Used with total_net_assets to detect filers that do not follow SFAS 117; see resolve_unrestricted_net_assets() .
total_net_assets	Total net assets, EOY. Substituted for unrestricted net assets when unrestricted and restricted net assets are both zero. Set to NULL to disable the fallback.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

Liquid unrestricted net assets (LUNA) measures how many months of operating expenses are covered by unrestricted net assets that are not tied up in illiquid fixed property (net of associated mortgage debt). It is a refinement of the operating reserve ratio ([get_operating_reserve_ratio\(\)](#)) that more precisely isolates liquid resources by netting out the value of fixed assets but adding back the portion financed by mortgage debt (since that debt reduces the net equity encumbrance of the fixed asset).

Formula variations and their sources:

$(\text{Unrestricted net assets} - \text{net fixed assets} + \text{mortgages payable}) / (\text{total expenses} / 12)$. The addition of mortgage debt back into the numerator reflects the logic that if the organization sold its building, it would receive the net proceeds (market value minus mortgage balance) - and the mortgage liability already reduces unrestricted net assets. Adding it back avoids double-counting the encumbrance.

This formulation follows Zietlow et al. (2007) and is cited by Calabrese (2013). It differs from [get_operating_reserve_ratio\(\)](#) in the addition of mortgages_payable to the numerator and in its units: LUNA divides by monthly expenses (months), while the operating reserve ratio divides by annual expenses (a multiple of annual expenses).

Canonical citations:

- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley. - Primary source for the LUNA definition.
- Calabrese, T.D. (2013). Running on empty. *Nonprofit Management and Leadership*, 23(3), 281-302. - Empirical application and validation.

Variables used::

- F9_10_NAFB_UNRESTRICT_EOY: Unrestricted net assets, EOY (unrestricted_net_assets)
- F9_10_ASSET_LAND_BLDG_NET_EOY: Net land, buildings, and equipment (net_fixed_assets)
- F9_10_LIAB_MTG_NOTE_EOY: Mortgages and notes payable (mortgages_payable)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)
- F9_10_NAFB_RESTRICT_EOY: Restricted net assets, EOY (restricted_net_assets)
- F9_10_NAFB_TOT_EOY: Total net assets, EOY (total_net_assets)

Unrestricted net assets fall back to total net assets for filers that do not follow SFAS 117 (see [resolve_unrestricted_net_assets\(\)](#)).

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `liquid\_assets\_months` - months of liquid unrestricted net assets (raw)
- `liquid\_assets\_months\_w` - winsorized version
- `liquid\_assets\_months\_z` - standardized z-score (based on winsorized values)
- `liquid\_assets\_months\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_liquid_assets_months( df = dat10k )
head( d[, c( "liquid_assets_months", "liquid_assets_months_w", "liquid_assets_months_z", "liquid_assets_months_p" )])
```

```
get_months_cash_operations
```

Months of Operating Cash on Hand

Description

Number of months the organization can operate using available liquid assets.

Formula:

$$\text{moch} = \text{liquid_assets} / \text{monthly_expenses}$$

$$\begin{aligned} \text{liquid_assets} &= \text{cash} + \text{savings} + \text{pledges_receivable} + \text{accounts_receivable} \\ \text{monthly_expenses} &= (\text{total_expenses} - \text{depreciation}) / 12 \end{aligned}$$
Definitional Range

Bounded below at zero; unbounded above. The typical operating range for nonprofits is approximately [0, 24] months, with values above 12 months uncommon for operating organizations (more typical for foundations).

Benchmarks and rules of thumb

- **Less than 1 month:** Acute liquidity risk.

- **1-3 months:** Below adequate; common guidance is at least 3 months.
- **3-6 months:** Generally considered healthy for most operating nonprofits.
- **6+ months:** Strong reserve position.
- The Nonprofit Finance Fund recommends 3-6 months as a target.

Calculated For: 990 filers only.

Usage

```
get_months_cash_operations( df,
    cash = "F9_10_ASSET_CASH_EOY",
    savings = "F9_10_ASSET_SAVING_EOY",
    pledges_receivable = "F9_10_ASSET_PLEDGE_NET_EOY",
    accounts_receivable = "F9_10_ASSET_ACC_NET_EOY",
    total_expenses = "F9_09_EXP_TOT_TOT",
    depreciation = "F9_09_EXP_DEPREC_TOT",
    numerator = NULL, denominator = NULL, winsorize = 0.98 ,
    range = "zp",
    sanitize = TRUE,
    summarize = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
cash	Cash on hand, EOY.
savings	Short-term investments (savings), EOY.
pledges_receivable	Pledges and grants receivable, EOY.
accounts_receivable	Accounts receivable, EOY.
total_expenses	Total functional expenses.
depreciation	Depreciation, depletion, and amortization.
numerator	Optional. A pre-aggregated column name for liquid assets. When supplied, cash, savings, pledges_receivable, accounts_receivable are ignored; it is an error to also set one of them explicitly.
denominator	Optional. A pre-aggregated column name for the denominator (annual non-depreciation expenses). The function divides this by 12 internally. When supplied, total_expenses, depreciation are ignored; it is an error to also set one of them explicitly.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.

`summarize` Logical. If TRUE, prints a `summary()` of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

Months of cash on hand is the monthly expression of `get_days_cash_operations()`, expressing the same liquidity concept in a unit that is more natural for annual budget planning and board reporting. It is one of the most commonly reported metrics in nonprofit financial dashboards and funder due diligence.

Formula variations and their sources:

The denominator uses $(\text{total expenses} - \text{depreciation}) / 12$. Subtracting depreciation follows standard practice (Zietlow et al. 2007) because the ratio measures cash coverage of cash expenses. Some formulations use total expenses without the depreciation adjustment; others use a rolling 12-month average of monthly expenses to smooth seasonal variation, which is not possible with annual 990 data.

Canonical citations:

- Nonprofit Finance Fund. *State of the Nonprofit Sector Survey* (annual).
 - Uses months of cash as a primary financial health indicator.
- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley.
- GuideStar/Candid. *Nonprofit Finance Indicators*. - Months of cash is one of the core metrics in the GuideStar financial health profile.

Variables used::

- F9_10_ASSET_CASH_EOY: Cash on hand (cash)
- F9_10_ASSET_SAVING_EOY: Savings (savings)
- F9_10_ASSET_PLEDGE_NET_EOY: Net pledges receivable (pledges_receivable)
- F9_10_ASSET_ACC_NET_EOY: Accounts receivable (accounts_receivable)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)
- F9_09_EXP_DEPREC_TOT: Depreciation and amortization (depreciation)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``months_cash_ops`` - months of operating cash on hand (raw)
- ``months_cash_ops_w`` - winsorized version
- ``months_cash_ops_z`` - standardized z-score (based on winsorized values)
- ``months_cash_ops_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_months_cash_operations( df = dat10k )
head( d[ , c( "months_cash_ops", "months_cash_ops_w", "months_cash_ops_z", "months_cash_ops_p" ) ] )
```

get_netassets_composition_ratio
Net Assets Composition Ratio

Description

Share of total net assets that are unrestricted and available for general use.

Formula:

$$\text{nacr} = \text{unrestricted_net_assets} / \text{total_net_assets}$$

Definitional Range

Bounded [0, 1] when both numerator and denominator are positive: zero means all net assets are restricted; one means all net assets are unrestricted. Values outside this range occur when either unrestricted or total net assets are negative.

Benchmarks and rules of thumb

- **Above 0.50:** At least half of net assets are unrestricted – generally healthy.
- **Below 0.20:** Heavy donor-restricted balance sheet; limited flexibility.
- Organizations with large permanent endowments may show low ratios without financial weakness.

Calculated For: 990 filers only.

Usage

```
get_netassets_composition_ratio( df,
    unrestricted_net_assets = "F9_10_NAFB_UNRESTRICT_EOY",
    total_net_assets        = "F9_10_NAFB_TOT_EOY",
    restricted_net_assets    = "F9_10_NAFB_RESTRICT_EOY",
    winsorize = 0.98 ,
    range      = "zo",
    sanitize   = TRUE,
    summarize  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
unrestricted_net_assets	Unrestricted net assets, EOY.
total_net_assets	Total net assets, EOY.
restricted_net_assets	Restricted net assets, EOY. When unrestricted and restricted net assets are both zero (filers that do not follow SFAS 117), all net assets are treated as unrestricted; see resolve_unrestricted_net_assets() .
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.

range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The net assets composition ratio measures what fraction of total net assets is unrestricted - available for general operations without donor-imposed constraints. A high ratio means the organization has substantial financial flexibility; a low ratio means most net assets are restricted and cannot be redirected to cover operating shortfalls or unexpected needs.

This metric is specific to the nonprofit sector: commercial firms have no equivalent concept because all equity is by definition "unrestricted." It is particularly relevant for organizations that receive large restricted gifts or endowments, where total net assets may look strong but the unrestricted portion available for operations may be minimal.

Formula variations and their sources:

Unrestricted net assets (Part X line 27B) / total net assets (Part X line 33B). An alternative formulation divides unrestricted net assets by total assets, which is more conservative and is sometimes used in rating agency analyses. The net assets denominator version is more common in the academic literature.

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOL-UNTAS: International Journal of Voluntary and Nonprofit Organizations*, 5(3), 273-290. - Discusses the role of unrestricted resources in financial flexibility.
- Calabrese, T.D. (2013). Running on empty. *Nonprofit Management and Leadership*, 23(3), 281-302. - Examines unrestricted reserve composition.

Variables used::

- F9_10_NAFB_UNRESTRICT_EOY: Unrestricted net assets, EOY (unrestricted_net_assets)
- F9_10_NAFB_TOT_EOY: Total net assets, EOY (total_net_assets)
- F9_10_NAFB_RESTRICT_EOY: Restricted net assets, EOY (restricted_net_assets)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `netassets\_comp` - net assets composition ratio (raw)
- `netassets\_comp\_w` - winsorized version
- `netassets\_comp\_z` - standardized z-score (based on winsorized values)
- `netassets\_comp\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_netassets_composition_ratio( df = dat10k )
head( d[, c( "netassets_comp", "netassets_comp_w", "netassets_comp_z", "netassets_comp_p" ) ] )
```

```
get_netassets_growth_ratio
```

Net Assets Growth Ratio

Description

Year-over-year change in total net assets, expressed as a proportion of beginning net assets.

Formula:

$$\text{or} = (\text{net_assets_eoy} - \text{net_assets_boy}) / \text{net_assets_boy}$$
Definitional Range

Unbounded in both directions. A value of 0 means net assets were unchanged. Values below -1.0 (a drop of more than 100% lost more than its entire starting equity base, resulting in negative net assets. The ratio is undefined (NA) when BOY net assets equal zero.

Benchmarks and rules of thumb

- Small positive values (0.01-0.10) are typical and healthy for operating nonprofits.
- **Below -0.10** in a single year is a warning sign; sustained negative growth indicates a structural financial problem.
- Very high positive values (above 0.50) often reflect one-time large gifts or asset revaluations rather than sustained operational performance.

Calculated For: 990 + 990EZ filers.

Usage

```
get_netassets_growth_ratio( df,
  net_assets_eoy = c( "F9_10_NAFB_TOT_EOY", "F9_01_NAFB_TOT_EOY" ),
  net_assets_boy = c( "F9_10_NAFB_TOT_BOY", "F9_01_NAFB_TOT_BOY" ),
  winsorize = 0.98 ,
  range      = "np",
  sanitize   = TRUE,
  summarize  = FALSE )
```

Arguments

df A data.frame containing the fields required for computing the metric.

net_assets_eoy Total net assets, end of year. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority over 990EZ.

net_assets_boy Total net assets, beginning of year. Accepts one or two column names.

winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The net assets growth ratio measures the year-over-year percentage change in total net assets - essentially the organizational equivalent of return on equity growth. Positive values indicate the organization grew its equity base during the year (surplus plus other balance sheet adjustments); negative values indicate net asset erosion (deficit or write-downs).

This ratio captures all changes to net assets, including unrealized investment gains, prior-period adjustments, and currency translation effects, not just the operating surplus. It is therefore a broader measure than `get_return_netassets_ratio()`, which uses only the revenues-less-expenses figure.

Formula variations and their sources:

$(\text{Net assets EOY} - \text{Net assets BOY}) / \text{Net assets BOY}$. This is the standard percentage change formula. Some studies use the absolute dollar change as the dependent variable rather than a ratio, especially when the BOY base is near zero or negative.

Using BOY net assets (rather than EOY or average) as the denominator is the standard practice for performance ratios: it measures the return generated on the starting equity base, avoiding circular dependency since the ending value is determined in part by the surplus being measured.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460. - Equity growth is a component of the financial vulnerability framework.
- Greenlee, J.S. & Trussel, J.M. (2000). Predicting the financial vulnerability of charitable organizations. *Nonprofit Management and Leadership*, 11(2), 199-210.

Variables used::

- F9_10_NAFB_TOT_EOY: Total net assets, EOY (net_assets_eoy, 990)
- F9_01_NAFB_TOT_EOY: Net assets from Part I, EOY (net_assets_eoy, 990EZ fallback)
- F9_10_NAFB_TOT_BOY: Total net assets, BOY (net_assets_boy, 990)
- F9_01_NAFB_TOT_BOY: Net assets from Part I, BOY (net_assets_boy, 990EZ fallback)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``netassets_growth`` - net assets growth ratio (raw)
- ``netassets_growth_w`` - winsorized version
- ``netassets_growth_z`` - standardized z-score (based on winsorized values)
- ``netassets_growth_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_netassets_growth_ratio( df = dat10k )
head( d[ , c( "netassets_growth", "netassets_growth_w", "netassets_growth_z", "netassets_growth_p" ) ] )
```

get_nteev2

Generate full NTEE v2 code

Description

Combines the normalized NTEE code, industry group, and organization type into the three-part structure: [INDUSTRY]-[NTEE]-[ORGTTYPE].

Usage

```
get_nteev2(x)
```

Arguments

`x` Character vector of raw NTEE codes.

Value

Character vector of NTEE v2 codes (e.g., "EDU-B29-RG").

Examples

```
get_nteev2( c("B29", "B8443", "E0521", "B01", "B0129") )
```

get_ntmaj12

*Convert NTEE V2 codes to NTMAJ12 major group labels***Description**

Convert NTEE V2 codes to NTMAJ12 major group labels

Usage

```
get_ntmaj12(nteelv2)
```

Arguments

nteelv2 Character vector of NTEE V2 codes.

Value

Factor vector with 12 levels (ARTS, EDUCATION, HIGHER ED, etc.).

Examples

```
get_ntmaj12( c("HOS-E05-RG", "EDU-B29-RG") )
```

get_operating_reserve_ratio

*Operating Reserves Ratio***Description**

Liquid unrestricted net assets available to cover operating expenses, expressed as a multiple of total annual expenses.

Formula:

$$\text{orr} = \text{operating_reserves} / \text{total_expenses}$$

$$\text{operating_reserves} = \text{unrestricted_net_assets} - \text{net_fixed_assets}$$
Definitional Range

Unbounded in both directions, expressed as a multiple of annual expenses (multiply by 12 to convert to months). Negative values occur when unrestricted net assets are negative or when net fixed assets exceed unrestricted net assets. Values above 2.0 (24 months) are uncommon for operating organizations and may indicate accumulation beyond what is needed for operating risk management.

Benchmarks and rules of thumb

- **Less than 0.083 (1 month):** Acute vulnerability.
- **0.083-0.25 (1-3 months):** Below the commonly recommended minimum.
- **0.25-0.50 (3-6 months):** Generally adequate for most operating nonprofits.
- **0.50-1.0 (6-12 months):** Strong; appropriate for volatile-revenue or capital-intensive organizations.
- The Nonprofit Finance Fund recommends a minimum of 3 months (0.25).

Calculated For: 990 filers only.

Usage

```
get_operating_reserve_ratio( df,
    unrestricted_net_assets = "F9_10_NAFB_UNRESTRICT_EOY",
    net_fixed_assets        = "F9_10_ASSET_LAND_BLDG_NET_EOY",
    total_expenses          = "F9_09_EXP_TOT_TOT",
    restricted_net_assets    = "F9_10_NAFB_RESTRICT_EOY",
    total_net_assets        = "F9_10_NAFB_TOT_EOY",
    winsorize = 0.98 ,
    range      = "np",
    sanitize   = TRUE,
    summarize  = FALSE )
```

Arguments

<code>df</code>	A data.frame containing the fields required for computing the metric.
<code>unrestricted_net_assets</code>	Unrestricted net assets, EOY.
<code>net_fixed_assets</code>	Net land, buildings, and equipment, EOY.
<code>total_expenses</code>	Total functional expenses.
<code>restricted_net_assets</code>	Restricted net assets, EOY. Used with <code>total_net_assets</code> to detect filers that do not follow SFAS 117; see resolve_unrestricted_net_assets() .
<code>total_net_assets</code>	Total net assets, EOY. Substituted for unrestricted net assets when unrestricted and restricted net assets are both zero. Set to NULL to disable the fallback.
<code>winsorize</code>	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
<code>summarize</code>	Logical. If TRUE, prints a <code>summary()</code> of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details**Primary uses and key insights:**

The operating reserve ratio answers a key sustainability question: what share of a year of operations can the organization fund from its unrestricted liquid assets, excluding illiquid fixed property? It is the recommended primary reserve adequacy metric of the Nonprofit Finance Fund and is widely used in financial health assessments, lender due diligence, and board reporting.

Unlike days/months of cash measures ([get_days_cash_operations\(\)](#), [get_months_cash_operations\(\)](#)), which count only cash and receivables, the operating reserve ratio starts from the full unrestricted

equity base and removes fixed assets - giving a broader picture of the liquid reserves available to sustain operations.

Formula variations and their sources:

(Unrestricted net assets - net fixed assets) / total expenses. This implementation follows the Non-profit Finance Fund and Zietlow et al. (2007) definition but expresses the result as a multiple of annual expenses rather than in months (multiply by 12 for months). An alternative adds back mortgage debt to fixed assets and divides by monthly expenses ([get_liquid_assets_months\(\)](#)) to more precisely isolate the unencumbered value of fixed property. Some formulations use unrestricted net assets alone (without subtracting fixed assets), which produces higher values.

Canonical citations:

- Nonprofit Finance Fund. *Operating Reserve Policy Toolkit for Nonprofits*.
– Primary source for the operating reserve ratio definition and benchmarks.
- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley.
- Calabrese, T.D. (2013). Running on empty. *Nonprofit Management and Leadership*, 23(3), 281-302. - Empirical analysis of operating reserve levels across the U.S. nonprofit sector.

Variables used::

- F9_10_NAFB_UNRESTRICT_EOY: Unrestricted net assets, EOY (unrestricted_net_assets)
- F9_10_ASSET_LAND_BLDG_NET_EOY: Net land, buildings, and equipment (net_fixed_assets)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)
- F9_10_NAFB_RESTRICT_EOY: Restricted net assets, EOY (restricted_net_assets)
- F9_10_NAFB_TOT_EOY: Total net assets, EOY (total_net_assets)

Filers that do not follow SFAS 117:

Organizations that do not follow SFAS 117 report equity on Part X lines 30-32 and leave lines 27-28 blank, so their unrestricted net assets read as zero. When unrestricted and restricted net assets are both zero, total net assets are used instead (see [resolve_unrestricted_net_assets\(\)](#)).

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `op\_reserve` - operating reserves ratio (raw)
- `op\_reserve\_w` - winsorized version
- `op\_reserve\_z` - standardized z-score (based on winsorized values)
- `op\_reserve\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_operating_reserve_ratio( df = dat10k )
head( d[ , c( "op_reserve", "op_reserve_w", "op_reserve_z", "op_reserve_p" ) ] )
```

get_org_type

Classify nonprofit organization type from NTEE code

Description

Determines the functional type of the organization (regular, advocacy, research, funding, etc.) based on digits 2-3 of the NTEE code.

Usage

```
get_org_type(x)
```

Arguments

x Character vector of NTEE codes.

Value

Character vector of organization type codes: AA (alliance), MT (management/technical), PA (planning/advocacy), RP (research), MS (single-org support), MM (multi-org support), NS (general non-profit support), RG (regular program organization).

Examples

```
get_org_type( c("B01", "B02", "B05", "B11", "B29") )
```

get_overhead_ratio

Overhead Ratio

Description

Combined management and fundraising expenses as a share of total expenses.

Formula:

$$\text{overhead} = (\text{mgmt_expenses} + \text{fundraising_expenses}) / \text{total_expenses}$$
Definitional Range

Bounded [0, 1]. Empirical range approximately [0.05, 0.50] for most operating nonprofits.

Benchmarks and rules of thumb

- **25% or less:** Charity Navigator threshold for a full efficiency score.
- **35% or less:** BBB Wise Giving Alliance standard.
- Low overhead can indicate underinvestment in organizational capacity (the "starvation cycle") rather than genuine efficiency.

Calculated For: 990 filers only.

Usage

```
get_overhead_ratio( df,
  mgmt_expenses      = "F9_09_EXP_TOT_MGMT",
  total_expenses     = "F9_09_EXP_TOT_TOT",
  fundraising_expenses = "F9_09_EXP_TOT_FUNDR",
  winsorize = 0.98 ,
  range       = "zo",
  sanitize    = TRUE,
  summarize   = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
mgmt_expenses	Management and general expenses.
total_expenses	Total functional expenses.
fundraising_expenses	Fundraising expenses.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Primary uses and key insights:

The overhead ratio measures the combined share of total expenses devoted to management and fundraising. It is the standard definition used in the nonprofit accountability literature and by charity watchdog ratings. The overhead ratio complements [get_program_expenses_ratio\(\)](#): the two sum to 1.0. However, its use as a primary quality indicator has been widely criticized.

Formula variations and their sources:

(Management and general + Fundraising) / Total expenses (Part IX lines 25C + 25D / 25A). The administrative-only version ([get_expenses_admin_ratio\(\)](#)) excludes fundraising.

Canonical citations:

- Charity Navigator. *Financial Health Methodology*. charitynavigator.org.
 - Uses overhead ratio as a primary rating factor (target: overhead 25% or less).
- Lecy, J.D. & Searing, E.A. (2015). Anatomy of the nonprofit starvation cycle. *Nonprofit and Voluntary Sector Quarterly*, 44(3), 539-563.
- Hager, M.A. & Flack, T. (2004). The pros and cons of financial efficiency standards. *Non-profit Overhead Cost Project Brief 5*. Urban Institute.

Variables used::

- F9_09_EXP_TOT_MGMT: Management and general expenses (mgmt_expenses)
- F9_09_EXP_TOT_FUNDR: Fundraising expenses (fundraising_expenses)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses)

Value

The original data.frame with four columns appended: overhead, overhead_w, overhead_z, overhead_p.

Examples

```
library( fiscal )
data( dat10k )
d <- get_overhead_ratio( df = dat10k )
head( d[ , c( "overhead", "overhead_w", "overhead_z", "overhead_p" ) ] )
```

get_panel

*Retrieve a multi-year IRS 990 efile panel***Description**

Fiscal convenience wrapper around `panel990::panelize()`, optionally attaching the current geocoded BMF master.

Usage

```
get_panel(
  years,
  tables = c("P00", "P01", "P08", "P09", "P10", "P11", "P12", "A01"),
  include_bmf = TRUE,
  efile_root = "https://nccs-efile.s3.us-east-1.amazonaws.com/public/efile_v2_1/",
  bmf_url = .BMF_URL,
  timeout = 300,
  retry_max = 3L,
  verbose = TRUE,
  backend = "memory",
  path = "efdata",
  cache = "retain",
  filters = NULL,
  columns = NULL,
  include_many = FALSE,
  collision = "error",
  overwrite = FALSE
)
```

Arguments

years	Tax years.
tables	Aliases or literal table names.
include_bmf	Attach BMF fields.
efile_root	Efile URL or local fixture directory.
bmf_url	BMF URL, local path, or in-memory data frame.
timeout	Download timeout.

retry_max	Download attempts.
verbose	Print progress.
backend	Retrieval backend; "memory" or "duckdb".
path	Retained CSV cache directory.
cache	"retain" or "temporary".
filters	Optional named source filters.
columns	Optional fields to retain.
include_many	Join one-to-many and supplemental tables.
collision	Non-key collision policy.
overwrite	Replace cached CSVs.

Value

A data frame with retrieval manifests attached as attributes.

get_pc_fields	<i>Return the vector of 990-only financial field names (PC scope)</i>
---------------	---

Description

Returns financial fields that appear only on the full Form 990 (Parts VIII, IX, and X). These fields are not present on the 990-EZ. Used by [sanitize_financials\(\)](#) to restrict zero-imputation to full 990 filers.

Usage

```
get_pc_fields()
```

Value

A character vector of column names.

Examples

```
get_pc_fields()
```

get_profit_margin_postdepr

Post-Depreciation Profitability Margin

Description

Operating surplus or deficit as a share of total revenue, after depreciation.

Formula:

$$\text{podpm} = (\text{total_revenue} - \text{total_expenses}) / \text{total_revenue}$$

Definitional Range

Bounded above at 1.0; unbounded below. The typical range for nonprofits is approximately [-0.30, 0.30]. Extreme negative values may reflect major write-downs or one-time capital expenses.

Benchmarks and rules of thumb

- **Near zero:** Normal for nonprofits operating close to break-even.
- **0.02-0.07** (2-7%): Healthy.
- **Below -0.05** for two or more consecutive years: Common vulnerability threshold.
- A persistent gap between post- and pre-depreciation margins signals capital consumption requiring future capital expenditure.

Calculated For: 990 filers only.

Usage

```
get_profit_margin_postdepr( df,
    expenses = "F9_09_EXP_TOT_TOT",
    revenue   = "F9_08_REV_TOT_TOT",
    winsorize = 0.98 ,
    range     = "np",
    sanitize  = TRUE,
    summarize = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
expenses	Total functional expenses.
revenue	Total revenue.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").

sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The post-depreciation profit margin measures the operating surplus or deficit as a share of total revenue, using expenses that include depreciation charges. It is the most conservative profit margin measure because depreciation, while non-cash, represents the consumption of capital assets that will eventually require replacement.

This measure is closely related to [get_surplus_margin_ratio\(\)](#) but uses Part VIII revenue and Part IX expenses directly (both 990-only, PC scope) rather than the Part I summary line, making it more precise but narrower in coverage.

Formula variations and their sources:

(Total revenue - Total expenses) / Total revenue, using Part VIII line 12A and Part IX line 25A. The pre-depreciation version ([get_profit_margin_predepr\(\)](#)) adds back depreciation to better approximate cash flow. The PODPM is also called the operating margin in some sources.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.
- Keating, E.K., Fischer, M., Gordon, T.P. & Greenlee, J. (2005). Assessing financial vulnerability in the nonprofit sector. *Harvard Business School Working Paper 04-016*.

Variables used::

- F9_09_EXP_TOT_TOT: Total functional expenses (expenses)
- F9_08_REV_TOT_TOT: Total revenue (revenue)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `profit\_postdepr` - post-depreciation profitability margin (raw)
- `profit\_postdepr\_w` - winsorized version
- `profit\_postdepr\_z` - standardized z-score (based on winsorized values)
- `profit\_postdepr\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_profit_margin_postdepr( df = dat10k )
head( d[ , c( "profit_postdepr", "profit_postdepr_w", "profit_postdepr_z", "profit_postdepr_p" ) ] )
```

get_profit_margin_predepr

Pre-Depreciation Profitability Margin

Description

Operating surplus or deficit as a share of total revenue, before depreciation.

Formula:

$$\text{predpm} = (\text{total_revenue} - (\text{total_expenses} - \text{depreciation})) / \text{total_revenue}$$

Definitional Range

Bounded above at 1.0; unbounded below. Always equal to or greater than the post-depreciation margin ([get_profit_margin_postdepr\(\)](#)) since adding back depreciation can only increase the margin. The typical range is approximately [-0.20, 0.40].

Benchmarks and rules of thumb

- A pre-depreciation margin above zero with a post-depreciation margin below zero is common and generally manageable for capital-intensive nonprofits.
- If both margins are negative, the organization is generating a true cash flow deficit, not just an accounting loss.

Calculated For: 990 filers only.

Usage

```
get_profit_margin_predepr( df,
    revenue      = "F9_08_REV_TOT_TOT",
    expenses     = "F9_09_EXP_TOT_TOT",
    depreciation = "F9_09_EXP_DEPREC_TOT",
    numerator    = NULL, denominator = NULL, winsorize = 0.98 ,
    range        = "np",
    sanitize     = TRUE,
    summarize    = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
revenue	Total revenue.
expenses	Total functional expenses.
depreciation	Depreciation, depletion, and amortization.
numerator	Optional. A pre-calculated column for the numerator (revenue minus non-depreciation expenses). When supplied, expenses and depreciation are ignored, and revenue is used only for the denominator; it is an error to also set expenses or depreciation explicitly.
denominator	Optional. A pre-calculated column for the denominator. When supplied, revenue is used only for the numerator. If both numerator and denominator are supplied, revenue is ignored and it is an error to set it explicitly.

winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The pre-depreciation profit margin approximates cash flow from operations by adding back non-cash depreciation charges to the surplus. Since depreciation is an accounting allocation of past capital expenditure rather than a current cash outflow, this measure better reflects the organization's actual cash-generating ability. A positive pre-depreciation margin alongside a negative post-depreciation margin indicates the organization is cash-flow positive but "booking" a loss due to large depreciation charges.

Formula variations and their sources:

$(\text{Total revenue} - (\text{Total expenses} - \text{Depreciation})) / \text{Total revenue}$. Equivalently, $(\text{revenue} - \text{expenses} + \text{depreciation}) / \text{revenue}$. This is the simplest cash flow approximation available from 990 data; it does not account for changes in working capital, capital expenditures, or debt service, which are not directly reported on the 990.

The EBITDA margin used in commercial analysis is similar in spirit but also adds back interest, taxes, and amortization. The pre-depreciation margin here adds only depreciation (and amortization where included), making it a more conservative approximation.

Canonical citations:

- Keating, E.K., Fischer, M., Gordon, T.P. & Greenlee, J. (2005). Assessing financial vulnerability in the nonprofit sector. *Harvard Business School Working Paper 04-016*.
- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley.

Variables used::

- F9_08_REV_TOT_TOT: Total revenue (revenue)
- F9_09_EXP_TOT_TOT: Total functional expenses (expenses)
- F9_09_EXP_DEPREC_TOT: Depreciation and amortization (depreciation)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `profit\_predepr` - pre-depreciation profitability margin (raw)
- `profit\_predepr\_w` - winsorized version
- `profit\_predepr\_z` - standardized z-score (based on winsorized values)
- `profit\_predepr\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_profit_margin_predepr( df = dat10k )
head( d[, c( "profit_predepr", "profit_predepr_w", "profit_predepr_z", "profit_predepr_p" ) ] )
```

```
get_program_expenses_ratio
```

Program Expense Ratio

Description

Share of total expenses devoted to program services.

Formula:

$$\text{per} = \text{program_expenses} / \text{total_expenses}$$
Definitional Range

Bounded zero to one. Zero means no spending on programs (all overhead); one means all spending is on programs (no administration or fundraising). In practice, values above 0.95 are unusual and may reflect accounting misallocation. The empirical range for most operating nonprofits is approximately 0.50 to 0.95.

Benchmarks and rules of thumb

- **>= 75%:** Charity Navigator threshold for a favorable efficiency score.
- **>= 65%:** BBB Wise Giving Alliance minimum standard.
- **Below 50%:** Commonly flagged as a concern, though legitimate for startup-phase organizations.
- **Caution:** high ratios achieved by underinvesting in administration (the "starvation cycle") can signal long-term organizational weakness (Lecy & Searing 2015).

Calculated For: 990 filers only.

Usage

```
get_program_expenses_ratio( df,
  program_expenses = "F9_09_EXP_TOT_PROG",
  total_expenses   = c( "F9_09_EXP_TOT_TOT", "F9_01_EXP_TOT_CY" ),
  winsorize        = 0.98 ,
  range            = "zo",
  sanitize         = TRUE,
  summarize        = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
program_expenses	Program service expenses.
total_expenses	Total functional expenses. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority over 990EZ.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The program expense ratio is arguably the most widely cited single metric in nonprofit financial analysis. It measures what fraction of total expenses is devoted directly to mission-related program services, as opposed to management, administration, and fundraising. It is the primary basis for charity rating systems (Charity Navigator, Better Business Bureau Wise Giving Alliance) and is commonly used by donors, funders, and board members as a shorthand efficiency measure.

However, it is also one of the most criticized metrics in the literature because it can be gamed (by misallocating shared costs to programs), because overhead is necessary for organizational effectiveness, and because the "ideal" program ratio varies substantially by mission type and operating model.

Formula variations and their sources:

Program service expenses / total functional expenses (Part IX line 25B / line 25A). For 990EZ filers, Part I total expenses is used as the denominator fallback; however, the 990EZ does not separately report program expenses, so this ratio is primarily meaningful for full-990 filers.

An alternative formulation uses total revenue rather than total expenses in the denominator, but the expense-to-expense version is standard and avoids distortions from surplus or deficit years.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.
- Charity Navigator. *Financial Health Methodology*. charitynavigator.org.
 - Uses program expense ratio as a primary rating component (target: 75 percent or more).
- Lecy, J.D. & Searing, E.A. (2015). Anatomy of the nonprofit starvation cycle. *Nonprofit and Voluntary Sector Quarterly*, 44(3), 539-563. - Critical analysis of program expense ratio benchmarks and their perverse incentives.

- Overhead Myth Campaign (GuideStar, BBB Wise Giving, Charity Navigator, 2013).
 - Industry statement cautioning against overreliance on overhead ratios.

Variables used::

- F9_09_EXP_TOT_PROG: Program service expenses (program_expenses, 990)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses, 990)
- F9_01_EXP_TOT_CY: Total expenses from Part I (total_expenses, 990EZ fallback)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``prog_exp`` - program expense ratio (raw)
- ``prog_exp_w`` - winsorized version
- ``prog_exp_z`` - standardized z-score (based on winsorized values)
- ``prog_exp_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_program_expenses_ratio( df = dat10k )
head( d[ , c( "prog_exp", "prog_exp_w", "prog_exp_z", "prog_exp_p" ) ] )
```

<code>get_pz_fields</code>	<i>Return the vector of 990 + 990-EZ financial field names (PZ scope)</i>
----------------------------	---

Description

Returns financial fields that appear on both the full Form 990 and the 990-EZ (Part I summary fields). Zero-imputation via [sanitize_financials\(\)](#) is applied to these fields for all filers.

Usage

```
get_pz_fields()
```

Value

A character vector of column names.

Examples

```
get_pz_fields()
```

get_quick_ratio	<i>Quick Ratio</i>
-----------------	--------------------

Description

Measures liquidity using only the most liquid assets, excluding inventory and prepaid expenses.

Formula:

$$qr = \text{quick_assets} / \text{current_liabilities}$$

$$\text{quick_assets} = \text{cash} + \text{savings} + \text{pledges_receivable} + \text{accounts_receivable}$$

$$\text{current_liabilities} = \text{accounts_payable} + \text{grants_payable}$$

Definitional Range

The quick ratio is bounded below at zero (no asset values are negative on a properly prepared balance sheet) and is unbounded above. In practice the empirical range for nonprofits is approximately [0, 10], with extreme values above 5 typically indicating cash hoarding, an unusually large receivables balance, or an organization with minimal current liabilities (e.g., a grant-making foundation with no accounts payable).

Values below zero can occur when net receivables are negative (allowances for uncollectible pledges exceed gross pledges) or when the denominator carries accounting adjustments. Very large values can result when current liabilities are near zero, producing ratios that are mathematically valid but practically uninterpretable. The default winsorization at the 1st/99th percentiles addresses both extremes.

Benchmarks and rules of thumb

- **Above 1.0:** Conventionally adequate; liquid assets cover current liabilities.
- **0.5-1.0:** Potential short-term stress.
- **Below 0.5:** Common vulnerability threshold (Tuckman & Chang 1991).
- Very high values (above 3-5) may signal excess cash hoarding rather than financial health.
- A declining trend over consecutive years is a stronger warning signal than any single year's value.

Calculated For: 990 filers only.

Usage

```
get_quick_ratio( df,
  cash           = "F9_10_ASSET_CASH_EOY",
  savings        = "F9_10_ASSET_SAVING_EOY",
  pledges_receivable = "F9_10_ASSET_PLEDGE_NET_EOY",
  accounts_receivable = "F9_10_ASSET_ACC_NET_EOY",
  accounts_payable   = "F9_10_LIAB_ACC_PAYABLE_EOY",
  grants_payable     = "F9_10_LIAB_GRANT_PAYABLE_EOY",
  numerator = NULL, denominator = NULL, winsorize = 0.98 ,
  range       = "zp",
  sanitize    = TRUE,
  summarize   = FALSE )
```

Arguments

<code>df</code>	A <code>data.frame</code> containing the fields required for computing the metric.
<code>cash</code>	Cash on hand, EOY.
<code>savings</code>	Short-term investments (savings), EOY.
<code>pledges_receivable</code>	Net pledges and grants receivable, EOY.
<code>accounts_receivable</code>	Accounts receivable, net, EOY.
<code>accounts_payable</code>	Accounts payable and accrued expenses, EOY.
<code>grants_payable</code>	Grants and similar amounts payable, EOY.
<code>numerator</code>	Optional. A pre-aggregated column for quick assets. When supplied, <code>cash</code> , <code>savings</code> , <code>pledges_receivable</code> , <code>accounts_receivable</code> are ignored; it is an error to also set one of them explicitly.
<code>denominator</code>	Optional. A pre-aggregated column for current liabilities. When supplied, <code>accounts_payable</code> , <code>grants_payable</code> are ignored; it is an error to also set one of them explicitly.
<code>winsorize</code>	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
<code>summarize</code>	Logical. If TRUE, prints a <code>summary()</code> of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details**Primary uses and key insights:**

The quick ratio (also called the acid-test ratio) is a short-term liquidity measure that asks whether an organization could pay all of its current obligations immediately, using only assets that can be converted to cash within days or weeks rather than months. Unlike the current ratio ([get_current_ratio\(\)](#)), it excludes inventory and prepaid expenses, which are less reliably liquid. For nonprofits this distinction is especially meaningful: prepaid expenses (insurance premiums, deposits) cannot be recovered quickly, so including them in a liquidity test is misleading.

The quick ratio is most useful for detecting near-term insolvency risk, stress-testing an organization's ability to absorb a sudden revenue disruption, and comparing liquidity positions across organizations of different sizes. It is a common component of multi-ratio financial health scoring models in the nonprofit literature.

Formula variations and their sources:

The standard for-profit formula (Brigham & Houston 2019) defines quick assets as cash + marketable securities + net receivables, and current liabilities as all obligations due within one year. Nonprofit applications require two adaptations:

- **Receivables:** Pledges receivable are included in the nonprofit version because they are a normal operating asset with a defined cash-in timeline. Some analysts exclude multi-year pledges (only the current-year portion is truly liquid), but the 990 Part X does not break pledges into current vs. long-term portions, so the full net amount is used here.
- **Current liabilities:** For-profit firms sum all current liabilities (accounts payable, short-term debt, accrued liabilities, current portion of long-term debt, deferred revenue, etc.). The 990 balance sheet does not label liabilities as current vs. long-term. This implementation uses accounts payable plus grants payable as the best available proxy for near-term obligations, following Tuckman & Chang (1991) and Greenlee & Trussel (2000). Mortgage notes payable (line 23B) are excluded because they are long-term in nature, and unsecured notes (line 24B) are excluded because they typically carry defined maturity dates outside a 12-month window.

Inventories for sale or use (F9_10_ASSET_INV_SALE_EOY, Part X line 8) and prepaid expenses are excluded, as in the standard quick-ratio definition, because they cannot be converted to cash quickly. `get_current_ratio()` includes both.

Why this formula was chosen:

The formula implemented here follows the approach used in the majority of empirical nonprofit financial health studies: a restricted numerator (cash, savings, and receivables only) divided by near-term payables. This is the most conservative defensible version given the ambiguity of 990 balance sheet categories. It is consistent with the operationalization in Tuckman & Chang (1991), the foundational paper establishing financial vulnerability indicators for nonprofits, and with the panel analyses in Greenlee & Trussel (2000) and Keating et al. (2005).

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460. - Introduced the four-indicator financial vulnerability framework for nonprofits; liquidity is a core component.
- Greenlee, J.S. & Trussel, J.M. (2000). Predicting the financial vulnerability of charitable organizations. *Nonprofit Management and Leadership*, 11(2), 199-210. - Applied and extended Tuckman & Chang using logistic regression on IRS 990 data.
- Keating, E.K., Fischer, M., Gordon, T.P. & Greenlee, J. (2005). Assessing financial vulnerability in the nonprofit sector. *Harvard Business School Working Paper 04-016*. - Comprehensive review of ratio-based vulnerability measures, including liquidity.
- Hager, M.A. (2001). Financial vulnerability among arts organizations: A test of the Tuckman-Chang measures. *Nonprofit and Voluntary Sector Quarterly*, 30(2), 376-392. - Tests alternative operationalizations including extended liquidity numerators in an arts sector context.
- Zietlow, J., Hankin, J.A. & Seidner, A. (2007). *Financial Management for Nonprofit Organizations*. Wiley. - Practitioner text with the most detailed discussion of quick ratio benchmarks specific to nonprofits.

Variables used::

- F9_10_ASSET_CASH_EOY: Cash on hand, EOY (cash)
- F9_10_ASSET_SAVING_EOY: Savings and short-term investments, EOY (savings)
- F9_10_ASSET_PLEDGE_NET_EOY: Net pledges receivable, EOY (pledges_receivable)
- F9_10_ASSET_ACC_NET_EOY: Accounts receivable, net, EOY (accounts_receivable)
- F9_10_LIAB_ACC_PAYABLE_EOY: Accounts payable and accrued expenses, EOY (accounts_payable)
- F9_10_LIAB_GRANT_PAYABLE_EOY: Grants and similar amounts payable, EOY (grants_payable)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `quick` - quick ratio (raw)
- `quick\_w` - winsorized version
- `quick\_z` - standardized z-score (based on winsorized values)
- `quick\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_quick_ratio( df = dat10k )
head( d[ , c( "quick", "quick_w", "quick_z", "quick_p" ) ] )
```

get_return_assets_ratio

Return on Assets

Description

Net surplus or deficit as a share of total assets.

Formula:

$$\text{roa} = \text{revenues_less_expenses} / \text{total_assets}$$
Definitional Range

ROA is unbounded in both directions. A ratio of zero means expenses exactly equalled revenues. Positive values indicate surplus years; negative values indicate deficit years. For nonprofits, where perpetual surpluses raise concerns about mission prioritization and perpetual deficits signal sustainability risk, the meaningful range is approximately [-0.20, 0.20] for most organizations in most years. Values outside this range are not impossible but typically reflect unusual circumstances (large asset disposals, one-time gifts, major capital campaigns, or accounting restatements) rather than stable operating performance.

Values can be distorted by very small total asset bases (newly formed organizations) or by organizations whose assets consist almost entirely of illiquid fixed property, making the denominator a poor proxy for the operational resource base.

Benchmarks and rules of thumb

- **Near zero:** Normal for nonprofits operating close to break-even by design.
- **0.02-0.07 (2-7%):** Generally considered healthy – building modest reserves without appearing to hoard resources.
- **Sustained below -0.05:** Common threshold for classifying an organization as at risk (Greenlee & Trussel 2000, Keating et al. 2005).
- Asset structure matters: capital-intensive organizations will mechanically produce low ROA even when financially healthy.

Calculated For: 990 + 990EZ filers.

Usage

```
get_return_assets_ratio( df,
  revenues_less_expenses = "F9_01_EXP_REV_LESS_EXP_CY",
  total_assets           = c( "F9_10_ASSET_TOT_EOY", "F9_01_NAFB_ASSET_TOT_EOY" ),
  winsorize = 0.98 ,
  range     = "np",
  sanitize  = TRUE,
  summarize = FALSE )
```

Arguments

<code>df</code>	A <code>data.frame</code> containing the fields required for computing the metric.
<code>revenues_less_expenses</code>	Revenues less expenses (net surplus or deficit) for the current year. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority over 990EZ.
<code>total_assets</code>	Total assets, EOY.
<code>winsorize</code>	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
<code>summarize</code>	Logical. If TRUE, prints a <code>summary()</code> of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details**Primary uses and key insights:**

Return on assets (ROA) measures how effectively an organization uses its entire asset base to generate a financial surplus. In the nonprofit context this is sometimes called the return on investment ratio, though "investment" here means the total asset base rather than financial securities. A positive ROA indicates the organization ended the year with more resources than it spent; a negative ROA indicates a deficit year.

ROA is most useful as a longitudinal indicator - watching whether an organization is becoming more or less efficient at generating surpluses per dollar of assets over time - and for cross-sectional comparisons within subsectors where asset structures are similar. It is frequently used as a dependent variable in financial vulnerability models and as a component of composite fiscal health scores.

Formula variations and their sources:

In commercial accounting, ROA is typically computed as net income divided by average total assets (the mean of beginning- and end-of-year values), to avoid the distortion of large mid-year asset changes. The nonprofit literature generally uses end-of-year assets as the denominator

(Greenlee & Trussel 2000, Keating et al. 2005) because beginning-of-year balance sheet data is not reliably available on 990 filings prior to recent years, and because the analytical focus is on the stock of resources at the time of reporting rather than a flow average.

The numerator also differs from the commercial definition. Commercial ROA uses net income after taxes. For nonprofits, the equivalent is revenues less expenses (F9_01_EXP_REV_LESS_EXP_CY, Part I line 19), which is the IRS-reported summary figure for the change in net assets before other adjustments. Some studies use the Part XI reconciliation value instead, but that field is 990-only and has higher rates of missing data. The Part I summary field is available on both 990 and 990EZ forms and is used here for broader coverage.

Why this formula was chosen:

End-of-year total assets in the denominator is the most common operationalization in the non-profit empirical literature and is the most reproducible given 990 data availability. The Part I revenues-less-expenses numerator was chosen over the Part XI reconciliation figure because it covers 990EZ filers, has lower rates of missingness, and is the value most analysts expect when referencing the "net income" line on the annual filing. The result is closely related to [get_surplus_margin_ratio\(\)](#) (which scales the same numerator by total revenue rather than total assets), and to [get_return_netassets_ratio\(\)](#) (which scales by beginning net assets).

Canonical citations:

- Greenlee, J.S. & Trussel, J.M. (2000). Predicting the financial vulnerability of charitable organizations. *Nonprofit Management and Leadership*, 11(2), 199-210. - One of the earliest systematic applications of ROA to nonprofit financial health prediction using 990 data.
- Keating, E.K., Fischer, M., Gordon, T.P. & Greenlee, J. (2005). Assessing financial vulnerability in the nonprofit sector. *Harvard Business School Working Paper 04-016*. - Provides comparative analysis of ROA alongside other fiscal health indicators.
- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460. - Foundational paper; ROA variants appear as components of composite vulnerability scores in subsequent applications.
- Wicker, P., Feiler, S. & Breuer, C. (2013). Organizational mission and financial vulnerability: A financial health analysis of German nonprofit sport clubs. *VOLUNTAS: International Journal of Voluntary and Nonprofit Organizations*, 24(4), 991-1013. - Applies ROA in a cross-national nonprofit context; demonstrates subsector variation.

Variables used::

- F9_01_EXP_REV_LESS_EXP_CY: Revenues less expenses, current year (revenues_less_expenses, 990 + 990EZ)
- F9_10_ASSET_TOT_EOY: Total assets, EOY (total_assets; 990 Part X line 16, 990EZ Part II line 25)
- F9_01_NAFB_ASSET_TOT_EOY: Total assets from 990 Part I line 20 (total_assets, fallback)

Value

Object of class data.frame: the original dataframe appended with four new columns:

- `return\_assets` - return on assets (raw)
- `return\_assets\_w` - winsorized version
- `return\_assets\_z` - standardized z-score (based on winsorized values)
- `return\_assets\_p` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_return_assets_ratio( df = dat10k )
head( d[ , c( "return_assets", "return_assets_w", "return_assets_z", "return_assets_p" ) ] )
```

get_return_netassets_ratio
Return on Net Assets

Description

Net surplus or deficit as a share of beginning-of-year net assets.

Formula:

$$\text{rona} = \text{revenues_less_expenses} / \text{net_assets_boy}$$
Definitional Range

Unbounded in both directions. A value of 0 means break-even. The typical range for nonprofits is approximately [-0.30, 0.30]. The ratio is undefined (NA) when beginning net assets equal zero. Extreme values occur when BOY net assets are near zero (small denominator) rather than when the surplus itself is large.

Benchmarks and rules of thumb

- Small positive values (0.02-0.10) are considered healthy.
- Sustained negative RONA over multiple years is a financial vulnerability indicator (Greenlee & Trussel 2000).
- Most useful in trend analysis and within-subsector comparisons.

Calculated For: 990 + 990EZ filers.

Usage

```
get_return_netassets_ratio( df,
  revenues_less_expenses = "F9_01_EXP_REV_LESS_EXP_CY",
  net_assets_boy          = c( "F9_10_NAFB_TOT_BOY", "F9_01_NAFB_TOT_BOY" ),
  winsorize = 0.98 ,
  range     = "np",
  sanitize  = TRUE,
  summarize = FALSE )
```

Arguments

df A data.frame containing the fields required for computing the metric.

revenues_less_expenses Revenues less expenses (net surplus or deficit) for the current year. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority over 990EZ.

net_assets_boy	Total net assets, beginning of year. Accepts one or two column names.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

Return on net assets (RONA) measures how effectively the organization used its equity base to generate a surplus. By using beginning-of-year net assets as the denominator, it expresses the annual surplus as a percentage return on the organizational equity in place at the start of the year - analogous to return on equity (ROE) in commercial finance.

RONA is closely related to [get_netassets_growth_ratio\(\)](#): both measure year-over-year equity change, but RONA uses the reported revenues-less-expenses figure while the growth ratio uses the direct balance sheet comparison (EOY - BOY). The two can differ when there are other net asset adjustments (Form 990 Part XI line 9; Form 990-EZ Part I line 20).

Formula variations and their sources:

Revenues less expenses / net assets BOY. Using beginning-of-year net assets in the denominator (rather than ending or average) avoids circular dependency: the ending value is determined partly by the surplus being measured. This is the standard approach in the nonprofit literature (Greenlee & Trussel 2000).

An alternative uses average net assets ((BOY + EOY)/2), which smooths distortions from large mid-year transactions, but requires two balance sheet fields and is not standard in the 990-based literature.

Canonical citations:

- Greenlee, J.S. & Trussel, J.M. (2000). Predicting the financial vulnerability of charitable organizations. *Nonprofit Management and Leadership*, 11(2), 199-210.
- Keating, E.K., Fischer, M., Gordon, T.P. & Greenlee, J. (2005). Assessing financial vulnerability in the nonprofit sector. *Harvard Business School Working Paper 04-016*.
- Nonprofit Finance Fund. *State of the Nonprofit Sector Survey* (annual).

Variables used::

- F9_01_EXP_REV_LESS_EXP_CY: Revenues less expenses, current year (revenues_less_expenses)
- F9_10_NAFB_TOT_BOY: Total net assets, BOY (net_assets_boy, 990)
- F9_01_NAFB_TOT_BOY: Net assets from Part I, BOY (net_assets_boy, 990EZ fallback)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``return_netassets`` - return on net assets (raw)
- ``return_netassets_w`` - winsorized version
- ``return_netassets_z`` - standardized z-score (based on winsorized values)
- ``return_netassets_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_return_netassets_ratio( df = dat10k )
head( d[ , c( "return_netassets", "return_netassets_w", "return_netassets_z", "return_netassets_p" ) ] )
```

```
get_revenue_fedcampaign_ratio
```

Federated Campaign Revenue Ratio

Description

Federated campaign contributions as a share of total revenue.

Formula:

$$\text{revenue_fedcampaign} = \text{federated_campaigns} / \text{total_revenue}$$
Definitional Range

Bounded [0, 1]. Zero means none of this revenue type was received; one means this channel accounted for all total revenue.

Calculated For: 990 filers only.

Usage

```
get_revenue_fedcampaign_ratio( df,
  federated_campaigns = "F9_08_REV_CONTR_FED_CAMP",
  total_revenue       = "F9_08_REV_TOT_TOT",
  winsorize           = 0.98 ,
  range               = "zo",
  sanitize            = TRUE,
  summarize           = FALSE )
```

Arguments

`df` A `data.frame` containing the fields required for computing the metric.

`federated_campaigns` Federated campaign contributions.

`total_revenue` Total revenue.

winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Revenue Fedcampaign Ratio - Revenue composition measure:

Formula: federated campaign contributions / total revenue. Bounded [0, 1].

Federated campaigns (United Way, Combined Federal Campaign) are a specific channel of collective giving. Organizations heavily funded through federated campaigns face dependency on a single intermediary.

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290.
- Carroll, D.A. & Stater, K.J. (2009). Revenue diversification in nonprofit organizations. *Journal of Public Administration Research and Theory*, 19(4), 947-966.

Variables used::

- F9_08_REV_CONTR_FED_CAMP: Numerator (federated_campaigns)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue)

Variables used:

- F9_08_REV_CONTR_FED_CAMP: Federated campaign contributions (federated_campaigns, 990)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue, 990)

Value

The original data.frame with four new columns:

- revenue_fedcampaign (raw ratio)
- revenue_fedcampaign_w (winsorized)
- revenue_fedcampaign_z (z-score)
- revenue_fedcampaign_p (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_revenue_fedcampaign_ratio( df = dat10k )
head( d[, c( "revenue_fedcampaign", "revenue_fedcampaign_w", "revenue_fedcampaign_z", "revenue_fedcampaign_p" )]
```

get_revenue_fundevents_ratio

Fundraising Events Revenue Ratio

Description

Fundraising event contributions as a share of total revenue.

Formula:

$$\text{revenue_fundevents} = \text{fundraising_event_revenue} / \text{total_revenue}$$

Definitional Range

Bounded [0, 1]. Zero means none of this revenue type was received; one means this channel accounted for all total revenue.

Calculated For: 990 filers only.

Usage

```
get_revenue_fundevents_ratio( df,
    fundraising_event_revenue = "F9_08_REV_CONTR_FUNDR_EVNT",
    total_revenue              = "F9_08_REV_TOT_TOT",
    winsorize                  = 0.98 ,
    range                      = "zo",
    sanitize                   = TRUE,
    summarize                  = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
fundraising_event_revenue	Contributions received through fundraising events (Part VIII line 1c).
total_revenue	Total revenue.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Revenue Fundevents Ratio - Revenue composition measure:

Formula: fundraising event contributions / total revenue. Bounded [0, 1].

Fundraising events (galas, walks, auctions) generate contributed revenue but typically carry high direct expenses. This ratio measures the contribution portion of event receipts (Part VIII line 1c) as a share of total revenue; gross event income (line 8a) and net event income after direct expenses (line 8c) are reported elsewhere.

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290.
- Carroll, D.A. & Stater, K.J. (2009). Revenue diversification in nonprofit organizations. *Journal of Public Administration Research and Theory*, 19(4), 947-966.

Variables used::

- F9_08_REV_CONTR_FUNDR_EVNT: Numerator (fundraising_event_revenue)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue)

Value

The original data.frame with four new columns:

- revenue_fundevents (raw ratio)
- revenue_fundevents_w (winsorized)
- revenue_fundevents_z (z-score)
- revenue_fundevents_p (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_revenue_fundevents_ratio( df = dat10k )
head( d[ , c( "revenue_fundevents", "revenue_fundevents_w", "revenue_fundevents_z", "revenue_fundevents_p" ) ] )
```

```
get_revenue_membdues_ratio
```

Membership Dues Revenue Ratio

Description

Membership dues as a share of total revenue.

Formula:

$$\text{revenue_membdues} = \text{membership_dues} / \text{total_revenue}$$
Definitional Range

Bounded [0, 1]. Zero means none of this revenue type was received; one means this channel accounted for all total revenue.

Calculated For: 990 filers only.

Usage

```
get_revenue_membdues_ratio( df,
  membership_dues          = "F9_08_REV_CONTR_MEMBSHIP_DUE",
  total_revenue             = "F9_08_REV_TOT_TOT",
  winsorize                 = 0.98 ,
  range                    = "zo",
  sanitize                  = TRUE,
  summarize                 = FALSE )
```

Arguments

<code>df</code>	A <code>data.frame</code> containing the fields required for computing the metric.
<code>membership_dues</code>	Membership dues received.
<code>total_revenue</code>	Total revenue.
<code>winsorize</code>	Winsorization proportion between 0 and 1 (default 0.98).
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
<code>summarize</code>	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Revenue Membdues Ratio - Revenue composition measure:

Formula: membership dues / total revenue. Bounded [0, 1].

Membership dues provide relatively predictable recurring revenue tied to member retention. A high ratio indicates a membership-model organization whose financial health depends on maintaining and growing the member base.

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290.
- Carroll, D.A. & Stater, K.J. (2009). Revenue diversification in nonprofit organizations. *Journal of Public Administration Research and Theory*, 19(4), 947-966.

Variables used::

- F9_08_REV_CONTR_MEMBSHIP_DUE: Numerator (membership_dues)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue)

Value

The original `data.frame` with four new columns:

- `revenue_membdues` (raw ratio)
- `revenue_membdues_w` (winsorized)
- `revenue_membdues_z` (z-score)
- `revenue_membdues_p` (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_revenue_membdues_ratio( df = dat10k )
head( d[, c( "revenue_membdues", "revenue_membdues_w", "revenue_membdues_z", "revenue_membdues_p" ) ] )
```

get_revenue_programs_ratio

Program Service Revenue Ratio

Description

Program service revenue as a share of total revenue.

Formula:

$$\text{revenue_programs} = \text{program_service_revenue} / \text{total_revenue}$$

Definitional Range

Bounded [0, 1]. Zero means none of this revenue type was received; one means this channel accounted for all total revenue.

Calculated For: 990 filers only.

Usage

```
get_revenue_programs_ratio( df,
    program_service_revenue = "F9_08_REV_PROG_TOT_TOT",
    total_revenue            = "F9_08_REV_TOT_TOT",
    winsorize                = 0.98 ,
    range                    = "zo",
    sanitize                 = TRUE,
    summarize                = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
program_service_revenue	Total program service revenue.
total_revenue	Total revenue.
winsorize	Winsorization proportion between 0 and 1 (default 0.98).
range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
summarize	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Revenue Programs Ratio - Revenue composition measure:

Formula: program service revenue / total revenue. Bounded [0, 1].

Pure program service share, distinct from get_earned_income_ratio which also includes dues, royalties, and misc.

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290.
- Carroll, D.A. & Stater, K.J. (2009). Revenue diversification in nonprofit organizations. *Journal of Public Administration Research and Theory*, 19(4), 947-966.

Variables used::

- F9_08_REV_PROG_TOT_TOT: Numerator (program_service_revenue)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue)

Value

The original data.frame with four new columns:

- revenue_programs (raw ratio)
- revenue_programs_w (winsorized)
- revenue_programs_z (z-score)
- revenue_programs_p (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_revenue_programs_ratio( df = dat10k )
head( d[ , c( "revenue_programs", "revenue_programs_w", "revenue_programs_z", "revenue_programs_p" ) ] )
```

get_revenue_reltdorgs_ratio

Related Organizations Revenue Ratio

Description

Revenue from related organizations as a share of total revenue.

Formula:

$$\text{revenue_reltdorgs} = \text{related_org_revenue} / \text{total_revenue}$$
Definitional Range

Bounded [0, 1]. Zero means none of this revenue type was received; one means this channel accounted for all total revenue.

Calculated For: 990 filers only.

Usage

```
get_revenue_reltdorgs_ratio( df,
  related_org_revenue      = "F9_08_REV_CONTR_RLTD_ORG",
  total_revenue            = "F9_08_REV_TOT_TOT",
  winsorize                = 0.98 ,
  range                    = "zo",
  sanitize                 = TRUE,
  summarize                = FALSE )
```

Arguments

<code>df</code>	A <code>data.frame</code> containing the fields required for computing the metric.
<code>related_org_revenue</code>	Contributions from related organizations.
<code>total_revenue</code>	Total revenue.
<code>winsorize</code>	Winsorization proportion between 0 and 1 (default 0.98).
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zo". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, imputes zero for NA financial fields before computing, respecting form scope.
<code>summarize</code>	Logical (default FALSE). If TRUE, prints summary statistics and density plots for all four output columns.

Details

Revenue Reltdorgs Ratio - Revenue composition measure:

Formula: related organization revenue / total revenue. Bounded [0, 1].

Revenue from related organizations (subsidiaries, affiliates, supporting organizations) may indicate resource sharing within a nonprofit family rather than arm's-length fundraising. High values warrant review of organizational structure and transfer pricing.

Canonical citations:

- Chang, C.F. & Tuckman, H.P. (1994). Revenue diversification among nonprofits. *VOLUNTAS*, 5(3), 273-290.
- Carroll, D.A. & Stater, K.J. (2009). Revenue diversification in nonprofit organizations. *Journal of Public Administration Research and Theory*, 19(4), 947-966.

Variables used::

- F9_08_REV_CONTR_RLTD_ORG: Numerator (related_org_revenue)
- F9_08_REV_TOT_TOT: Total revenue (total_revenue)

Value

The original `data.frame` with four new columns:

- `revenue_reltdorgs` (raw ratio)
- `revenue_reltdorgs_w` (winsorized)
- `revenue_reltdorgs_z` (z-score)
- `revenue_reltdorgs_p` (percentile rank, 1-100)

Examples

```
library( fiscal )
data( dat10k )
d <- get_revenue_reltdorgs_ratio( df = dat10k )
head( d[, c( "revenue_reltdorgs", "revenue_reltdorgs_w", "revenue_reltdorgs_z", "revenue_reltdorgs_p" ) ] )
```

get_self_sufficiency_ratio
Self Sufficiency Ratio

Description

Measures whether program service revenue is sufficient to cover total expenses.

Formula:

$$\text{ssr} = \text{program_service_revenue} / \text{total_expenses}$$

Definitional Range

Bounded below at zero; values above 1.0 are possible (and indicate program revenue exceeds total expenses - a surplus from earned income alone). The typical range for nonprofits is approximately [0, 1.5], with most values below 1.0 since most nonprofits depend on some contributed income.

Benchmarks and rules of thumb

- **SSR > 1.0:** Fully self-sufficient from program revenue; characteristic of mature social enterprises and fee-based service providers.
- **SSR 0.50-1.0:** Majority of costs covered by program revenue; moderate philanthropy dependence.
- **SSR < 0.25:** Heavily dependent on contributed income; common for advocacy organizations, arts nonprofits, and grant-funded research entities.
- High SSR is not universally better – organizations serving low-income populations often require philanthropic subsidy by design.

Calculated For: 990 + 990EZ filers.

Usage

```
get_self_sufficiency_ratio( df,
    program_service_rev = c( "F9_08_REV_PROG_TOT_TOT", "F9_01_REV_PROG_TOT_CY" ),
    total_expenses      = c( "F9_09_EXP_TOT_TOT",      "F9_01_EXP_TOT_CY"      ),
    winsorize = 0.98 ,
    range     = "zp",
    sanitize  = TRUE,
    summarize = FALSE )
```

Arguments

df	A data.frame containing the fields required for computing the metric.
program_service_rev	Program service revenue. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority over 990EZ.
total_expenses	Total functional expenses. Accepts one or two column names.
winsorize	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.

range	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "zp". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
sanitize	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
summarize	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The self-sufficiency ratio (SSR) measures whether an organization's earned program service revenue is sufficient to cover its total expenses. A ratio of 1.0 means the organization could fully sustain itself from program fees and earned income alone, without any reliance on donations, grants, or investment income. Values below 1.0 indicate the degree of dependence on contributed revenue.

SSR is particularly important in the nonprofit financial sustainability literature as a measure of earned income dependency and mission-related revenue generation. High SSR organizations (social enterprises, fee-for-service providers) are sometimes considered more financially resilient because earned revenue is more predictable and controllable than philanthropic support.

Formula variations and their sources:

Program service revenue / total expenses. Some studies (Young 2007) use total earned income (including membership dues and investment income) in the numerator for a broader self-sufficiency concept. This implementation uses program service revenue only, consistent with the most common definition in the financial vulnerability literature.

Canonical citations:

- Young, D.R. (2007). Financing nonprofits: Putting theory into practice. AltaMira Press. - Extensive discussion of earned income and self-sufficiency concepts.
- Weisbrod, B.A. (1998). The nonprofit mission and its financing. *Journal of Policy Analysis and Management*, 17(2), 165-174. - Examines the tension between earned income and mission.
- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.

Variables used::

- F9_08_REV_PROG_TOT_TOT: Program service revenue (program_service_rev, 990)
- F9_01_REV_PROG_TOT_CY: Program revenue from Part I (program_service_rev, 990EZ fallback)
- F9_09_EXP_TOT_TOT: Total functional expenses (total_expenses, 990)
- F9_01_EXP_TOT_CY: Total expenses from Part I (total_expenses, 990EZ fallback)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``self_suff`` - self sufficiency ratio (raw)
- ``self_suff_w`` - winsorized version
- ``self_suff_z`` - standardized z-score (based on winsorized values)
- ``self_suff_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_self_sufficiency_ratio( df = dat10k )
head( d[ , c( "self_suff", "self_suff_w", "self_suff_z", "self_suff_p" ) ] )
```

```
get_surplus_margin_ratio
```

Surplus Margin

Description

Net surplus or deficit as a share of total revenue.

Formula:

$$sm = \text{revenues_less_expenses} / \text{total_revenue}$$
Definitional Range

Bounded above at 1.0 (expenses cannot be negative). Bounded below at negative infinity in theory, but in practice the empirical range for nonprofits is approximately [-0.30, 0.30] in most years. Values below -0.50 or above 0.50 typically reflect unusual one-time events (large gifts, major write-downs, asset sales).

Benchmarks and rules of thumb

- **Near zero:** Normal and expected for nonprofits operating close to break-even.
- **0.02-0.07** (2-7%): Commonly considered a healthy surplus range.
- **Below -0.05** for two or more consecutive years: Common vulnerability threshold (Greenlee & Trussel 2000).

Calculated For: 990 + 990EZ filers.

Usage

```
get_surplus_margin_ratio( df,
  revenues_less_expenses = "F9_01_EXP_REV_LESS_EXP_CY",
  total_revenue           = c( "F9_08_REV_TOT_TOT", "F9_01_REV_TOT_CY" ),
  winsorize = 0.98 ,
  range     = "np",
  sanitize  = TRUE,
  summarize = FALSE )
```

Arguments

<code>df</code>	A data.frame containing the fields required for computing the metric.
<code>revenues_less_expenses</code>	Revenues less expenses (net surplus or deficit) for the current year. Accepts one or two column names; if two are provided they are coalesced with the 990 value taking priority over 990EZ.
<code>total_revenue</code>	Total revenue. Accepts one or two column names.
<code>winsorize</code>	The winsorization value (between 0 and 1), defaults to 0.98, which winsorizes at the 1st and 99th percentiles.
<code>range</code>	Character string specifying the theoretical range of the ratio, used to set winsorization bounds. Default "np". Options: "np" (negative to positive), "zp" (zero to positive), "zo" (zero to one), "nz" (negative to zero), or a custom "lo;hi" pair (e.g. "0;10").
<code>sanitize</code>	Logical (default TRUE). If TRUE, NA values in the financial input columns are imputed to zero before the ratio is computed, respecting form scope: Part X and VIII/IX fields (990 only) are imputed only for 990 filers; Part I summary fields (990 + 990EZ) are imputed for all filers. The returned dataframe always contains the original unmodified input columns.
<code>summarize</code>	Logical. If TRUE, prints a summary() of the results and plots density curves for all four output columns (raw, winsorized, z-score, percentile). Defaults to FALSE.

Details

Primary uses and key insights:

The surplus margin (also called the operating margin or profit margin) measures what fraction of total revenue remains after all expenses. It is the most direct measure of annual financial performance: positive values indicate a surplus year (more revenue than expenses); negative values indicate a deficit year.

Unlike `get_return_assets_ratio()` (which scales by assets) or `get_return_netassets_ratio()` (which scales by equity), the surplus margin scales by revenue - answering the question: "of each dollar raised, how much is left after all expenses?" It is the nonprofit equivalent of the commercial net profit margin.

Formula variations and their sources:

Revenues less expenses / total revenue. The numerator uses the Part I summary line (F9_01_EXP_REV_LESS_EXP_CY), which is available on both 990 and 990EZ and may include adjustments not captured in a simple revenue-minus-expenses calculation from Parts VIII and IX. An alternative uses (Part VIII total revenue - Part IX total expenses) / Part VIII revenue, which is more precise for full-990 filers but unavailable for 990EZ filers. The Part I version is used here for maximum coverage.

Canonical citations:

- Tuckman, H.P. & Chang, C.F. (1991). A methodology for measuring the financial vulnerability of charitable nonprofit organizations. *Nonprofit and Voluntary Sector Quarterly*, 20(4), 445-460.
- Greenlee, J.S. & Trussel, J.M. (2000). Predicting the financial vulnerability of charitable organizations. *Nonprofit Management and Leadership*, 11(2), 199-210.
- Tuckman, H.P. & Chang, C.F. (1992). Nonprofit equity: A behavioral model and its policy implications. *Journal of Policy Analysis and Management*, 11(1), 76-87.

Variables used::

- F9_01_EXP_REV_LESS_EXP_CY: Revenues less expenses, current year (revenues_less_expenses)
- F9_08_REV_TOT_TOT: Total revenue from Part VIII (total_revenue, 990)
- F9_01_REV_TOT_CY: Total revenue from Part I (total_revenue, 990EZ fallback)

Value

Object of class `data.frame`: the original dataframe appended with four new columns:

- ``surplus_margin`` - surplus margin (raw)
- ``surplus_margin_w`` - winsorized version
- ``surplus_margin_z`` - standardized z-score (based on winsorized values)
- ``surplus_margin_p`` - percentile rank (1-100)

Examples

```
library( fiscal )
data( dat10k )

d <- get_surplus_margin_ratio( df = dat10k )
head( d[ , c( "surplus_margin", "surplus_margin_w", "surplus_margin_z", "surplus_margin_p" ) ] )
```

get_top_n	<i>Compute the upper winsorization threshold for a given range specification</i>
-----------	--

Description

Compute the upper winsorization threshold for a given range specification

Usage

```
get_top_n(x, range, winsorize = 0.98, offset = 0.001)
```

Arguments

- | | |
|-----------|--|
| x | Numeric vector (NAs are ignored). |
| range | Character range code ("np", "zp", "zo", "nz", or "lo;hi"). |
| winsorize | Winsorization proportion (default 0.98). |
| offset | Sentinel offset for fixed bounds (default 0.001). |

Value

A single numeric threshold value, or NA when no upper clipping applies.

guess_normalize_type	<i>Guess the best normalization type from the stable interior of a distribution</i>
----------------------	---

Description

Applies heuristics based on the support range, boundary mass, and zero inflation to select among "asinh", "logit", and "rank_normal". Proportions with boundary mass above boundary_mass_cutoff (at zero or one) get "rank_normal"; "hurdle" is never auto-selected and must be requested through vtype.

Usage

```
guess_normalize_type(
  x,
  range,
  boundary_mass_cutoff = 0.1,
  zero_tol = 1e-08,
  one_tol = 1e-08
)
```

Arguments

x	Numeric vector of stable (non-sentinel, non-NA) winsorized values.
range	Character range code used to clip the vector to its theoretical support before assessment.
boundary_mass_cutoff	Proportion of values at or near zero (or one) that triggers the "rank_normal" path. Default 0.10.
zero_tol	Tolerance for identifying values near zero. Default 1e-8.
one_tol	Tolerance for identifying values near one. Default 1e-8.

Value

A named list with elements type (character) and reason (character explanation).

impute_zero	<i>Impute zero for NA values in financial fields, respecting form scope</i>
-------------	---

Description

For a given set of column names and a data frame, imputes zero for NA values in financial fields while respecting whether a field is available to 990-EZ filers. PC-scope fields (990 only) are imputed only for rows that are not 990-EZ filers. PZ-scope fields (990 + 990-EZ) are imputed for all rows. Rows where every financial variable is NA are left untouched.

Usage

```
impute_zero(dat, vars, ez_rows)
```

Arguments

<code>dat</code>	A <code>data.frame</code> (the working subset, not the full dataset).
<code>vars</code>	Character vector of financial column names to consider.
<code>ez_rows</code>	Logical vector (length <code>nrow(dat)</code>) marking 990-EZ rows, as returned by <code>detect_ez_rows()</code> .

Value

The modified `data.frame` with zeros imputed where appropriate.

Examples

```
data( dat10k )
ez <- detect_ez_rows( dat10k )
dat_imp <- impute_zero( dat10k, vars = get_pc_fields(), ez_rows = ez )
```

<code>inspect_duplicates</code>	<i>Inspect duplicate filings before deduplication</i>
---------------------------------	---

Description

Diagnoses duplicated organization-year filings using the same ranking logic as `deduplicate()`. For each duplicated `by_id` x `by_year` group, the function identifies the row that would be retained and classifies each candidate filing by filing type.

Filing types are:

- "normal": not amended, not group, not partial
- "partial": partial only
- "amended": amended only
- "group": group only
- "amended_partial": amended + partial
- "amended_group": amended + group
- "group_partial": group + partial
- "amended_group_partial": amended + group + partial

The retained filing is indicated separately by the logical column `retained`.

Usage

```
inspect_duplicates(
  df,
  by_id = "EIN2",
  by_year = "TAX_YEAR",
  col_group = "RETURN_GROUP_X",
  col_partial = "RETURN_PARTIAL_X",
  col_amended = "RETURN_AMENDED_X",
  col_stamp = "RETURN_TIME_STAMP",
  return_candidates = FALSE
)
```

Arguments

df	A data.frame containing IRS 990 e-file data.
by_id	Name of the organization identifier column. Default "EIN2".
by_year	Name of the tax year column. Default "TAX_YEAR".
col_group	Name of the group return flag column. Default "RETURN_GROUP_X".
col_partial	Name of the partial-year return flag column. Default "RETURN_PARTIAL_X".
col_amended	Name of the amended return flag column. Default "RETURN_AMENDED_X".
col_stamp	Name of the filing timestamp column. Default "RETURN_TIME_STAMP".
return_candidates	Logical. If TRUE, return the full duplicate candidate table with ranking columns. Default is FALSE.

Value

A list with:

summary_types Count of duplicated candidate rows by filing type, with total, kept, and dropped counts.

summary_sizes Count of duplicated org-years by number of filings.

summary_size_by_type Count of duplicated candidate rows by number of filings and filing type, with total, kept, and dropped counts.

examples One representative duplicated org-year for each number of filings, including all candidate rows in that example group.

candidates Optional full duplicate candidate table, returned only if return_candidates = TRUE.

merge_bmf	<i>Merge current geocoded BMF fields onto a panel</i>
-----------	---

Description

Compatibility wrapper around `panel1990::bmf_merge()`.

Usage

```
merge_bmf(df, bmf_path = .BMF_URL, timeout = 300, verbose = TRUE)
```

Arguments

df	Panel data containing EIN2.
bmf_path	BMF URL, local CSV path, or data frame.
timeout	Retained for compatibility; URL timeout uses the active R download settings.
verbose	Print progress.

Value

Panel rows with native BMF fields and diagnostics attached.

normalize_x

Normalize a numeric vector for multivariate analysis

Description

Convenience wrapper that calls `find_best_normalization()` and `apply_normalization()` in a single step. For repeated application of the same fitted model use those functions directly.

The function winsorizes `x` according to `range` and `winsorize`, isolates the stable interior (non-sentinel, non-NA values) to fit transformation parameters, then scores all observations – including sentinels – using those fitted parameters. This prevents boundary pile-up at winsorization bounds from distorting the centering and spread estimates (for `"rank_normal"` the reference distribution keeps the winsorized tails; see below).

Four transformation types are supported:

- `"asinh"`: inverse hyperbolic sine, scaled by the median absolute nonzero value of the stable interior. Defined at zero, accommodates negatives, and behaves like `log` in the tails.
- `"logit"`: maps the stable interior to `[0, 1]` relative to its theoretical support, then applies `qlogis()`. Suited to variables bounded in `[0, 1]` without heavy boundary mass.
- `"rank_normal"`: rank-based inverse normal transform. Preserves ordering, not distances. Robust to irregular shapes and mixed support.
- `"hurdle"`: structural-zero mass receives 0; positive values are transformed separately (rank or logit) and rescaled to a positive interval.

When `vtype = NULL` the type is auto-detected from the stable interior using heuristics based on the support range and boundary mass (`"hurdle"` is only used when requested explicitly).

Scores stay bounded at the edges of the distribution so that they preserve the rank correlations of the raw data:

- `"rank_normal"` uses mid-ranks, so a pile-up of tied values (e.g. exact zeros, or program-expense ratios of exactly one) is scored at the middle of its block, and winsorized tails are scored at their true quantile.
- `"logit"` clamps to the $(1 - \text{winsorize}) / 2$ quantiles of the strictly interior values, so values at 0 or 1 are not sent to $\pm \log(2n)$.
- When more than half the transformed values are tied, the MAD is zero and robust standardization falls back to mean/SD instead of returning NA.

Usage

```
normalize_x(
  x,
  range,
  vtype = NULL,
  winsorize = 0.98,
  offset = 0.001,
  standardize = TRUE,
  robust = TRUE,
  zero_tol = 1e-08,
  one_tol = 1e-08,
  boundary_mass_cutoff = 0.1,
```

```

    hurdle_trans = c("rank", "logit"),
    hurdle_range = c(1, 100),
    verbose = TRUE
  )

```

Arguments

x	A numeric vector.
range	Character range code ("np", "zp", "zo", "nz", or "lo;hi"). Required.
vtype	Optional transformation override. One of NULL, "asinh", "logit", "rank_normal", "hurdle".
winsorize	Winsorization proportion. Default 0.98.
offset	Sentinel offset for fixed bounds. Default 0.001.
standardize	Logical. Standardize after transformation. Default TRUE.
robust	Logical. Use median/MAD standardization. Default TRUE.
zero_tol	Tolerance for zero detection. Default 1e-8.
one_tol	Tolerance for near-one detection. Default 1e-8.
boundary_mass_cutoff	Share of values at zero (or one) above which a [0, 1] variable is given "rank_normal" instead of "logit". Default 0.10.
hurdle_trans	Positive-part method for hurdle variables ("rank" or "logit"). Default "rank".
hurdle_range	Rescaled range for hurdle positive component. Default c(1, 100).
verbose	Print diagnostic summaries. Default TRUE.

Value

A numeric vector of the same length as x, with transformation metadata attached as attributes (see [apply_normalization\(\)](#)).

See Also

[find_best_normalization\(\)](#), [apply_normalization\(\)](#), [plot_normalize_x\(\)](#)

Examples

```

x <- c( NA, rnorm(200), rep(0, 20), 50, -50 )

# one-step normalize with auto-detected transformation
z <- normalize_x( x, range = "np", verbose = FALSE )
hist( z, breaks = 30, main = "Normalized" )

# force a specific transformation
z_asinh <- normalize_x( x, range = "np", vtype = "asinh", verbose = FALSE )

```

panel_composition	<i>Summarize panel composition over time</i>
-------------------	--

Description

Compatibility wrapper around panel990's classification verbs. Classification uses the persistent, entrant, exit, transient, and empty boundary types and the independent seamless/segmented spell dimension.

Usage

```
panel_composition(
  df,
  time = "TAX_YEAR",
  id = "EIN2",
  append_classification = FALSE,
  return_classification = TRUE,
  print_table = TRUE
)
```

Arguments

df	A panel data frame.
time	Name of the panel-time column.
id	Name of the panel-ID column.
append_classification	Append classification fields to every row (delegates to panel990::panel_label()).
return_classification	Return the per-ID classification.
print_table	Print the year-by-type summary (delegates to panel990::panel_describe()).

Value

An enriched panel, a per-ID classification, or (invisibly) the input.

panel_filter_types	<i>Filter a panel by composition type and spell balance</i>
--------------------	---

Description

Compatibility wrapper around panel990::panel_filter().

Usage

```
panel_filter_types(
  df,
  panel_types,
  keep = .PANEL_TYPES,
  spell_balance = .PANEL_SPELL_BALANCE,
  id = "EIN2"
)
```

Arguments

df	A panel data frame.
panel_types	A per-ID classification, classification summary, or data with appended panel_* columns.
keep	Panel types to retain: "persistent", "entrant", "exit", "transient", or "empty".
spell_balance	Spell values to retain: "seamless" and/or "segmented".
id	Name of the ID variable.

Value

Filtered input rows.

panel_impute	<i>Impute missing panel years</i>
--------------	-----------------------------------

Description

Compatibility wrapper around `panel990::panel_impute()`.

Usage

```
panel_impute(
  df,
  panel_types = NULL,
  types = "persistent",
  max_gap_size = Inf,
  max_gap_count = Inf,
  vars = NULL,
  time = "TAX_YEAR",
  id = "EIN2",
  as_integers = FALSE
)
```

Arguments

df	Panel data frame.
panel_types	Optional panel classification.
types	Eligible panel boundary types (default "persistent").
max_gap_size	Maximum single gap length.
max_gap_count	Maximum gaps per organization.
vars	Numeric fields to fill.
time	Panel-time column.
id	Panel-ID column.
as_integers	Round integer field imputations.

Value

Panel data with inserted rows flagged by `imputed_row`.

panel_smooth	<i>Smooth financial variables within panel organizations</i>
--------------	--

Description

Compatibility wrapper around `panel990::panel_smooth()`. Fiscal retains the PZ, PC, and ALL field shortcuts.

Usage

```
panel_smooth(  
  df,  
  vars = "PZ",  
  window = 3,  
  time = "TAX_YEAR",  
  id = "EIN2",  
  weights = "equal",  
  verbose = TRUE,  
  engine = "memory"  
)
```

Arguments

df	Panel data frame.
vars	"PZ", "PC", "ALL", or numeric field names.
window	Odd rolling-window width.
time	Panel-time column.
id	Panel-ID column.
weights	Weighting mode.
verbose	Print progress.
engine	Must currently be "memory".

Value

Smoothed panel data.

panel_summary	<i>Print a panel composition summary from appended classification columns</i>
---------------	---

Description

Produces a year-by-type count table from a data frame that already has `panel_*` columns appended by `panel_composition(append_classification = TRUE)`. Useful for quickly re-checking composition after imputation or other panel manipulations without re-running the full classification.

Usage

```
panel_summary(df, time = "TAX_YEAR", id = "EIN2", print_table = TRUE)
```

Arguments

df	A data frame with panel_type and the time column present.
time	Name of the time variable column. Default "TAX_YEAR".
id	Name of the organization identifier column. Default "EIN2".
print_table	Logical. If TRUE, print the formatted table. Default TRUE.

Value

A data frame with one row per panel year and one column per panel type, invisibly. Same structure as the summary table returned by `panel_composition()`.

See Also

[panel_composition\(\)](#)

Examples

```
## Not run:
d <- panel_composition(d, append_classification = TRUE)
d <- panel_impute(d)
panel_summary(d)

## End(Not run)
```

plot_normalize_x	<i>Plot raw and normalized versions of a numeric vector</i>
------------------	---

Description

Diagnostic companion to [normalize_x\(\)](#). Plots side-by-side histograms of the raw and normalized distributions.

Usage

```
plot_normalize_x(
  x,
  range = NULL,
  normalize = TRUE,
  breaks = 40,
  col_raw = "steelblue",
  col_norm = "firebrick",
  border = "white",
  main_raw = "Raw",
  main_norm = NULL,
  ...
)
```

Arguments

x	A numeric vector (raw input, or already-normalized if <code>normalize = FALSE</code>).
range	Character range code passed to <code>normalize_x()</code> when <code>normalize = TRUE</code> . Required unless <code>normalize = FALSE</code> .
normalize	Logical. If <code>TRUE</code> (default), normalize x inside the function. If <code>FALSE</code> , treat x as already normalized.
breaks	Number of histogram breaks. Default 40.
col_raw	Fill color for the raw histogram. Default "steelblue".
col_norm	Fill color for the normalized histogram. Default "firebrick".
border	Bar border color. Default "white".
main_raw	Title for the raw panel. Default "Raw".
main_norm	Optional title for the normalized panel. If <code>NULL</code> , uses the <code>transform_label</code> attribute when available.
...	Additional arguments passed to <code>normalize_x()</code> when <code>normalize = TRUE</code> .

Value

Invisibly returns a list with elements `raw` and `normalized`.

See Also

`normalize_x()`, `find_best_normalization()`

resolve_col

Resolve one or two candidate column names to a single numeric vector

Description

Given one or two candidate column names and a data frame, returns a single numeric vector. If both columns are present, they are coalesced with the first (990 PC) taking priority and the second (990-EZ) filling in where the first is NA.

Usage

```
resolve_col(dat, cols)
```

Arguments

dat	A <code>data.frame</code> .
cols	Character vector of one or two column names.

Value

A numeric vector of length `nrow(dat)`.

resolve_unrestricted_net_assets

Resolve unrestricted net assets with a fallback for non-SFAS 117 filers

Description

Part X lines 27-28 (net assets without / with donor restrictions) are completed only by organizations that follow SFAS 117 (ASC 958). Other filers report equity on lines 30-32 (capital stock, paid-in surplus, retained earnings), so their unrestricted net assets read as zero even when total net assets (line 33) are large. Without donor-restriction reporting, that equity is effectively unrestricted.

This helper returns unrestricted net assets, substituting total net assets for rows where unrestricted **and** restricted net assets are both zero. Rows where unrestricted net assets are NA (e.g. 990-EZ filers) are left NA.

Usage

```
resolve_unrestricted_net_assets(
  dat,
  unrestricted = "F9_10_NAFB_UNRESTRICT_EOY",
  restricted = "F9_10_NAFB_RESTRICT_EOY",
  total = "F9_10_NAFB_TOT_EOY",
  verbose = TRUE
)
```

Arguments

dat	A data.frame.
unrestricted	Column name(s) for unrestricted net assets (line 27).
restricted	Column name(s) for restricted net assets (line 28), or NULL. When NULL or absent from dat, restricted net assets are treated as zero, so the fallback applies wherever unrestricted net assets are zero.
total	Column name(s) for total net assets (line 33), or NULL to disable the fallback.
verbose	Logical. Print the number of substituted rows. Default TRUE.

Value

A numeric vector of length nrow(dat).

Examples

```
d <- data.frame( F9_10_NAFB_UNRESTRICT_EOY = c( 100, 0, 0 ),
                 F9_10_NAFB_RESTRICT_EOY   = c( 50, 0, 80 ),
                 F9_10_NAFB_TOT_EOY        = c( 150, 900, 80 ) )
resolve_unrestricted_net_assets( d ) # 100 900 0
```

retrieve_efile_data	<i>Retrieve and merge IRS 990 efile tables for one tax year</i>
---------------------	---

Description

Compatibility wrapper around `panel990::panelize()`. It returns a data frame and attaches structured download, table, join, and BMF diagnostics.

Usage

```
retrieve_efile_data(
  year,
  tables = c("P00", "P01", "P08", "P09", "P10"),
  include_bmf = TRUE,
  bmf_vars = .BMF_VARS,
  join_1xm = FALSE,
  efile_root = "https://nccs-efile.s3.us-east-1.amazonaws.com/public/efile_v2_1/",
  bmf_url = .BMF_URL,
  timeout = 300,
  retry_max = 3L,
  path = "efdata",
  keep_files = TRUE,
  verbose = TRUE,
  backend = "memory",
  cache = NULL,
  filters = NULL,
  columns = NULL,
  collision = "error",
  overwrite = FALSE
)
```

Arguments

year	One tax year.
tables	Aliases or literal canonical table names.
include_bmf	Attach current geocoded BMF fields.
bmf_vars	Native BMF fields to retain; NULL retains all available.
join_1xm	Join one-to-many and supplemental tables.
efile_root	Efile URL or local source directory.
bmf_url	BMF URL, path, or data frame.
timeout	Download timeout.
retry_max	Download attempts.
path	Retained CSV cache directory.
keep_files	Retain downloaded CSVs. When false, use a temporary cache.
verbose	Print progress.
backend	"memory" or "duckdb".

cache	Optional explicit cache mode: "retain", "temporary", or "none". By default it is derived from keep_files.
filters	Optional named source filters.
columns	Optional source fields to retain.
collision	Non-key collision policy.
overwrite	Replace cached files.

Value

A data frame with structured retrieval attributes.

sanitize_financials	<i>Sanitize Financial Variables in an IRS 990 Dataset</i>
---------------------	---

Description

Impute zero for NA values in IRS 990 financial fields, respecting the distinction between fields available to all filers (990 + 990EZ) and fields that only appear on the full 990 form.

Usage

```
sanitize_financials(df, pz_vars = NULL, pc_vars = NULL)
```

Arguments

df	A data.frame containing IRS 990 efile financial fields. Should contain a RETURN_TYPE column (values "990" or "990EZ") for accurate filer-type detection. If absent, filer type is inferred from field availability.
pz_vars	Optional character vector of PZ-scope column names (present on both 990 and 990EZ). When supplied, the local scope-aware imputer is used with exactly these fields. When NULL (default), scope is resolved from panel990 via panel990::panel_normalize
pc_vars	Optional character vector of PC-scope column names (full 990 only). When supplied, imputation is restricted to 990 filers for these fields. When NULL (default), scope is resolved from panel990.

Details

In IRS 990 efile data, organizations often leave financial line items blank when the value is zero rather than explicitly reporting zero. These blanks are typically encoded as NA in processed datasets. For ratio calculations this is problematic: a nonprofit with no investment income genuinely has a zero in that field, not a missing value.

sanitize_financials() corrects this by imputing zero for NA values in financial fields, subject to an important constraint: fields that only exist on the full 990 form (PC scope; most of the Part VIII, IX, and X detail, see [get_pc_fields\(\)](#)) should not be imputed to zero for 990EZ filers, since those fields simply don't exist on the EZ form. Fields that appear on both forms (PZ scope; the Part I summary plus lines with an EZ counterpart such as total assets, see [get_pz_fields\(\)](#)) can be safely imputed for all filers. Rows in which every financial field is NA are left untouched.

Filer type is determined from the RETURN_TYPE column when present ("990" = full filer, "990EZ" = short-form filer). If RETURN_TYPE is absent, filer type is inferred: rows with Part I data but missing Part VIII data are treated as 990EZ filers.

As of the panel990 portage, the default path delegates to `panel990::panel_normalize()`, which applies the same form-scoped, non-filer- protected zero imputation over the shared concordance-derived field scopes. Supplying `pz_vars/pc_vars` explicitly falls back to the local imputer for backward compatibility.

Before imputing, bit64 integer64 columns are converted to double. Columns holding integer64 bytes with the class attribute dropped (every nonzero value smaller than 1e-300 in absolute value) are repaired with a warning; imputing them first would zero out negative amounts, which such columns read as NaN.

Value

A `data.frame` identical in structure to `df`, with NA values replaced by zero in the applicable financial columns. The original `df` is not modified.

Examples

```
library( fiscal )
data( dat10k )

# Sanitize the full dataset before computing ratios
dat_clean <- sanitize_financials( dat10k )

# All ratios can then be computed without worrying about NA/zero ambiguity
dat_clean <- get_grants_govt_ratio( dat_clean )
dat_clean <- get_program_expenses_ratio( dat_clean )
```

validate_inputs

Validate numerator and denominator inputs for get_ functions*

Description

Checks that `winsorize` is in `[0, 1]` and that numerator and denominator arguments are not in an inconsistent NULL state.

Usage

```
validate_inputs(
  winsorize,
  num_args,
  den_args,
  num_name = "numerator",
  den_name = "denominator"
)
```

Arguments

<code>winsorize</code>	Numeric scalar.
<code>num_args</code>	Numerator argument value (may be NULL).
<code>den_args</code>	Denominator argument value (may be NULL).
<code>num_name</code>	Name of numerator argument used in error messages.
<code>den_name</code>	Name of denominator argument used in error messages.

Value

Invisibly NULL; called for its side-effect of stopping on invalid input.

winsorize_var	<i>Backward-compatible alias for <code>apply_transformations()</code></i>
---------------	---

Description

`winsorize_var()` is a deprecated alias retained for backward compatibility. New code should use [apply_transformations\(\)](#) directly.

Usage

```
winsorize_var(
  x,
  winsorize = 0.98,
  offset = 0.001,
  range = "np",
  normalize_type = NULL
)
```

Arguments

x	Numeric vector (the computed ratio, before any transformation).
winsorize	Winsorization proportion between 0 and 1 (default 0.98, which clips at the 1st and 99th percentiles for "np" range).
offset	Sentinel offset applied to fixed bounds (default 0.001). Observations clipped to a fixed bound are stored just outside it (a lower bound minus offset, an upper bound plus offset) so they remain identifiable in the <code>_w</code> column.
range	Character string describing the theoretical range of the ratio. Controls how the lower and upper winsorization bounds are determined: "np" Negative to positive (unbounded both directions). Winsorizes symmetrically at the $(1 - \text{winsorize})/2$ and $1 - (1 - \text{winsorize})/2$ percentiles. Default behaviour. "zp" Zero to positive. The lower bound is fixed at <code>-offset</code> (flagging truncated-at-zero values) and the upper bound is the <code>winsorize</code> percentile of the full distribution. "zo" Zero to one. Both bounds are fixed (<code>-offset</code> and <code>1+offset</code>), flagging values outside <code>[0, 1]</code> . No percentile-based clipping is applied. "nz" Negative to zero. The upper bound is fixed at <code>+offset</code> and the lower bound is the <code>1 - winsorize</code> percentile. "lo;hi" Custom numeric range, e.g. "0;10". The lower bound is fixed at <code>lo - offset</code> and the upper bound at <code>hi + offset</code> .
normalize_type	Transformation type override passed to find_best_normalization() . One of NULL (auto-detect among "asinh", "logit", and "rank_normal"), "asinh", "logit", "rank_normal", or "hurdle" (never auto-selected). Default NULL.

Value

Same as [apply_transformations\(\)](#).

winsorize_x

Winsorize a numeric vector according to a range specification

Description

Clips x to $[\text{bottom}, \text{top}]$ bounds determined by `range` and `winsorize`, flags which observations were reassigned to a sentinel bound, and returns a list suitable for downstream normalization.

Usage

```
winsorize_x(x, range, winsorize = 0.98, offset = 0.001)
```

Arguments

<code>x</code>	Numeric vector (NAs are ignored).
<code>range</code>	Character range code ("np", "zp", "zo", "nz", or "lo;hi").
<code>winsorize</code>	Winsorization proportion (default 0.98).
<code>offset</code>	Sentinel offset for fixed bounds (default 0.001).

Value

A named list with elements:

- `x_w` — winsorized vector
- `top, bottom` — the computed bounds
- `is_top_sentinel, is_bottom_sentinel, is_sentinel` — logical vectors
- `support` — parsed range specification list
- `winsorize, offset` — echoed inputs

Examples

```
x <- c( NA, rnorm(200), 100, -100 )

# winsorize with symmetric np bounds
w <- winsorize_x( x, range = "np", winsorize = 0.98 )
range( w$x_w, na.rm = TRUE )
sum( w$is_sentinel )
```

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